

0011007

E.C.JORDAN CO.

ENGINEERS &
SCIENTISTS



APPENDIX A

ARAR IDENTIFICATION CHECKLISTS

EPA Region 5 Records Ctr.



234879

APPENDIX E

SUMMARY TABLES OF PERTINENT MONITORING WELL INFORMATION

APPENDIX E
MONITORING WELL INSTALLATION SPECIFICATIONS

SOURCE LOCATION MONITORING WELLS

Monitoring Well	ELEVATION		DEPTH				VOCs at Screen
	Ground	TOC	Bottom of Boring	Upper Aquitard	Lower Aquitard	Screen Interval	
SLW-1	1298.2	1300.17	45	NP	--	38-43	ND
SLW-2	1298.9	1298.75	89	(25-26)	--	54-59	57
SLW-3	1296.4	1298.72	89	(37-39)	--	54-59	Tr
SLW-4	1297.3	1296.92	89	NP	--	54-59	34
SLW-5	1297.0	1296.78	89	(30-30.5)	--	54-59	Tr
SLW-6	1295.1	1296.69	64	64+	--	34-39	5,754
SLW-7	1299.6	1301.14	89	NP	--	55-60	13
SLW-8	1296.5	1298.32	77.5	77.5+	--	45-50	483
SLW-9	1295.3	1296.49	74	74+	--	55-60	10,079
SLW-10S	1296.7	1298.61	47	47+	--	42-47	4,059
SLW-10D	1296.7	1299.60	177	(48-56)	NE	102-107	35,407
SLW-11	1295.0	1295.03	84	84+	--	44-49	279
SLW-12S	1291.5	1295.05	40	40+	--	35-40	5,968
SLW-12D	1291.5	1294.47	163	(40-62)(69-70) (82-83)	NE	78-83	345
SLW-13	1295.9	1295.69	132	NP	--	127-132	171
SLW-14	1296.4	1295.90	89	NP	--	54-59	Tr
SLW-15	1295.8	1297.18	194	(53-54)(63-74)	194	83-88	6,330
SLW-16	1288.8	1288.99	89	(45-61)89+	--	84-89	Tr
SLW-17	1296.2	1298.44	89	NP	--	53-58	Tr
SLW-18	1294.1	1297.20	191	(33-36)(46-49) (50-63)	189	150-155	Tr
SLW-19	1295.6	1295.33	89	NP	--	54-59	Tr
SLW-20	1299.7	1301.18	178	NP	NE	143-148	139
SLW-21S	1299.6	1301.32	106	106+	--	101-106	Tr
SLW-21D	1299.5	1302.57	173	(106-121)	NE	168-173	Tr
SLW-22	1296.0	1297.74	121	(75-92)	--	115-120	1,970
SLW-23	1297.9	1301.70	89	(55-58)	--	50-55	ND
SLW-24	1297.1	1300.90	113	113+	--	95-100	5
SLW-25	1296.8	1300.09	92	91+	--	43-48	11,521
SLW-26	1291.9	1293.60	94	(24-30)	--	39-44	6
SLW-27	1293.6	1296.27	89	NP	--	84-89	51
SLW-28	1299.1	1301.56	116	116+	--	111-116	Tr
SLW-29	1297.4	1299.36	90	(24-26)	--	85-90	Tr
SLW-30	1299.4	1301.87	68	64.5+	--	49-54	694
SLW-31	1297.5	1299.21	49	47.5+	--	41-46	246
SLW-32	1298.2	1300.28	69	68.5+	--	62-67	Tr
SLW-33	1296.6	1298.41	94	NP	--	44-49	331
SLW-34	1295.3	1295.08	95	NP	--	90-95	3

APPENDIX E (PAGE 2)

SOURCE LOCATION MONITORING WELLS

Monitoring Well	ELEVATION		DEPTH					
	Ground	TOC	Bottom of Boring	Upper Aquitard	Lower Aquitard	Screen Interval	VOCs at Screen	
SLW-35	1297.9	1300.05	94	(14-16.5)	(65-70)	--	87-92	Tr
SLW-36	1296.4	1298.52	94	NP		--	44-49	Tr
SLW-37	1299.9	1299.83	114	NP		--	44-49	1,894
SLW-38	1299.6	1301.72	64	(29-32)	63+	--	54-59	2,780
SLW-39	1295.6	1298.92	94	NP		--	60-65	141
SLW-40	1295.7	1300.19	88	(39-41)		--	51-56	50
SLW-41	1296.7	1299.57	145	NP		NE	133-138	9,934
SLW-42	1297.7	1300.44	133	NP		NE	124-129	6,230
SLW-43	1297.6	1300.22	150	NP		NE	42-47	5,447
SLW-44	1297.8	1300.83	150	NP		150	49-54	1,000
SLW-45	1297.8	1300.45	140	NP		140	48-53	5
SLW-46	1296.0	1298.78	153	NP		NE	48-53	20
SLW-47	1297.8	1301.16	137	NP		NE	69-74	2,735
SLW-48	1297.1	1299.90	123	NP		NE	48-53	25
SLW-49	1293.7	1296.86	148	(60-65)		NE	75-80	405
SLW-50	1297.5	1300.24	138	NP		NE	98-103	223,416
SLW-51	1297.3	1300.01	125	NP		NE	118-123	86
SLW-52	1297.8	1300.34	143	NP		NE	105-110	7
SLW-53	1294.2	1297.28	173	NP		NE	62-67	

TOC = Top of Casing

NP = Not Present

NE = Not Encountered (but may be present at greater depth)

VOC = Results in parts per billion (ppb) and based on screened auger analyses

ND = Not Detected

Tr = Trace

APPENDIX E (PAGE 3)

GENERAL MONITORING WELLS

Monitoring Well	ELEVATION		Bottom of Boring	DEPTH			VOCs at Screen
	Ground	TOC		Upper Aquitard	Lower Aquitard	Screen Interval	
GW-1S	1293.3	1295.51	57	(14-17)(18-42)	--	52-57	3
GW-1D	1293.3	1296.34	178	(14-17)(18-42) (57-63)	NE	165-170	ND
GW-2	1293.2	1295.65	168	(17-19)(24-26)	NE	162-167	Tr
GW-3	1295.3	1297.09	183	NP	NE	166-171	Tr
GW-6S	1298.1	1299.26	39	(14-18)(24-26) (29-30)	NE	34-39	163
GW-6D	1298.1	1300.25	188	(14-17)(25-28) (89-92)(94-103)	NE	181-186	Tr
GW-7	1297.4	1300.00	173	NP	NE	165-170	Tr
GW-8	1291.0	1294.07	203	(42-46)(104-115)	NE	177-182	Tr
GW-9S	1291.6	1292.84	101	(29-44) 103+	NE	96-101	ND
GW-9D	1291.6	1293.24	203	(31-44)(103-106)	NE	161-166	201
GW-10S	1288.6	1291.60	28	28+	--	23-28	816
GW-10D	1289.1	1292.33	168	(32-34)(47-55) (67-70)(74-80)	NE	163-168	439
GW-11		1301.92	173	(22-23)	NE	54-59	Tr
GW-13	1296.2	1299.22	178	(90-94)	178	163-168	ND
GW-14S	1295.2	1298.35	35	(19-25)	--	30-35	10
GW-14D	1295.2	1297.53	198	(19-25)(73-77) (93-111)	198	165-170	ND
GW-16	1295.5	1297.66	203	NP	NE	70-75	ND
GW-17	1297.5	1297.10	188	NP	NE	173-178	ND
GW-18	1296.4	1298.67	187	(73-74)	NE	163-168	ND
GW-20	1295.8	1298.31	148	(51-72)	NE	141-146	Tr

TOC = Top of Casing

NP = Not Present

NE = Not Encountered (but may be present at greater depth)

VOC = Results in parts per billion (ppb) and based on screened auger analyses

ND = Not Detected

Tr = Trace

APPENDIX E (PAGE 4)
PIEZOMETERS AND MUD BORINGS

Monitoring Well	ELEVATION		DEPTH			
	Ground	TOC	Bottom of Boring	Upper Aquitard*	Lower Aquitard	Screen Interval
P-1	1299.3	1301.83	53	(32-33.5)	--	45-50
P-2	--	1301.04	53	NP	--	44-49
P-3	1287.5	1289.65	90	NP	--	34-36
P-4	1302.2	1304.40	47	(16-20)47+	--	34-36
P-5	1347.0	1349.31	54	(12-14.5)	--	52-54
MB-1	--	--	266	(68-73) (127-129)	NE	--
MB-2	1297.9	--	316	(7-8)(14-18) (26-33)(65-67) (70-73)(154-157) (197-199)(201-203)	(232-236) (240-244)	--
MB-3	--	--	220	(29-32)(38-54) (107-118)	218+	--
MB-4	1296.4	--	316	(13-14)(31-32)	(231-234)	--
MB-5	1297.5	--	300	(30-31)(75-77)	(217-222) (238-240)	-- --
MB-6	1296.3	--	300	(65-67)(70-75)	(197-200) (205-209) (232-236)	--
MRW-1	1293.5	1296.52	298	(12-14)(74-75)	(175-178) (204-205) (225-227) (228-235)	292-297
MRW-2	1297.2	1300.11	300	(3-7)(128-129)	(210-212) (214-216) (244-246)	295-300
MRW-3	1294.5	1297.70	300	(10-13)(17-18) (49-52)(143-148)	(219-221) (235-241)	295-300

NP = Not Present

NE = Not Encountered (but may be present at greater depth)

*Clay intervals listed for MB and MRW borings are based on geophysical logging.

APPENDIX E (PAGE 5)

NORTHERNAIRE MONITORING WELLS

Monitoring Well	ELEVATION		DEPTH				TCE at Screen 12/85
	Ground	TOC	Bottom of Boring	Upper Aquitard	Lower Aquitard	Screen Interval	
MW-1		bent	64	63+	--	59-64	--
MW-1S	1294.0	1295.18	40	--	--	35-40	--
MW-2	1294.8	1297.83	69	67+	--	48-53	--
MW-3	1295.3	1297.41	51	50+	--	38-48	--
MW-4	1293.5	1295.06	33	--	--	26-31	--
MW-5	1295.0	1297.41	51	50+	--	38-48	--
MW-6	1295.7	1298.02	55	54+	--	44-54	--
MW-7	1299.3	1301.32	171.5	NP	169	158-163	--
MW-8	1295.4	1297.99	190	(51-61)	188	183-188	--
MW-9	1299.1	1301.86	174.5	(58-61)	173	162-167	47
MW-10	1298.4	1301.40	160	NP	156	150-155	41
MW-11	1299.4	1301.34	172	NP	169	161-166	807
MW-12	1297.1	1299.60	176	(36-40) (49-58)(60-67)	173	149-154	569
MW-13	1299.2	1301.01	161	NP	159	154-159	118
MW-14	1298.7	1301.06	184	NP	183	153-158	17
MW-15	1297.4	1299.60	183	(42-49)(59-73)	180	160-165	>724
MW-16	1298.5	1301.17	178	(54-63)	175	164-169	0
MW-17	1295.8	1297.81	178	(40-49)	177	144-149	0
MW-18	1298.3	1300.97	157	NP	155	149-154	93

NP = Not Present

NE = Not Encountered (but may be present at greater depth)

VOC = Results in parts per billion (ppb) and based on screened auger analyses

ND = Not Detected

Tr = Trace

APPENDIX E (PAGE 6)

DEPARTMENT OF NATURAL RESOURCES MONITORING WELLS

Monitoring Well	ELEVATION		DEPTH				TCE/PCE at Screen, 7/85
	Ground	TOC	Bottom of Boring	Upper Aquitard	Lower Aquitard	Screen Interval	
DNR-1	1295.8	1297.97	140	(43-45)	NE	38-40	4
DNR-2	1298.9	1300.36	144	NP	NE	50-52	2,300
DNR-3	1298.8	1300.92	65	NP	--	65-67	7,400
DNR-4	1298.2	1299.72	65	NP	--	65-67	3,480
DNR-5	1298.2	1299.80	65	NP	--	65-67	45
DNR-6	--	1297.24	65	NP	--	65-67	24
DNR-7	1298.1	1299.97	65	NP	--	37-39	ND
DNR-8	1298.6	1302.42	65	NP	--	37-39	ND
DNR-9	1297.1	1299.56	65	NP	--	65-67	33
DNR-10S	1298.1	1299.03	45	NP	--	40-42	15
DNR-10D	1298.1	1300.77	65	NP	--	68-70	15
DNR-11	1298.3	1301.68	65	(48-50)	--	46-48	200
DNR-13	1297.7	1298.65	42	41+	--	38-40	86
DNR-14S	1298.0	1299.85	48	NP	--	46-48	--
DNR-14D	1298.0	1300.62	80	NP	--	78-80	--
DNR-15	1297.7	1298.89	50	(47-48)	--	52-55	--

NP = Not Present

NE = Not Encountered (but may be present at greater depth)

ND = Not Detected

APPENDIX E (PAGE 7)
FOUR STAR MONITORING WELLS

Monitoring Well	ELEVATION		DEPTH				TCE/PCE at Screen, 12/83
	Ground	TOC	Bottom of Boring	Upper Aquitard	Lower Aquitard	Screen Interval	
FS-1-10	1298.5	1300.66	60	(43-57)	--	40-42	37.7
FS-2-10	1298.2	1200.12	50	(34-37)	--	40-42	ND
FS-3-10	1295.3	1297.36	40	37+	--	40-42	0.01
FS-4-0	1294.9	1296.83	35	--	--	30-32	0.484
FS-5-5	1293.5	1294.88	35	--	--	35-37	23.4
FS-6-5	1295.9	1297.80	35	--	--	35-37	--
FS-7-0	1298.7	1302.14	40	--	--	30-32	.457
FS-7-10	1299.0	1301.36	50	NP	--	40-42	6.9
FS-7-20	1299.1	1301.35	60	NP	--	50-52	0.507
FS-8-10	1298.9	1301.42	50	NP	--	40-42	3.4
FS-9-10	1298.8	1302.65	50	NP	--	40-42	0.152
FS-10-5	1297.5	1299.72	50	(37-42)	--	35-37	0.018
FS-11-10	1299.5	1301.69	50	NP	--	40-42	0.046
FS-12-10	1298.0	1299.92	50	NP	--	40-42	ND
FS-13-10	1298.4	1300.41	50	NP	--	40-42	ND
FS-13-20	1298.5	1300.98	55	NP	--	50-52	ND
FS-14-10	1298.2	1300.02	50	NP	--	40-42	ND
FS-14-20	1298.2	1300.68	55	NP	--	50-52	ND
FS-15-10	1298.6	1301.50	50	NP	--	40-42	ND
FS-16-10	1298.5	removed	50	NP	--	40-42	ND
FS-17-10	1297.3	removed	50	NP	--	40-42	ND
FS-18-10	1298.7	1300.94	50	NP	--	40-42	ND
FS-18-20	1298.5	1300.38	60	NP	--	50-52	ND
FS-19-20	1297.8	1300.42	45	NP	--	40-42	0.037
FS-19-50	1297.9	1300.48	55	NP	--	50-52	--
FS-20-50	1298.1	1300.42			--	80-82	--

NP = Not Present
ND = Not Detected

APPENDIX E (PAGE 8)

KYSOR MONITORING WELLS

Monitoring Well	ELEVATION		DEPTH			
	Ground	TOC	Bottom of Boring	Upper Aquitard	Lower Aquitard	Screen Interval
K-1	1299.3	1300.05	43	--	--	36-39
K-2	--	removed	(40)			(27-30)
K-3	--	removed	(103)	(90-103+)		--
K-4	--	1298.29	(103)	(90-103+)		(28-38)
K-5S	1294.1	1297.72	31	31+	--	26-31
K-5D	1294.1	1295.15	188	(32-40)	186	148-151
K-6S	1293.8	1294.86	35	--	--	30-35
K-6D	1293.8	1297.02	150	(60-62)(70-72)	--	98-101
K-7S	1297.4	1300.03	35	--	--	28-33
K-7D	1297.4	1298.91	103	(42-47)(84-103+)	--	70-72
K-8	1295.2	1296.64	150	(55-58)	NE	98-101
K-9	1296.0	1297.94	103	(57-60)(75-103+)	--	70-72
K-10	1292.7	1294.35	102	(37-42)	--	70-72
K-11	1295.0	1295.84	148	NP	147	48-51
K-12	1298.2	1299.88	150	NP	NE	146-149
K-13	1298.1	1299.13	148	NP	147	48-51
K-14	1296.4	1297.82	150	(38-50)	NE	59-62
K-15	1298.2	1299.82	179	(50-61)	174	147-150
K-16S		1299.96	54	--	--	48-51
K-16D	1298.8	1299.99	150	(68-72)	NE	99-102
K-17	1298.9	1300.52	98	98+	--	41-44
K-18	1297.1	1298.41	77	(39-41)77+	--	36-39
K-19	1296.9	1298.06	99	(68-75)98+	--	92-95
K-20	1294.0	1294.99	150	(29-50)	NE	50-53
K-21	1292.7	1293.48	150	edge	--	48-51
K-22	1300.2	1301.73	106	106+	--	48-51
K-23	1294.6	1296.26	98	NP	--	95-98
K-24	1298.1	1299.60	127	NP	--	97-100

NP = Not Present

NE = Not Encountered (but may be present at greater depth)

APPENDIX G

FOUR WINNS BOATS AND ASI INVESTIGATION DATA

FOUR WINNS, INCORPORATED
HYDROGEOLOGICAL INVESTIGATION REPORT

January 1988

Prepared for: Four Winns, Inc.
925 Frisbie Street
Cadillac, Michigan 49601

Prepared by: ASI
Environmental Technologies Division
1100 Conrad Industrial Drive
PO Box 647
Ludington, Michigan 49431

11.0 CONCLUSIONS

CONCLUSIONS

- The soils beneath the Four Winns site are predominantly fine to medium grained sands with variable clay lenses deposited in a glacial outwash environment.
- The clay lenses are of varying thicknesses and are mostly found at different depths. Generally the clays tend to dip upward to the north.
- Groundwater flow for the majority of the site is from the upper water table aquifer and ranges from north to north 23 degrees east.
- All monitoring wells show contamination from chlorinated solvents and the contamination is located in the upper water table aquifer.
- The principle contaminants of concern are TCA, PCE, TCE, and acetone.
- Analysis of present groundwater data indicates that Four Winns is partially responsible for a plume of TCA migrating from the Frisbie Street property and centered approximately at MW9.
- Four Winns is responsible for a contaminant plume of acetone moving off their property.
- Analysis of present groundwater data indicates that Four Winns is not responsible for plumes of PCE and TCE leaving their property. A

possible source of upgradient contamination is the Ingraham property where leaking barrels of solvents were known to have been stored.

- The main TCA and acetone plume has migrated approximately 900 feet downgradient where acetone was detected at .063 ppm in MW17 on July 9, 1987.
- Significant amounts of TCA, acetone, and some Scan 2 contaminants have been found in the soils at the Four Winns barrel storage and glue handling and storage areas.

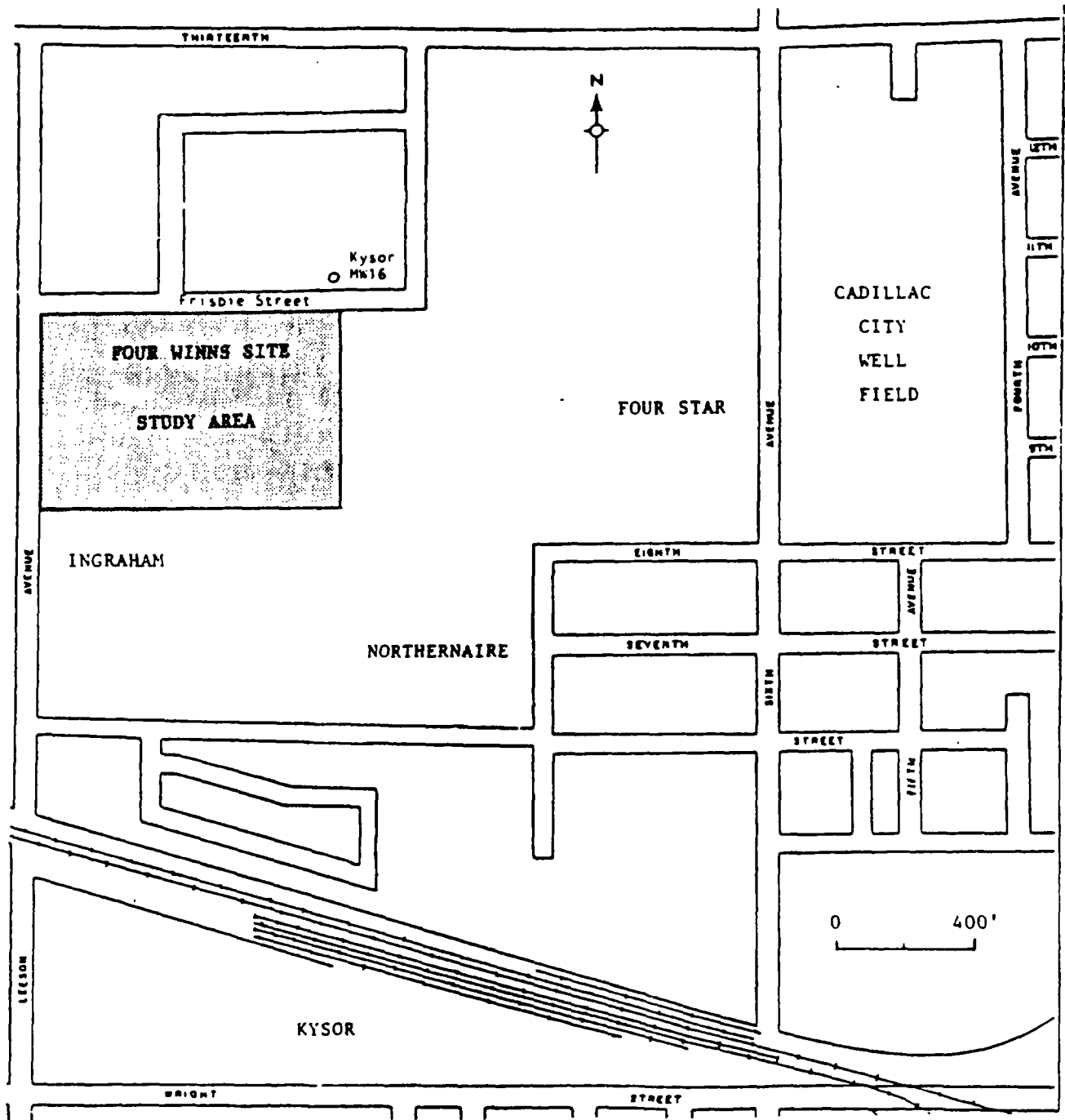


FIGURE 1. Site location map of Cadillac Industrial Park area.

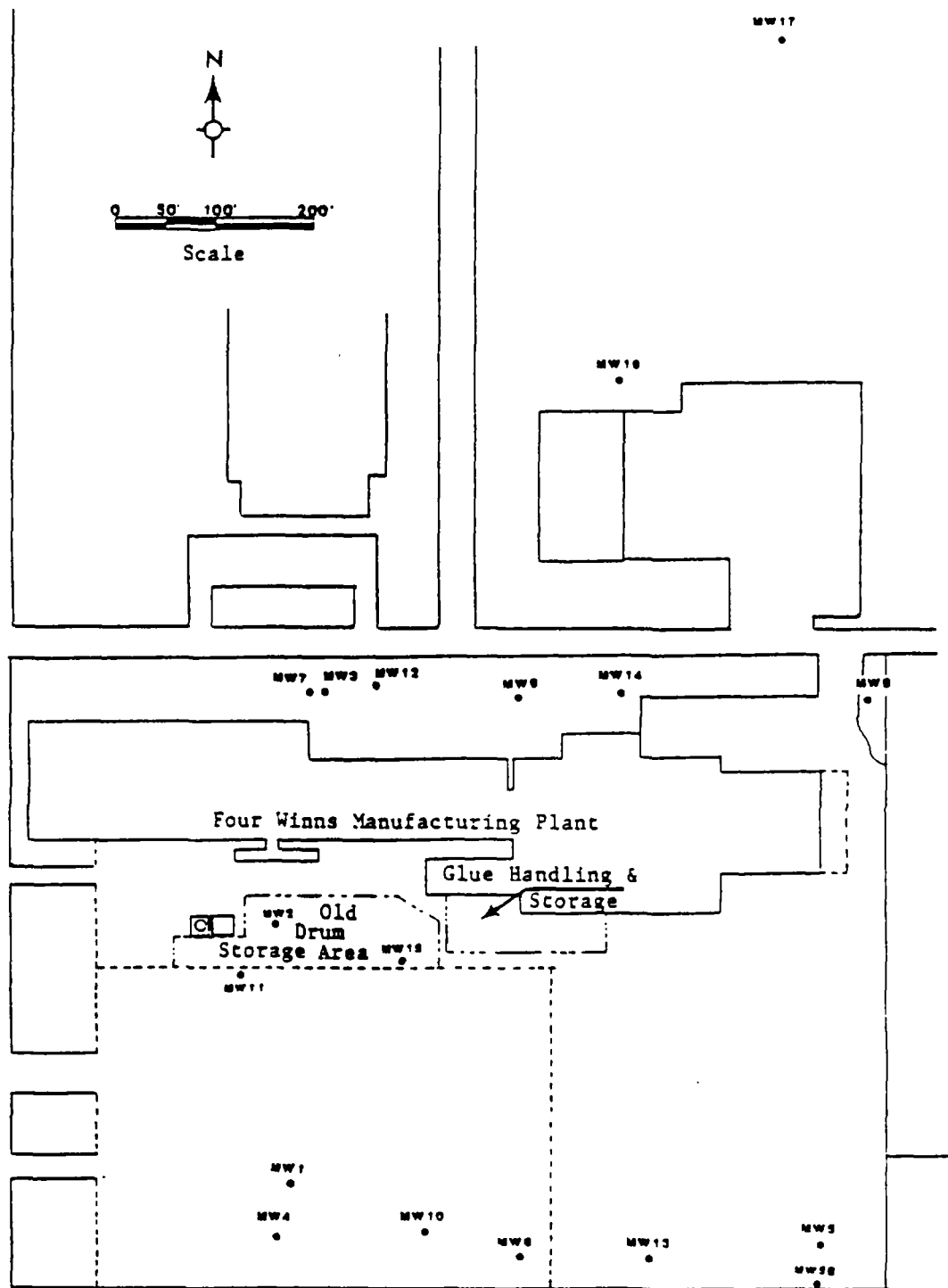


Figure 2
Four Winns Boats
Monitoring Well Location Map - 10/20/87

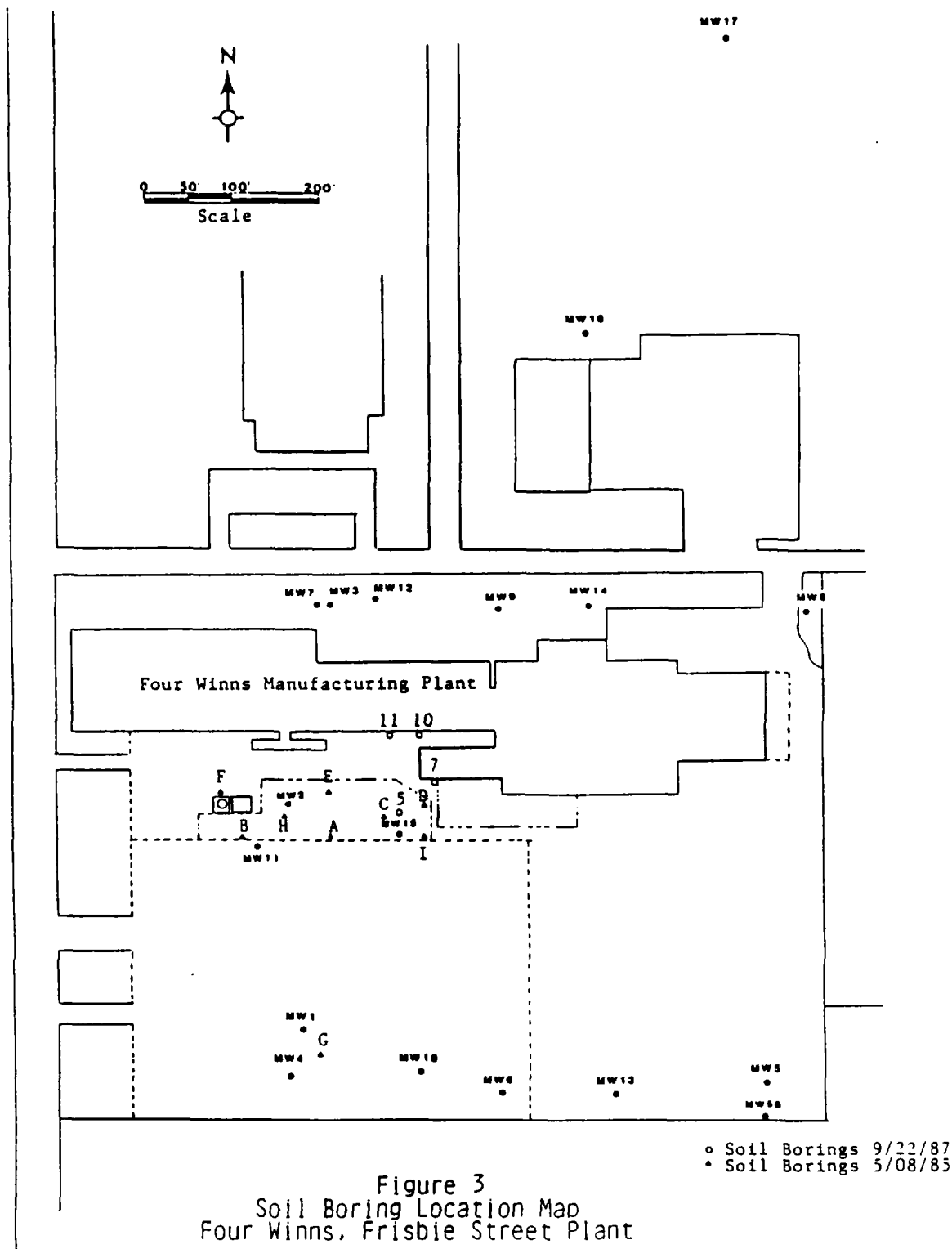


TABLE 5. SUMMARY OF ANALYTICAL DATA, SOUTHERN PROPERTY BOUNDARY AND CENTRAL MONITORING WELLS, FOR FOUR WINNS, INC., FRISBIE STREET PLANT, 1986-1987.

Monitoring Well	Date Sampled	Depth Sampled	TCA Mg/L	PCE Mg/L	TCE Mg/L	Total Scan 1 Contamination Mg/L
MW4	10/15/86	26-31'	.008	.019	ND	.027
MW5	10/20/86	43-48'	.240	ND	.054	.294
MW5B	10/16/87	25-30'	ND	ND	ND	ND
MW6	11/2/86	23-28'	.492	.089	.170	.787
MW10	2/3/87	26-31'	.060	.029	.014	.103
MW11	2/4/87	44-49'	.047	.390	.006	.445
MW13	3/9/87	26-31'	.542	.005	.003	.550
MW15	3/20/87	35-40'	.576	.368	.141	1.085

ND = Nondetectable

TABLE 6. SUMMARY OF ANALYTICAL DATA, ADDITIONAL MONITORING WELL
SAMPLE DATA, FOR FOUR WINNS, INC., FRISBIE STREET PLANT,
1987.

Monitoring Well	Date Sampled	Depth Sampled	TCA Mg/L	PCE Mg/L	TCE Mg/L	Total Scan 1 Contamination Mg/L
MW5	7/9/87	29-34'	.075	.043	.058	.177
MW6	7/9/87	26-31'	.003	.003	ND	.006
MW9	7/9/87	53-58'	1.080	.145	.345	1.997
MW10	7/9/87	35-40'	.004	.029	.005	.038
MW12	7/9/87	53-58'	.001	.006	.001	.008
MW13	7/9/87	44-49'	.002	ND	ND	.002
MW14	7/9/87	53-58'	.795	.056	.144	1.486
MW15	7/9/87	53-58'	.009	.003	.001	.013
MW16	7/9/87	56-61'	4.150	.317	.887	7.287

ND = Nondetectable

E.C. JORDAN CO. ENGINEERS &
SCIENTISTS

APPENDIX A
ARAR IDENTIFICATION CHECKLISTS

APPENDIX A: TABLE 1
LOCATION-SPECIFIC ARARS CHECKLIST
CADILLAC AREA RI

FEDERAL REGULATORY REQUIREMENTS

REQUIREMENT	Does this requirement apply to site conditions?		COMMENTS
	YES	NO	
1. Clean Water Act - 40 CFR Part 404		X	No area wetlands are believed to be affected by the site.
2. National Environmental Policy Act - 40 CFR Part 6		X	No area wetlands are believed to be affected by the site.
3. Endangered Species Act (16 U.S.C. 1531, 50 CFR Parts 8, 225, 402)		X	No endangered species have been detected at the site.
4. River and Harbors Act (33 CFR Parts 320 - 329)		X	No Rivers or Harbors are located in the area believed to be affected by the site.
5. Wild and Scenic Rivers Act (16 U.S.C. 1271)		X	There are no rivers in the vicinity of the site.
6. Fish and Wildlife Coordination Act (16 U.S.C. 661)		X	If a remedial alternative proposes to alter any body of water, the EPA would be required to notify various federal agencies under this act.
7. Fish and Wildlife Conservation Act (16 U.S.C. 2901 50 CFR Part 83)		X	The area that may require remedial action is not a conservation area.
8. Coastal Zone Management Act (16 U.S.C. 1451, (15 CFR 930, 923.45)		X	The site is not located in a coastal area.
9. Historical Sites, Buildings, and Antiquities Act (16 U.S.C. 461)		X	No Historical Buildings or sites are located at the Cadillac site.
10. National Historic Preservation Act (16 U.S.C. 470)		X	
11. Protection of Archaeological Resources (32 CFR Part 229, 229.4; 43 CFR Parts 107, 171.1 - 171.500)		X	If archaeological items are encountered during any actions requiring excavations, work must be stopped until a permit is obtained. However, no known archaeological items are located onsite.
12. RCRA - Standards for Owners and Operators of Permitted Hazardous Waste facilities (40 CFR 264.10 - 264.18)			Because there are RCRA facilities onsite, RCRA requirements are applicable. If waste onsite is RCRA listed, these requirements may have jurisdiction. Therefore, during remedial action planning and design, RCRA requirements 12 - 16 may have to be attained when possible.
13. RCRA - Preparedness and Prevention (40 CFR 264.30 - 264.37)			
14. RCRA - Contingency Plan and Emergency Procedures (40 CFR 264.50 - 264.56)	X		
15. RCRA - Groundwater Protection (40 CFR 264.90 - 264.109)		X	
16. RCRA - Closure and Postclosure (40 CFR 264.110 - 264.120)		X	

APPENDIX A: TABLE 1 (continued)
LOCATION-SPECIFIC ARARS CHECKLIST
CADILLAC AREA RI

FEDERAL REGULATORY REQUIREMENTS

REQUIREMENT	Does this requirement apply to site conditions?		COMMENTS
	YES	NO	
17. RCRA - Surface Impoundments (40 CFR 264.220 - 264.249)		X	RCRA Surface Impoundments do not exist onsite.
18. RCRA - Waste Piles (40 CFR 264.250 - 264.269)		X	RCRA Waste Piles do not exist onsite.
19. RCRA - Landfills (40 CFR 264.300 - 264.339)		X	A RCRA Landfill does not exist onsite.
20. RCRA - Underground Storage Tanks (40 CFR Part 264.190)	X		Because there are underground storage tanks onsite, these requirements should be considered.

APPENDIX A: TABLE 1 (continued)
LOCATION-SPECIFIC CHECKLIST
CADILLAC RI
FEDERAL CRITERIA, GUIDANCE, AND ADVISORIES TO BE CONSIDERED

GUIDELINE	Does this requirement apply to site conditions?		COMMENTS
	YES	NO	
1. FWS, MWFS, Advisories (Under Fish and Wildlife Coordination Act)		X	Because the requirements of these advisories are covered under CWA Section 404, they do not have to be considered separately.
2. Floodplain Executive Order (EO 11988)		X	The site is not located within the 100 year floodplain.
3. Wetlands Executive Order (EO 11990)		X	There are no area wetlands believed to be affected by the site.
4. EPA Regulations Sole Source Aquifer Subpart A	X		The aquifers located within the vicinity of the site supply the water for the city well field.
5. EPA Office of Water-Wasteload Allocation		X	Not applicable.
6. EPA Office of Water Guidance Technical Support Manual: Water Body Surveys and Assessments for Conducting Use Attainability Analyses.		X	Not applicable.

APPENDIX A: TABLE 1 (continued)
LOCATION-SPECIFIC ARARA CHECKLIST
CADILLAC, RI
MICHIGAN STATE REQUIREMENTS

REQUIREMENT	Does this requirement apply to site conditions?		COMMENTS
	YES	NO	
1. Hazardous Waste Management Act (Act No. 64, P.A. 1979)	X		Compliance with RCRA requirements should also comply with this act.
2. Michigan Underground Storage Tank Act (Act No. 423, P.A. 1984)	X		These requirements may be applicable when developing source control measures that address underground storage tanks.
3. Michigan Resource Recovery Commission Rules		X	These requirements pertain to the conservation of natural resources for energy production and to provide a coordinated statewide management and resource recovery program.
4. Michigan Shoreland Protection Act		X	The Cadillac site is not located on a shoreland.
5. Michigan Wetlands Protection Act		X	The Cadillac site is not located on a wetland.

APPENDIX A: TABLE 2
CONTAMINANT-SPECIFIC ARARS CHECKLIST
CADILLAC AREA RI

FEDERAL REGULATORY REQUIREMENTS

REQUIREMENT	Does this requirement apply to site conditions?		COMMENTS
	YES	NO	
1. SDWA - MAXIMUM CONTAMINANT LEVELS (MCLs) (40 CFR 141.11 - 141.16)	X		The risks to human health due to consumption of groundwater should be assessed.
2. CAA - NATIONAL AIR QUALITY STANDARDS (NAQS) (40 CFR 50)	X		Standards primarily developed to control stack and automobile emissions. Air monitoring and modeling data is not yet available. These standards will be considered for potential alternatives that involve air emissions.
3. MAXIMUM CONTAMINANT LEVEL GOALS (MCLGs)	X		The risks to human health due to consumption of groundwater should be assessed. For compounds without MCLs, MCLGs should be considered.
4. RCRA - SUBPART F GROUNDWATER PROTECTION STANDARDS ALTERNATIVE CONCENTRATION LIMITS (ACLs)	X		ACLs are applicable for setting a cleanup level for remediating contamination from a RCRA facility.
5. CWA - Pretreatment Standards (40 CFR 403)	X		Discharges to a POTW must comply with standards set by the POTW's EPA-approved pretreatment requirements.

MICHIGAN STATE REQUIREMENTS

REQUIREMENT	Does this requirement apply to site conditions?		COMMENTS
	YES	NO	
1. MDNR - WATER RESOURCES COMMISSION (ACT 245, P.A. 1929, AS AMENDED) R 323, PART 22, GROUNDWATER DISCHARGE RULES.	X		Requirements should be considered due to proximity to city well field.
2. MDNR - WATER RESOURCES COMMISSION (ACT 245, P.A. OF 1929, AS AMENDED) R 323, PART 4, WATER QUALITY STANDARDS.	X		These are surface water requirements, but may be relevant and appropriate for Cadillac.
3. AIR POLLUTION CONTROL COMMISSION GENERAL RULES (R 336 PART 9).	X		Standards for emissions of contaminants. Should be considered when airborne risks are characterized.

APPENDIX A: TABLE 2 (continued)
CONTAMINANT-SPECIFIC ARARS CHECKLIST
CADILLAC AREA RI

FEDERAL CRITERIA, ADVISORIES, AND GUIDANCE

REQUIREMENT	Does this requirement apply to site conditions?		COMMENTS
	YES	NO	
1. FEDERAL AMBIENT WATER QUALITY CRITERIA (AWQC)	X		Ambient Water Quality Criteria should be considered.
2. TSCA SECTION 8D HEALTH AND SAFETY DATA		X	An adequate toxicological data base exists for the contaminants of concern identified for the Cadillac Site. Thus, TSCA §8D files were not consulted.
3. HEALTH ADVISORIES (EPA OFFICE OF DRINKING WATER)	X		Health advisories should be considered for contaminants in groundwater that may be used for drinking water.
4. TSCA CHEMICAL ADVISORIES/FACT SHEETS		X	An adequate toxicological data base exists for the contaminants of concern. The TSCA fact sheets provide qualitative information and were not considered here.
5. EPA RISK REFERENCE DOSES (RfD)	X		EPA RfDs should be used to characterize risks due to noncarcinogens in various media.
6. EPA CARCINOGEN ASSESSMENT GROUP POTENCY FACTORS	X		EPA Carcinogen Potency factors should be used to compute the individual incremented cancer risk resulting from exposure to compounds found on site.
7. ACCEPTABLE INTAKE - CHRONIC (AIC) AND SUBCHRONIC (AIS) - EPA HEALTH ASSESSMENT DOCUMENT	X		AIC and AIS Values should be used to characterize the risks due to several noncarcinogens in various media.
8. PESTICIDE REGISTRATION DATA		X	Pesticides were not identified as contaminants on site. Registration data therefore will not need to be considered.
9. EPA OFFICE OF WATER GUIDANCE - WATER-RELATED ENVIRONMENTAL FATE OF 129 PRIORITY POLLUTANTS (1979)	X		The transport and fate of contaminants should be assessed.
10. TSCA - POLYCHLORINATED BIPHENYLS SPILL CLEANUP POLICY (40 CFR 761, SUBPART G; 761.120 - 761.135)		X	PCBs were not identified as contaminants of concern at the site.

APPENDIX A: TABLE 3
ACTION-SPECIFIC ARARS CHECKLIST
CADILLAC AREA RI

FEDERAL REQUIREMENTS

REQUIREMENT	Does this Requirement Apply to Site Conditions?		COMMENTS
	YES	NO	
1. RCRA - STANDARDS FOR OWNERS AND OPERATORS OF PERMITTED HAZARDOUS WASTE FACILITIES (40 CFR 264.10 - 264.18)	X		RCRA requirements 1-6 should be considered for all alternatives that involve storage, transport, or disposal of RCRA hazardous waste.
2. RCRA - PREPAREDNESS AND PREVENTION (40 CFR 264.30 - 264.37)	X		
3. RCRA - CONTINGENCY PLAN AND EMERGENCY PROCEDURES (40 CFR 264.50 - 264.56)	X		
4. RCRA - MANIFESTING, RECORD KEEPING, AND REPORTING (40 CFR 264.70 - 264.77)	X		
5. RCRA - GROUNDWATER PROTECTION (40 CFR 264.90 - 264.109)	X		
6. RCRA - CLOSURE AND POST-CLOSURE (40 CFR 264.110 - 264.120)	X		
7. RCRA - SURFACE IMPOUNDMENTS (40 CFR 264.220 - 264.249)	X		Applicable for long-term site maintenance and monitoring.
8. RCRA - WASTE PILES (40 CFR 264.250 - 264.269)		X	Applicable when remedial alternatives include a leachate collection system and for closure options.
9. RCRA - LAND TREATMENT (40 CFR 264.270 - 264.299)		X	Not applicable.
10. RCRA - LANDFILLS (40 CFR 264.300 - 264.339)	X		Not applicable.
11. RCRA - INCINERATORS (40 CFR 264.340 - 264.999)		X	Applicable for disposal of contaminated materials to a RCRA Landfill.
12. RCRA - UNDERGROUND STORAGE TANKS (40 CFR 264.190)	X		Will only be considered if a remedial alternative includes incineration.
13. CWA - 40 CFR PARTS 122, 125, 403	X		Applicable when remedial alternatives include removal or installation of underground storage tanks.
14. TSCA - MANUFACTURING, PROCESSING, DISTRIBUTION IN COMMERCE, USE OF PCBs AND PCB ITEMS (40 CFR 761.60 - 761.79)		X	Will be considered when remedial alternatives include discharges to groundwater or surface water.
15. TSCA - MARKINGS OF PCBs AND PCB ITEMS (40 CFR 761.40 - 761.79)		X	TSCA requirements 14-17 are not applicable since PCBs were not found on-site.
16. TSCA - STORAGE AND DISPOSAL (40 CFR 761.60 - 761.79)		X	
17. TSCA - RECORDS AND REPORTS (40 CFR 761.180 - 761.185, ALSO 40 CFR 129.105, 750)		X	

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APPENDIX A: TABLE 3 (continued)

ACTION-SPECIFIC ARARS CHECKLIST
CADILLAC AREA RI

FEDERAL REQUIREMENTS

REQUIREMENT	Does this Requirement Apply to Site Conditions?		COMMENTS
	YES	NO	
18. CAA - NATIONAL AIR QUALITY STANDARDS FOR TOTAL SUSPENDED PARTICULATES (40 CFR 129.105, 750)	X		Applicable for soil removal activities and air stripping.
19. OSHA - GENERAL INDUSTRY STANDARDS (29 CFR PART 1910)	X		OSHA requirements 18-20 must be considered during implementation of all remedial alternatives.
20. OSHA - SAFETY AND HEALTH STANDARDS FOR FEDERAL SERVICE CONTRACTS (29 CFR PART 1926)	X		
21. OSHA - RECORDKEEPING, REPORTING, AND RELATED REGULATIONS	X		
22. DOT RULES FOR THE TRANSPORTATION OF HAZARDOUS MATERIALS (49 CFR PARTS 107, 171.1 - 171.500)	X		Applicable to the transport of contaminated materials.
23. RIVERS AND HARBORS ACT (33 CFR PARTS 320-329)		X	
24. GUIDELINES FOR SPECIFICATIONS OF DISPOSAL SITES FOR DREDGED OR FILL MATERIAL (40 CFR 230)		X	
25. DENIAL OR RESTRICTION OF DISPOSAL SITES (40 CFR 231)		X	
26. OCEAN DUMPING REGULATIONS (40 CFR 220-229)		X	
27. DISPOSAL OF WASTE MATERIAL CONTAINING TCDD (40 CFR 775.180 - 775.197)		X	
28. NATIONAL ENVIRONMENTAL POLICY ACT (40 CFR PART 6)	X		Applicable to areas of the site not covered by CERCLA that potentially impact the environment.

MICHIGAN REQUIREMENTS

REQUIREMENT	Does this Requirement Apply to Site Conditions?		EXPLANATIONS
	YES	NO	
1. HAZARDOUS WASTE MANAGEMENT ACT (ACT NO. 64, P.A. 1979)	X		Consistent with RCRA, applicable to all potential alternatives.
2. UNDERGROUND STORAGE TANK ACT (ACT NO. 423, P.A. 1984)	X		Requires registration and permitting of underground storage tanks.
3. AIR POLLUTION CONTROL COMMISSION GENERAL RULES (R 336 PARTS 2,3,6,9)	X		Applicable to soil removal activities and potential air stripping.

APPENDIX A: TABLE 3 (continued)

ACTION-SPECIFIC ARARS CHECKLIST
CADILLAC AREA RI

FEDERAL CRITERIA, ADVISORIES, AND GUIDANCE

REQUIREMENT	Does this Requirement Apply to Site Conditions?		COMMENTS
	YES	NO	
1. EPA'S GROUNDWATER PROTECTION STRATEGY	X		To be considered during remedial planning and design.
2. TSCA COMPLIANCE POLICY		X	
3. WASTE LOAD ALLOCATION PROCEDURES (EPA OFFICE OF WATER)		X	
4. FEDERAL SOLE SOURCE AQUIFER REQUIREMENTS	X		Because the aquifers within the vicinity of the site supply the water for the city well field, these requirements should be considered.
<u>U.S. EPA RCRA DESIGN GUIDELINES</u>			
A. EPA'S RCRA DESIGN GUIDELINES			
1. SURFACE IMPOUNDMENTS, LINER SYSTEMS, FINAL COVER AND FREEBOARD CONTROL	X		To be considered for leachate collection systems.
2. WASTE PILE DESIGN - LINER SYSTEMS		X	Remaining RCRA Guidelines will not be considered, since these facilities will not be constituted on-site. However, off-site disposal facilities for excavated soil must be RCRA permitted.
3. LAND TREATMENT UNITS		X	
4. LANDFILL DESIGN - LINER SYSTEMS AND FINAL COVER		X	
B. PERMITTING GUIDANCE MANUALS			
1. PERMIT APPLICANTS' GUIDANCE MANUAL FOR HAZARDOUS WASTE LAND TREATMENT, STORAGE AND DISPOSAL FACILITIES	X		These permit guidance manuals are not directly applicable to the part of the site covered by CERCLA, since permits need not be secured under CERCLA. However, for the non-superfund parts of the site, permits are required, and contents of these manuals shall be used if guidance in a particular area is necessary.
2. PERMIT WRITERS' GUIDANCE MANUAL FOR HAZARDOUS WASTE LAND TREATMENT, STORAGE AND DISPOSAL FACILITIES	X		
3. PERMIT WRITERS' GUIDANCE MANUAL FOR SUBPART F	X		
4. PERMIT APPLICANTS' GUIDANCE MANUAL FOR GENERAL FACILITY STANDARDS	X		
5. WASTE ANALYSIS PLAN GUIDANCE MANUAL	X		
6. PERMIT WRITERS' GUIDANCE MANUAL FOR HAZARDOUS WASTE TANKS	X		

APPENDIX A: TABLE 3 (continued)

ACTION-SPECIFIC ARARS CHECKLIST
CADILLAC AREA RI

FEDERAL CRITERIA, ADVISORIES, AND GUIDANCE

GUIDELINE	Does this Requirement Apply to Site Conditions?		COMMENTS
	YES	NO	
7. MODEL PERMIT APPLICATION FOR EXISTING INCINERATORS		X	
8. GUIDANCE MANUAL FOR EVALUATING PERMIT APPLICATIONS FOR THE OPERATION OF HAZARDOUS WASTE INCINERATOR UNITS	X		
9. A GUIDE FOR PREPARING RCRA PERMIT APPLICATIONS FOR EXISTING STORAGE FACILITIES		X	
10. GUIDANCE MANUAL ON CLOSURE AND POST-CLOSURE INTERIM STATUS STANDARDS		X	
C. TECHNICAL RESOURCE DOCUMENTS			
1. EVALUATING COVER SYSTEMS FOR SOLID AND HAZARDOUS WASTE		X	
2. HYDROLOGIC SIMULATION OF SOLID WASTE DISPOSAL SITES		X	
3. LANDFILL AND SURFACE IMPOUNDMENT PERFORMANCE EVALUATION	X		To be considered for leachate collection systems.
4. LINING OF WATER IMPOUNDMENTS AND DISPOSAL FACILITIES		X	
5. TEST METHODS FOR EVALUATING SOLID WASTES	X		To be considered.
6. A METHOD FOR DETERMINING THE COMPATIBILITY OF HAZARDOUS WASTES	X		To be considered.
7. GUIDANCE MANUAL ON HAZARDOUS WASTE COMPATIBILITY	X		To be considered.
<u>EPA OFFICE OF WATER GUIDANCE DOCUMENTS</u>			
A. PRETREATMENT GUIDANCE DOCUMENTS			
1. 304(G) GUIDANCE DOCUMENT REVISED PRETREATMENT GUIDELINES	X		Appropriate for discharges to the public sewer.
B. NPDES GUIDANCE DOCUMENTS			
1. NPDES BEST MANAGEMENT PRACTICES GUIDANCE MANUAL (6/81)	X		Appropriate for discharges to surface water.
2. CASE STUDIES ON TOXICITY REDUCTION EVALUATION (5/83)	X		Appropriate for discharges to surface water.
C. GROUNDWATER/UIC GUIDANCE DOCUMENT		X	
<u>USEPA MANUALS FROM THE OFFICE OF RESEARCH AND DEVELOPMENT</u>			
1. SW 846 METHODS - LABORATORY ANALYTICAL METHODS	X		Lab method protocols should be followed during any chemical analysis.
2. LAB PROTOCOLS DEVELOPED PURSUANT TO CLEAN WATER ACT §304(b)	X		

APPENDIX A: TABLE 3 (continued)
ACTION-SPECIFIC ARARS CHECKLIST
CADILLAC AREA RI

MICHIGAN, REQUIREMENTS

REQUIREMENT	Does this requirement apply to site conditions?		COMMENTS
	YES	NO	
4. Michigan Solid Waste Management Act (Act No. 641, P.A. 1978)		X	Not applicable.
5. MDNR - Water Resources Commission (Act 245, P.A. 1929, as amended) R 323, Parts 9, 21. Wastewater Reporting and Surveillance Fees Rules, and Wastewater Discharge Permits (MWA NPDES)	X		This regulation outlines the requirements for reporting discharges of wastewater to waters of the state or to POTWs. The discharge permit requirements are compatible with NPDES requirements.
6. Soil and Erosion Control (Act No. 347, P.A. 1972)		X	Not applicable.
7. Liquid Industrial Waste Disposal Act (Act No. 136, P.A. 1984)	X		This regulation addresses the removal of liquid industrial waste and outlines requirements for licenses, recordkeeping and transportation.
8. MDNR - Water Resources Commission (Act 245, P.A. 1929, as amended) R 323, Part 22, Ground Water Discharge Rules	X		Applicable to alternatives that involve discharges to groundwater.
9. MDNR - Water Resources Commission (Act 245, P.A. 1929, as amended) R 323, Part 4. Water Quality Standards.	X		Applicable to alternatives that involve discharges to surface water.
10. Environmental Response Act - Act No. 307, P.A. 1982.	X		Requires risk assessments for hazardous waste sites.

APPENDIX B
CHEMICAL-SPECIFIC ARARS COMPARISON

APPENDIX B
CADILLAC AREA GROUNDWATER STUDY: GROUNDWATER
CHEMICAL-SPECIFIC ARARs COMPARISON

CADILLAC FEASIBILITY STUDY

Compound	ARARs (mg/l)			Existing Maximum Concentrations (mg/l)			
	MCL	MCLG	AWQC*	Rexair	Upper Aquifer	Intermediate Aquifer	Northernaire Plume ¹
1,1,1-Trichloroethane	0.2	0.2	18.4	.017	32	1.2	0.44
Trans-1,2-dichloroethylene	--	0.07**	--	.048	--	4.6	.04
1,1-Dichloroethylene	0.007	0.007	0.00033	--	1.0	.099	.026
1,2-Dichloroethane	0.005	0	0.00094	--	2.0	--	--
Methylene chloride	--	--	0.00019	.078	0.69	0.24	--
Tetrachloroethylene	--	0**	0.0008	--	0.72	0.17	0.003
Trichloroethylene	0.005	0	0.0028	153	78	7.5	2.5
Chromium (hexavalent)	0.05	0.12**	0.05	--	--	--	0.773
Xylene	--	0.44**	--	--	1.7	--	--
Toluene	--	2.0**	14.3	.042	3.2	.001	--

* AWQC are adjusted for drinking water exposure only (USEPA, 1986) or calculated for water and fish ingestion if drinking water exposure value does not exist.

** Proposed levels.

¹ The VOCs present within the boundaries of the Northernaire plume are not derived from Northernaire; the chromium from Northernaire and VOCs from other sources have mixed in the aquifer.

NOTES: AWQC = Ambient Water Quality Criteria; MCL = Maximum Contaminant Level; MCLG = Maximum Contaminant Level Goal.

APPENDIX C
SOIL GAS METHODS AND PROCEDURES

The following are descriptions provided by Keck Consulting Services of the soil-gas methods and protocols used for the Cadillac Area RI.

Soil Gas Sampling Method

The following outlines the rationale, theory and field procedures for the soil gas sampling method. The method is described in detail in a paper presented at a 1983 National Water Well Association conference ("Detection of Groundwater Contamination by Shallow Soil Gas Sampling in the Vadose Zone," Eric G. Lappala and Glen M. Thompson).

Groundwater occurs as four zones within soils. The deepest zone occurs beneath the water table is called the phreatic zone. The phreatic zone is saturated with water at a pressure greater than atmospheric pressure. Above the phreatic zone is the capillary zone where water is drawn into the soil voids by surface tension. Soil within the capillary zone voids by surface tension. Soil within the capillary zone ranges from being saturated with water at a pressure less than atmospheric pressure to having a water content nearly equal to the specific moisture capacity of the soil. The tension saturated portion of the capillary zone is called the capillary fringe. The portion of the capillary zone above the capillary fringe where water content grades from 100 percent to residual water is called the funicular zone. Above the capillary zone lies the unsaturated vadose zone with residual moisture equal to the specific moisture capacity of the soil.

When a product loss occurs in granular soils above an unconfined aquifer, vertical migration via gravity through the vadose zone develops. A lighter than water liquid contaminant (gasoline) will migrate through the unsaturated soils until it reaches the capillary zone where it will begin to flow outward. Lateral flow occurs through the unoccupied voids in the funicular zone. A lense of nearly pure product develops with a primary direction and rate of movement governed by the hydraulic gradient, soil permeability and rate of product loss. A gasoline product plume will move slightly faster than one-half the velocity of the groundwater ("Practical Geophydrological Aspects of Groundwater Contamination," William D. Shepard). Some of the product components will be solubilized into the groundwater and will travel at a greater velocity than the free product plume. Therefore, given a sufficient product spill, a plume of product will develop within the capillary zone with a halo of soluble components preceding it all flowing in the general direction of groundwater flow.

Water table fluctuations along with wicking actions of other transport mechanisms (hydrodynamic dispersion, molecular diffusion) cause contaminants to be introduced into the unsaturated portions of the capillary zone. Once volatile contaminants (gasoline) are present in and above the capillary fringe, they will volatilize (separate into liquid and gas phases). The more volatile the compound, the more tendency there is for it to go into the gaseous phase. Subsequent upward movement of the gas through the air filled voids of the vadose zone occurs through gaseous diffusion. Gas concentrations within the vadose zone are related to the concentration in the capillary zone and the distance from the source. Soil gas concentration over a free product plume will be higher than those over a soluble product plume and concentrations two

feet above a plume will be higher than concentrations 50 feet above a plume. A small spill on the surface which decrease with depth. This decrease results from the product being tied up the soil's specific moisture retention capacity. The result is less and less product with depth that is available to partition into the gaseous phase. Conversely, if the soil gas is being introduced into the vadose zone from the groundwater, soil gas concentrations will increase with depth until the groundwater source is encountered.

Soil gas sampling field procedures start by establishing a survey grid that extends upgradient of suspected sources and off to the sides of any suspicioned plume. This grid will usually be extended to include any known contaminated groundwater points. Soil gas samples are obtained at each grid point by driving a hollow metal probe to a desired reference elevation into the soil. Work by Eric Lappala et al has shown that a depth of approximately 10 percent of the vertical distance to the water table is sufficient. A gas sample is obtained by attaching Tygon tubing to the probe and applying a vacuum to the system. A sample is drawn directly from the tubing by a needle syringe which is then injected into a gas chromatograph for analysis and quantification.

Soil gas concentrations are entered on a grid map of the area at each respective sample point and subsequently contoured. The contour map will outline the contamination plume by depicting the areas of highest concentrations. In this manner, plume contours can be used to define the contamination origin, lateral extent and direction of movement.

Based on the soil gas sample contour map, monitor wells are installed upstream of the pinpointed source, on either side of the outlined plume, and within the plume. During the drilling process to install the monitor wells, vertical soil gas samples are obtained to confirm that the mapped soil gas concentrations are the result of a groundwater contamination plume and not from a minor localized surface spill. Vertical sampling is conducted by driving the sampling probe ahead of the hollow stem lead auger which has been advanced to just above the sampling point. Samples are taken at regular intervals from the surface down to the water table.

PROTOCOLS FOR SOIL GAS ANALYSIS

The analysis of soil gas for organic constituents requires the collection of a representative sample followed by accurate, precise detection. Because many variables are involved in the process, Keck Consulting Services, Inc. has developed a precise analytical protocol to minimize the influence of these variables.

The detection component of the technique is accomplished using either a Photovac 10A10 or a AID 210 portable gas chromatograph (GC). Each instrument is designed to provide based on the operation of chromatographic separation techniques and the differential photoionization potential of organic compounds. The sample is injected into a nitrogen gas stream which carries the sample through a column packed with a porous solid coated with a stationary liquid phase. The volume provides a surface which attracts and adsorbs molecules. The extent of attraction and retention is determined by the size and polarity of the molecules, and the liquid solvent. As a result of these differential attractions and retentions each organic compound completes passage through the column at a different reproducible rate. Thus, this process functions to effectively separate each compound within the gas stream.

As the sample stream is eluted from the column it enters a detector chamber. An adjacent ultraviolet lamp emits photons at an energy of 10.0 or 10.6 electron volts (interchangeable lamps are available) which are directed into the gas stream. A percentage of the organic molecules in the gas stream having sufficient low ionization potentials will be ionized by this energy source. The ions produced are collected, forming a small current which is amplified and eventually measured by a potentiometric recorder. Table 1 contains a list of selected organic compounds having a photoionization potential less than 12eV.

Through this combination of separation and ionization it is possible to detect the presence of numerous organic compounds. When organic compounds of known concentrations are similarly analyzed it becomes possible to quantify the presence of organic compounds in unknown samples through comparison of their relative potentiometric responses.

The operation of the GC is largely independent of sampling techniques. To insure optimal operation the method of injection, injection volume, flow rate and temperature are identical for samples and standards. Fresh standards are prepared daily using the headspace vapor techniques from pure liquid standards.

Glass Teflon syringes are used for all standards and samples. Quality Control is assured by the injection of blank samples with syringes prior to their use in sampling or with standards.

The collection of samples requires the ability to withdraw representative soil gas from a known depth in the soil column and maintain the integrity of the sample during transport to the GC. This process is accomplished with a soil gas probe in conjunction with a vacuum pump. This probe is constructed of hollow 7/8-inch O.D. stainless steel with a solid, screw-on tip. The lower one-foot section contains 16 1/6-inch holes which permit entry to the soil gas.

The upper section has a 1/16-inch hose barb to which 1/16-inch I.D. urethane tubing is attached.

The probe is driven to the appropriate depth and the tubing attached to a battery operated vacuum pump. The pump design limits air contact to the urethane tubing and the stainless steel probe. The vacuum pump is operated for 30 seconds to purge the probe casing and insure a representative sample. The sample is collected by inserting a Pressure-Lock syringe into the air vacuum tubing and withdrawing a one ml sample. The sample is secured in the syringe with the locking mechanism and transported to the GC for injection. The section of tubing perforated by the syringe is removed and the line is reconnected to the probe for the next sample.

Kysor Soil Study

The following tables and figure contain the analytical results and location of the soil-gas survey conducted by Keck behind the Kysor Facility in the Cadillac industrial park.

APPENDIX C (PAGE 1)

KYSOR SOIL STUDY
RESULTS OF SOIL-GAS ANALYSES - CADILLAC AREA RI

<u>Date</u>	<u>Line#</u>	<u>Location</u>	<u>Trans-1,2-Dichloroethene</u>	<u>Benzene</u>	<u>Trichloroethene</u>	<u>Toluene</u>	<u>Tetrachloroethene</u>	<u>Ethyl Benzene</u>	<u>Total Xylenes</u>
2/23/87	1	#1*	ND	ND	8.5	K	1.5	K	K
	1	#2*	ND	ND	35	ND	4.5	ND	ND
	1	#3*	ND	ND	161	ND	2.0	ND	ND
	1	#4*	ND	ND	115	ND	1.0	ND	ND
	1	#5*	ND	ND	4.0	K	K	ND	ND
	1	#6*	ND	ND	6.0	ND	K	ND	ND
	1	#7	ND	ND	K	ND	K	ND	ND
	1	#8	ND	ND	K	ND	K	ND	ND
2/24/87	1	#9	ND	ND	22	ND	2.5	ND	ND
	1	#10	ND	ND	23	ND	K	ND	ND
	1	#11	ND	ND	3.0	ND	K	ND	ND
	1	#12	ND	ND	3.0	ND	K	ND	ND
	1	#13	ND	ND	2.5	ND	K	ND	ND
	1	#14	ND	ND	K	ND	K	ND	ND
	2	#1*	ND	ND	1.0	ND	K	ND	ND
	2	#2*	ND	ND	1.0	ND	ND	ND	ND
	2	#3*	ND	ND	ND	ND	ND	ND	ND
	2	#4*	ND	ND	ND	ND	ND	ND	ND
	2	#5*	ND	ND	ND	ND	ND	ND	ND
	2	#6*	ND	ND	ND	ND	ND	ND	ND
	2	#7	ND	ND	ND	ND	ND	ND	ND
	2	#8	ND	ND	ND	ND	ND	ND	ND
	3	#1	ND	ND	ND	ND	ND	ND	ND
	3	#2	ND	ND	3.0	ND	K	ND	ND
	3	#3	ND	ND	ND	ND	ND	ND	ND
	3	#4	ND	ND	2.7	ND	.5	ND	ND
	3	#5	ND	ND	.5	ND	K	ND	ND
2/25/87	1	#2	ND	ND	6.5	ND	ND	ND	ND
	3	#6	ND	ND	ND	ND	ND	ND	ND
	3	#7	ND	ND	ND	ND	ND	ND	ND

ND = Not Detectable

K = Less than 1 ppb

* = Unidentified Peaks

All results in ppb

8.87.31T

0037.0.0

TABLE 7. SUMMARY OF ANALYTICAL DATA, NORTHERN PROPERTY BOUNDARY AND
DOWNGRADIENT MONITORING WELLS, FOR FOUR WINNS, INC.,
FRISBIE STREET PLANT, 1986-1987.

Monitoring Well	Date Sampled	Depth Sampled	TCA Mg/L	PCE Mg/L	TCE Mg/L	Total Scan 1 Contamination Mg/L
MW7	11/13/86	62-67'	.300	.240	.110	.683
MW8	12/1/86	44-49'	.280	ND	.048	.339
MW9	12/4/86	53-58'	24.00	ND	2.20	28.60
MW12	2/6/87	62-67'	.415	.001	.004	.766
MW14	3/19/87	53-58'	4.710	.137	.222	5.188
MW16	4/16/87	43-48'	7.640	.331	.439	9.293
MW17	6/29/87	25-30'	ND	ND	ND	ND

ND = Nondetectable

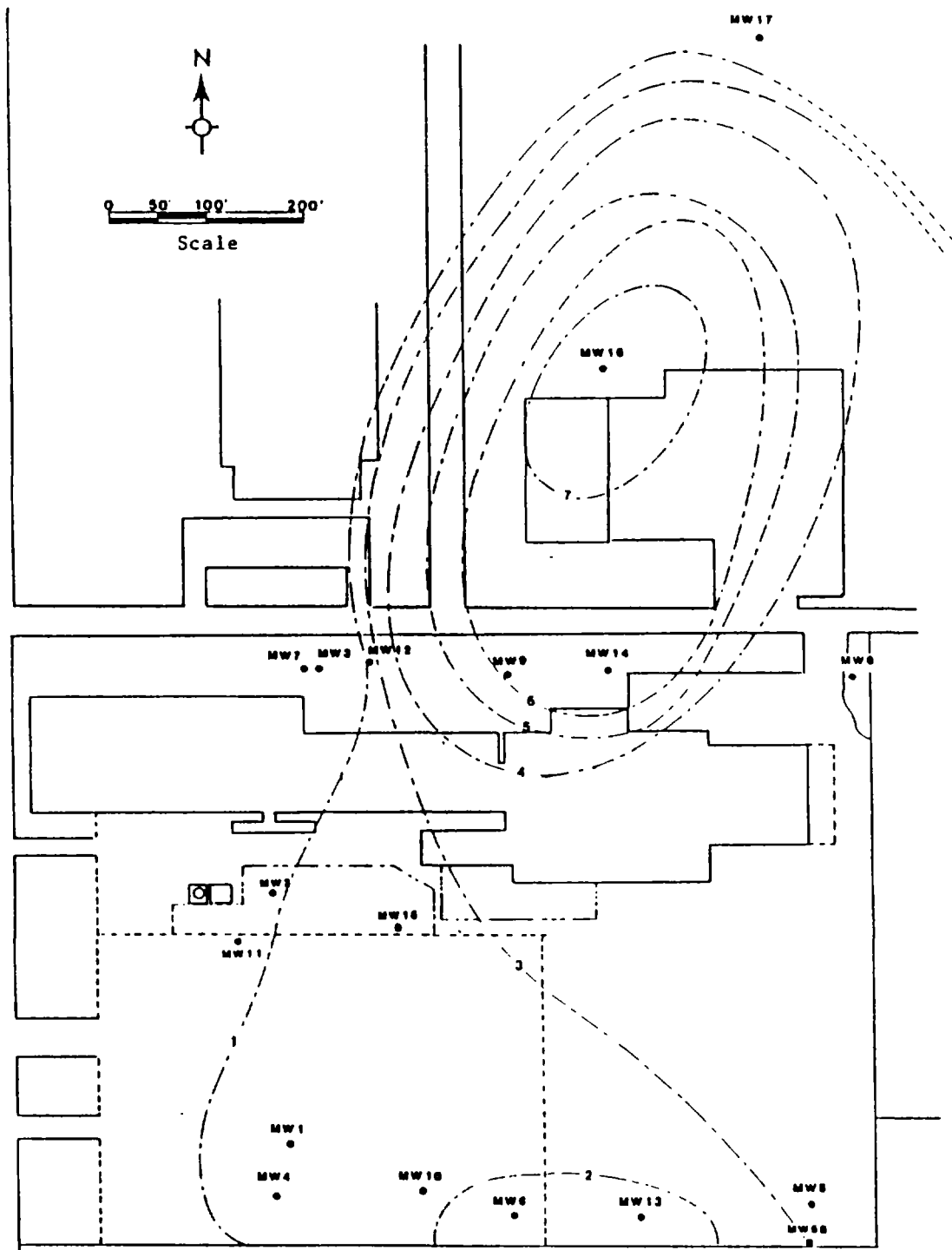


Figure 19
Four Winns Boats
Total Scan 1 Contamination - 7/9/87

- KEY
1. 0-1
 2. 1-10
 3. 10-100
 4. 100-500
 5. 500-1,000
 6. 1,000-5,000
 7. 5,000-10,000

in ppb (parts per billion)
--- Estimation of contamination Boundaries

TABLE 8. SUMMARY OF ANALYTICAL DATA, ACETONE CONTAMINATION,
FOR FOUR WINNS, INC., FRISBIE STREET PLANT, 1987

Monitoring Well	Date Sampled	Depth Sampled	Acetone Mg/L
MW2	3/23/87	27-32'	ND
MW7	4/16/87	44-49'	ND
MW9	3/23/87	53-58'	4.840
	7/9/87	53-58'	2.930
MW12	4/16/87	62-67'	ND
	7/9/87	53-58'	.042
MW14	3/19/87	53-58'	3.910
	7/9/87	53-58'	ND
MW15	3/20/87	44-49'	1.100
	7/9/87	53-38'	ND
MW16	4/16/87	43-48'	ND
	4/16/87	57-62'	ND
	7/9/87	56-61'	6.400
MW17	6/29/87	25-30'	.063

ND = Nondetectable

TABLE 9. COMPARISON OF CHEMICAL COMPOUNDS DOWNGRAIDENT VRS.
UPGRAIDENT MONITORING WELLS, FOUR WINNS FRISBIE STREET
PLANT SITE, 1986/1987.

MONITORING WELL	DATE SAMPLED	DEPTH (FEET)	TCE MG/L	1,1-DCE MG/L	1,2-DCE MG/L	1,1-DCA MG/L	1,2-DCA MG/L	1,1,2-TCA MG/L
=====								
South and central monitoring wells:								
MW5	7/9/87	29-34	.058	ND	ND	ND	ND	.001
MW6	7/9/87	26-31	ND	ND	ND	ND	ND	ND
MW10	7/9/87	35-40	.005	ND	ND	ND	ND	ND
MW13	7/9/87	44-49	ND	ND	ND	ND	ND	ND
MW15	7/9/87	53-58	.001	ND	ND	ND	ND	ND
Northern and downgradient monitoring wells:								
MW9	7/9/87	53-58	.345	.051	ND	.245	.121	.010
MW12	7/9/87	53-58	.001	ND	ND	ND	ND	ND
MW14	7/9/87	53-58	.144	.047	ND	.414	.025	.005
MW16	7/9/87	56-61	.887	.218	.019	.750	.930	.016

ND = Non-detectable

1,1-DCE = 1,1-Dichloroethene

1,2-DCE = 1,2-Dichloroethene

1,1-DCA = 1,1-Dichloroethane

1,2-DCA = 1,2-Dichloroethane

1,1,2-TCA = 1,1,2-Trichloroethane

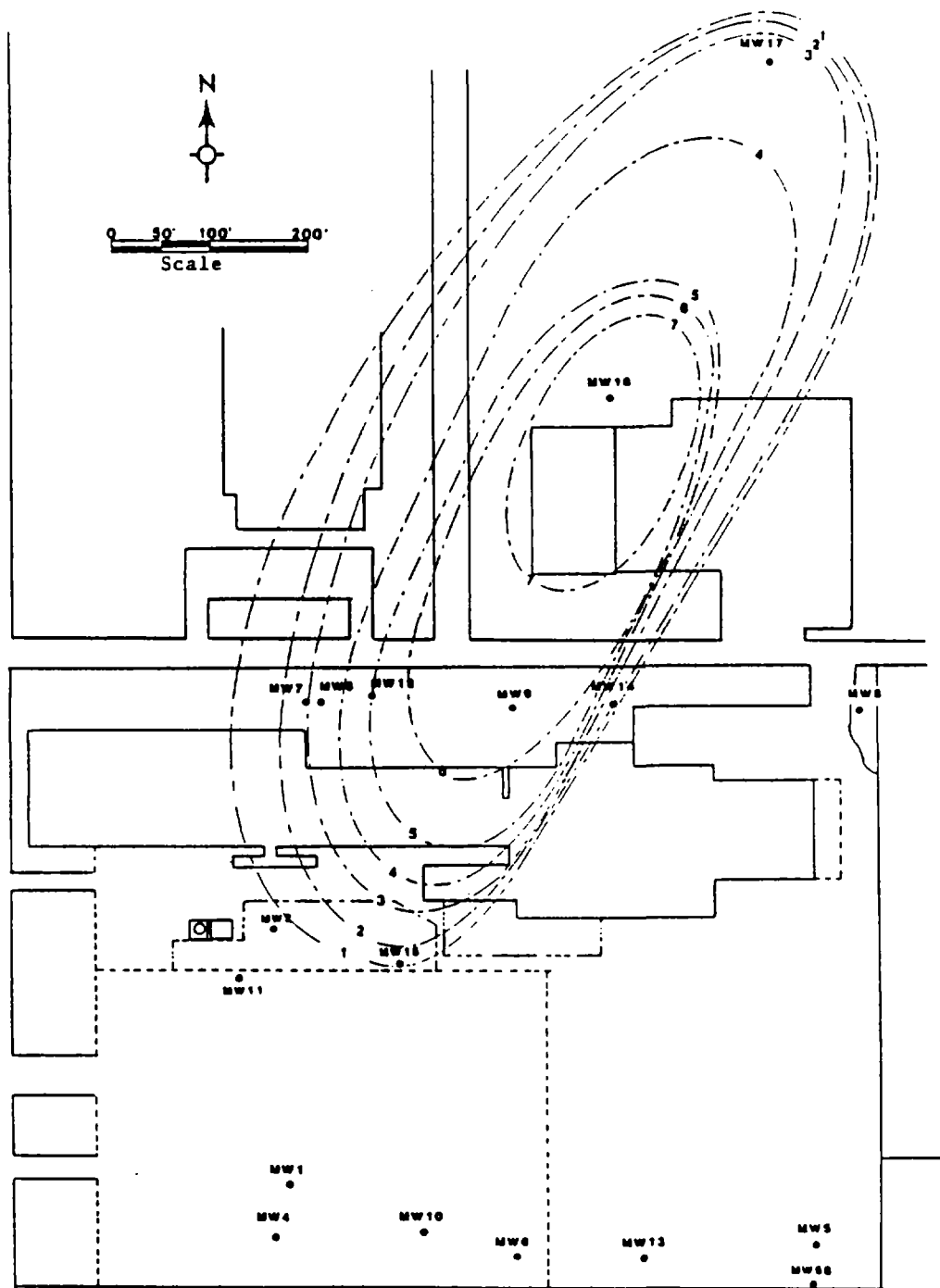


Figure 20
Four Winns Boats
Acetone Contamination - 7/9/87

KEY
 1. 0-1
 2. 1-10
 3. 10-100
 4. 100-500
 5. 500-1,000
 6. 1,000-5,000
 7. 5,000-10,000
 in ppb (parts per billion)
 - - - Estimate of Contamination Boundaries

TABLE 10. SUMMARY OF SCAN 2 ANALYTICAL DATA, FOR FOUR WINNS, INC.,
FRISBIE STREET PLANT, 1986-1987.

Monitoring Well	Date Sampled	Depth Sampled	Toluene Mg/L	Styrene Mg/L	Xylene Mg/L	Total Scan 2 Contamination Mg/L
MW4	10/15/86	26-31'	ND	ND	ND	ND
MW5	10/20/86	43-48'	ND	ND	ND	ND
MW6	11/12/86	23-28'	ND	ND	ND	ND
MW7	11/13/86	44-49'	.063	ND	ND	.063
MW8	12/1/86	50-55'	.005	ND	ND	.005
MW9	12/4/86	53-38'	ND	ND	ND	ND
MW11	2/4/87	62-67'	.002	.003	.005	.010
MW12	2/6/87	53-58'	ND	.004	ND	.004

ND = Nondetectable

TABLE 11. SUMMARY OF SOIL ANALYTICAL DATA
FOUR WINNS FRISBIE STREET PLANT,
1985-1987.

SOIL BORING	DATE SAMPLED	DEPTH SAMPLED	TOTAL SCAN 1	TOTAL SCAN 2	TCA MG/L	METHYLENE MG/L	TCE MG/L	PCE MG/L	STYRENE MG/L	ETHYLBENZENE MG/L	XYLENE MG/L
			CONTAMINATION MG/L	CONTAMINATION MG/L							
=====											
MDNR samples											
47367-47373	4/23/87	surface	*	*	*(7)	*(3)	*(1)	*(0)	*(7)	*(7)	*(7)
SB-A	5/8/85	10'	ND	.064	ND	ND	ND	ND	.029	ND	.035
SB-B	5/8/85	10'	ND	.089	ND	ND	ND	ND	.058	ND	.025
SB-C	5/8/85	10'	.098	.066	ND	.027	.040	ND	.033	ND	.014
SB-D	5/8/85	7'	ND	.254	ND	ND	ND	ND	ND	.073	.154
SB-E	5/8/85	7'	5.182	.103	ND	4.70	ND	ND	.023	ND	.045
SB-F	5/8/85	10'	.161	.218	ND	.083	.014	ND	.105	.045	.036
SB-G	5/8/85	10'	ND	1.093	ND	ND	.040	ND	.111	ND	.017
SB-H	5/8/85	10'	.021	ND	ND	.021	ND	ND	ND	ND	ND
SB-I	5/8/85	5'	ND	ND	ND	ND	ND	ND	ND	ND	ND
SB-2	9/22/87	2'	ND	---	ND	ND	ND	ND	---	---	---
SB-7	9/22/87	2'	ND	---	ND	ND	ND	ND	---	---	---
SB-10	9/22/87	2'	ND	---	ND	ND	ND	ND	---	---	---
SB-11	9/22/87	2'	.189	---	.127	ND	ND	ND	---	---	---
=====											

* = Not quantified

*(#) = Number of times compound was identified

ND = Non-detectable

--- = Not analyzed for

APPENDIX K

1987 KB BORING LOGS AND SOIL ANALYSES FOR VOCs BEHIND KYSOR

APPENDIX K

SUMMARY TABLES, KYSOR BORING VOC ANALYSES
IN SOILS

JULY 1987

	KB-1 S-3	KB-1 S-5	KB-1 S-5D	KB-2 S-4	KB-2 S-5	KB-3 S-3	KB-3 S-4	KB-3 S-4R	KB-4 S-3
Acetone	10	15	26	18	900	2,000	--	--	13
2-Butanone	75	--	--	--	2,100	5,700	--	--	11
1,1,1-Trichloroethane	28	21	64	25	--	920	14,000	24,000	17
Trichloroethene	--	22	110	80	690	12,000	58,000	74,000	69
1,1,2-Trichloroethane	7	27	55	--	--	--	--	--	--
Tetrachloroethene	31	7	5	53	--	2,200	--	--	--
Toluene	16	12	38	--	730	840	92,000	95,000	29
Ethylbenzene	--	7	8	--	--	--	68,000	59,000	15
Total Xylenes	--	40	43	6	1,900	17,000	520,000	420,000	73
TICs	--	<u>17</u>	<u>25</u>	<u>39</u>	<u>--</u>	<u>1,800</u>	<u>--</u>	<u>--</u>	<u>9</u>
Total	99	151	349	134	6,320	40,660	752,000	672,000	227

All results in parts per billion (ppb).

TICs = Tentatively Identified Compounds

-- = Not Detected

APPENDIX K (PAGE 2)

SUMMARY TABLES, KYSOR BORING VOC ANALYSES
IN SOILS

JULY 1987

	KB-4 S-3R	KB-4 S-5	KB-5 S-4	KB-5 S-5	KB-6 S-1	KB-6 S-3	KB-6 S-3R	KB-7 S-3	KB-7 S-5
Acetone	15	27	22	24	--	20	15	--	--
2-Butanone	11	58	12	11	--	--	--	--	--
1,1,1-Trichloroethane	32	26	10	5	14	8	7	--	8,000
Trichloroethene	85	110	38	28	72	31	32	65,000	21,000
Tetrachloroethene	--	5	--	3	--	--	--	--	--
Toluene	35	36	1	65	45	15	15	7,800	6,000
Ethylbenzene	14	12	8	5	3	6	5	17,000	68,000
Total Xylenes	54	53	31	18	17	33	24	410,000	190,000
TICs	<u>14</u>	<u>17</u>	<u>7</u>	<u>--</u>	<u>6</u>	<u>11</u>	<u>6</u>	<u>--</u>	<u>--</u>
Total	246	327	137	99	151	125	115	499,800	293,000

All results in parts per billion (ppb).

TICs = Tentatively Identified Compounds

-- = Not Detected

APPENDIX K (PAGE 3)

SUMMARY TABLES, KYSOR BORING VOC ANALYSES
IN SOILS

JULY 1987

	KB-7 S-3R	KB-8 S-3	KB-8 S-5D/L	KB-8 S-5	KB-8 S-5D/L	KB-9 S-2	KB-9 S-5	KB-9 S-5D	KB-10 S-2
Acetone	--	8	39	23	19	7	18	18	11
1,2-Dichloroethane	--	--	--	5	--	--	--	--	--
2-Butanone	--	--	--	--	--	--	--	--	9
1,1,1-Trichloroethane	--	320	11	300	34	--	37	11	5
Trichloroethene	58,000	530	51	370	67	4	58	36	21
1,1,2-Trichloroethane	--	9	--	28	19	--	--	--	--
Tetrachloroethene	--	45	--	96	31	--	4	--	--
Toluene	--	100	34	110	26	5	60	19	56
Ethylbenzene	11,000	5	--	5	4	--	4	6	--
Total Xylenes	300,000	20	19J	45	50	13	20	36	--
TICs	--	--	--	--	--	--	--	15	--
Total	369,000	1,037	206	982	250	29	201	141	102

All results in parts per billion (ppb).

TICs = Tentatively Identified Compounds

-- = Not Detected

APPENDIX K (PAGE 4)

SUMMARY TABLES, KYSOR BORING VOC ANALYSES
IN SOILS

JULY 1987

	KB-10 S-2D	KB-10 S-5	KB-10 S-5R
Acetone	--	22	13
2-Butanone	--	16	10
1,1,1-Trichloroethane	18	10	43
Trichloroethene	48	39	170
Tetrachloroethene	--	--	4
Toluene	32	10	47
Ethylbenzene	5	3	10
Total Xylenes	29	33	59
TICs	<u>15</u>	<u>7</u>	<u>12</u>
Total	140	133	356

All results in parts per billion (ppb).

TICs = Tentatively Identified Compounds

-- = Not Detected

APPENDIX L

PHOTOVAC RESULTS OF SCREENED AUGER GROUNDWATER
SAMPLES TAKEN FOR CADILLAC RI

APPENDIX L

SUMMARY TABLES OF FIELD SCREENING DATA, SCREENED AUGER SAMPLES CADILLAC AREA GROUNDWATER STUDY CADILLAC, MICHIGAN

MONITORING WELL

Approximate Depth	SLW-1	SLW-2	SLW-3	SLW-4	SLW-5	SLW-6	SLW-7	SLW-8	SLW-9
20	--	--	--	--	--	--	--	--	--
30	ND	--	--	--	--	1,297	--	10	571
40	ND	3	Tr	5	Tr	5,754	13	58	3,238
50	Clay	42	Tr	28	3	3,461	9	483	7,646
60	--	57	Tr	34	Tr	4,292	13	261	10,079
70	--	46	Tr	27	Tr	1,663	12	79	5,263
80	--	13	Tr	31	Tr	Clay	17	11	2,654
90	--	6	Tr	5	--	--	21	Clay	Clay
100	--	--	--	--	--	--	--	--	--
110	--	--	--	--	--	--	--	--	--
120	--	--	--	--	--	--	--	--	--
130	--	--	--	--	--	--	--	--	--
140	--	--	--	--	--	--	--	--	--
150	--	--	--	--	--	--	--	--	--
160	--	--	--	--	--	--	--	--	--
170	--	--	--	--	--	--	--	--	--
180	--	--	--	--	--	--	--	--	--
190	--	--	--	--	--	--	--	--	--
200	--	--	--	--	--	--	--	--	--

All results in parts per billion (ppb)

ND = Not Detected

Tr = Trace (less than 2 ppb)

APPENDIX L (PAGE 2)

SUMMARY TABLES OF FIELD SCREENING DATA,
SCREENED AUGER SAMPLES
CADILLAC AREA GROUNDWATER STUDY
CADILLAC, MICHIGAN

MONITORING WELL

Approximate Depth	SLW-10D	SLW-11	SLW-12D	SLW-13	SLW-14	SLW-15	SLW-16	SLW-17	SLW-18	SLW-19
20	--	--	--	--	--	--	Tr	--	--	--
30	--	30	510	--	--	Tr	Tr	--	24	--
40	668	49	5,968	10	Tr	628	Tr	Tr	8	Tr
50	4,059	279	Clay	10	Tr	6,994	Clay	Tr	17	Tr
60	1,724	102	11	21	Tr	8,471	Clay	Tr	Clay	Tr
70	14,257	30	101	44	Tr	Clay	Tr	Tr	Tr	Tr
80	18,575	Tr	345	32	Tr	2,460	Tr	Tr	Tr	Tr
90	25,949	Clay	367	85	Tr	6,330	Clay	Tr	Tr	Tr
100	33,787	--	298	91	--	684	--	--	Tr	--
110	35,407	--	313	133	--	4	--	--	Tr	--
120	31,890	--	209	171	--	Tr	--	--	Tr	--
130	28,812	--	15	--	--	Tr	--	--	Tr	--
140	31,081	--	6	--	--	Tr	--	--	Tr	--
150	36,522	--	8	--	--	Tr	--	--	Tr	--
160	30,549	--	3	--	--	Tr	--	--	Tr	--
170	26,326	--	--	--	--	Tr	--	--	Tr	--
180	14,844	--	--	--	--	--	--	--	--	--
190	--	--	--	--	--	Clay	--	--	Clay	--
200	--	--	--	--	--	--	--	--	--	--

All results in parts per billion (ppb)

ND = Not detected

Tr = Trace (less than 2 ppb)

APPENDIX L (PAGE 3)

SUMMARY TABLES OF FIELD SCREENING DATA,
SCREENED AUGER SAMPLES
CADILLAC AREA GROUNDWATER STUDY
CADILLAC, MICHIGAN

MONITORING WELL

Approximate Depth	SLW-20	SLW-21	SLW-22	SLW-23	SLW-24	SLW-25	SLW-26	SLW-27	SLW-28	SLW-29
20	--	--	--	--	--	--	Clay	--	--	--
30	--	Tr	Tr	ND	12	3,688	5	5	Tr	Tr
40	Tr	Tr	1554	ND	Tr	9,700	6	5	Tr	Tr
50	Tr	Tr	234	ND	Tr	11,521	Tr	12	Tr	Tr
60	5	Tr	447	Clay	Tr	580	Tr	5	Tr	Tr
70	17	3	Tr	ND	Tr	140	Tr	6	Tr	Tr
80	24	Tr	Clay	Tr	Tr	67	Tr	6	Tr	Tr
90	49	Tr	Clay	--	Tr	49	Tr	51	Tr	--
100	59	Tr	63	--	5	Clay	--	--	Tr	--
110	75	Tr	125	--	ND	--	--	--	Tr	--
120	77	Tr	1,970	--	Clay	--	--	--	Clay	--
130	93	Tr	--	--	--	--	--	--	--	--
140	74	Tr	--	--	--	--	--	--	--	--
150	139	Tr	--	--	--	--	--	--	--	--
160	94	Tr	--	--	--	--	--	--	--	--
170	92	Tr	--	--	--	--	--	--	--	--
180	81	--	--	--	--	--	--	--	--	--
190	--	--	--	--	--	--	--	--	--	--
200	--	--	--	--	--	--	--	--	--	--

All results in parts per billion (ppb)

ND = Not Detected

Tr = Trace (less than 2 ppb)

APPENDIX L (PAGE 4)

SUMMARY TABLES OF FIELD SCREENING DATA,
SCREENED AUGER SAMPLES
CADILLAC AREA GROUNDWATER STUDY
CADILLAC, MICHIGAN

MONITORING WELL

Approximate Depth	SLW-30	SLW-31	SLW-32	SLW-33	SLW-34	SLW-35	SLW-36	SLW-37	SLW-38	SLW-39
20	--	--	--	--	--	--	--	--	--	--
30	50	4	Tr	30	3	--	Tr	148	--	54
40	260	246	Tr	318	Tr	Tr	Tr	1,894	632	148
50	694	Clay	Tr	331	3	Tr	Tr	1,849	2,526	90
60	281	--	Tr	240	3	Tr	Tr	1,484	2,780	141
70	Clay	--	Tr	76	3	Tr	Tr	752	Clay	131
80	--	--	Clay	9	Tr	Tr	Tr	787	--	76
90	--	--	--	10	3	Tr	Tr	865	--	38
100	--	--	--	--	--	--	--	807	--	--
110	--	--	--	--	--	--	--	836	--	--
120	--	--	--	--	--	--	--	--	--	--
130	--	--	--	--	--	--	--	--	--	--
140	--	--	--	--	--	--	--	--	--	--
150	--	--	--	--	--	--	--	--	--	--
160	--	--	--	--	--	--	--	--	--	--
170	--	--	--	--	--	--	--	--	--	--
180	--	--	--	--	--	--	--	--	--	--
190	--	--	--	--	--	--	--	--	--	--
200	--	--	--	--	--	--	--	--	--	--

All results in parts per billion (ppb)

ND = Not detected

Tr = Trace (less than 2 ppb)

APPENDIX L (PAGE 5)

FIELD SCREENING DATA
 SCREENED AUGER SAMPLES
 CADILLAC AREA GROUNDWATER STUDY
 CADILLAC, MICHIGAN

MONITORING WELL

Approximate Depth	SLW-40	SLW-41	SLW-42	SLW-43	SLW-44	SLW-45	SLW-46	SLW-47	SLW-48	SLW-49
20	--	--	--	--	--	--	--	--	--	5
30	4	8	--	3,888	6	8	10	4	25	138
40	3	52	--	5,314	488	48	4	357	17	822
50	32	40	--	5,447	1,000	5	20	694	25	4,911
60	50	3,183	--	2,685	265	3	15	1,462	6	2,561
70	9	83	--	1,108	210	8	4	2,735	Tr	Clay
80	8	88	--	89	165	11	39	649	Tr	405
90	3	13	--	9	8	Tr	10	36	Tr	268
100	--	Tr	--	12	9	3	10	5	4	4
110	--	10	4,807	25	7	4	15	4	Tr	3
120	--	1,492	6,230	20	4	6	10	Tr	3	3
130	--	9,162	3,070	44	147	9	5	Tr	--	2
140	--	9,934	--	10	102	Tr	10	--	--	2
150	--	--	--	4	45	--	5	--	--	Tr
160	--	--	--	--	Clay	--	--	--	--	--
170	--	--	--	--	--	--	--	--	--	--
180	--	--	--	--	--	--	--	--	--	--
190	--	--	--	--	--	--	--	--	--	--
200	--	--	--	--	--	--	--	--	--	--

All results in parts per billion (ppb)

ND = Not Detected

Tr = Trace (less than 2 ppb)

APPENDIX L (PAGE 6)

FIELD SCREENING DATA
 SCREENED AUGER SAMPLES
 CADILLAC AREA GROUNDWATER STUDY
 CADILLAC, MICHIGAN

MONITORING WELL

Approximate Depth	SLW-50	SLW-51	SLW-52	SLW-53
20	--	--	--	--
30	10	Tr	4	4
40	1,259	39	Tr	Tr
50	8,906	70	3	Tr
60	9,387	77	4	Tr
70	13,310	47	3	Tr
80	14,307	88	5	Tr
90	81,507	100	5	ND
100	223,416	28	Tr	ND
110	14,947	100	7	ND
120	9,687	86	4	ND
130	11,441	--	5	Tr
140	8,231	--	4	ND
150	--	--	--	Tr
160	--	--	--	ND
170	--	--	--	ND
180	--	--	--	--
190	--	--	--	--
200	--	--	--	--

All results in parts per billion (ppb)

ND = Not Detected

Tr = Trace

APPENDIX L (PAGE 7)

FIELD SCREENING DATA
 SCREENED AUGER SAMPLES
 CADILLAC AREA GROUNDWATER STUDY
 CADILLAC, MICHIGAN

MONITORING WELL

Approximate Depth	GW-1D	GW-2	GW-3	GW-6D	GW-7	GW-8	GW-9	GW-10D	GW-11	GW-13
20	--	--	--	--	--	ND	ND	355	--	--
30	Clay	ND	Tr	--	Tr	ND	Clay	816	Tr	ND
40	Clay	ND	Tr	163	Tr	ND	ND	2	Tr	ND
50	3	ND	Tr	50	Tr	ND	ND	21	Tr	ND
60	ND	ND	Tr	Tr	Tr	ND	ND	12	Tr	ND
70	ND	3	Tr	8	Tr	ND	ND	Clay	Tr	Tr
80	ND	Tr	Tr	7	Tr	ND	ND	Clay	Tr	Tr
90	ND	Tr	Tr	Tr	Tr	ND	ND	Tr	Tr	Tr
100	ND	Tr	Tr	Clay	Tr	Tr	ND	68	Tr	Tr
110	ND	Tr	Tr	16	Tr	Clay	10	97	Tr	ND
120	ND	Tr	Tr	Tr	Tr	Tr	29	169	Tr	Tr
130	Tr	Tr	Tr	Tr	Tr	Tr	12	248	Tr	Tr
140	13	ND	Tr	Tr	Tr	Tr	Tr	273	Tr	ND
150	ND	ND	Tr	Tr	Tr	Tr	114	364	Tr	ND
160	ND	Tr	Tr	4	Tr	Tr	201	439	Tr	ND
170	ND	Tr	Tr	31	Tr	Tr	128	386	Tr	ND
180	ND	--	Tr	Tr	--	Tr	--	--	--	--
190	--	--	--	--	--	--	--	--	--	--
200	--	--	--	--	--	--	--	--	--	--

All results in parts per billion (ppb)

ND = Not Detected

Tr = Trace (less than 2 ppb)

APPENDIX L (PAGE 8)

FIELD SCREENING DATA
SCREENED AUGER SAMPLES
CADILLAC AREA GROUNDWATER STUDY
CADILLAC, MICHIGAN

MONITORING WELL

Approximate Depth	GW-14D	GW-16	GW-17	GW-18	GW-20
20	Tr	--	--	--	--
30	10	--	--	--	--
40	6	--	6	Tr	Tr
50	3	ND	Tr	Tr	4
60	6	ND	Tr	Tr	Clay
70	Tr	ND	Tr	Tr	Clay
80	Tr	ND	Tr	Tr	Tr
90	Tr	ND	Tr	Tr	4
100	Clay	ND	Tr	Tr	Tr
110	Clay	ND	Tr	Tr	4
120	ND	ND	Tr	Tr	Tr
130	ND	ND	Tr	Tr	3
140	ND	ND	Tr	Tr	Tr
150	ND	ND	Tr	Tr	Tr
160	ND	ND	Tr	Tr	--
170	ND	ND	Tr	Tr	--
180	ND	ND	Tr	Tr	--
190	--	ND	--	--	--
200	Clay	ND	--	--	--

All results in parts per billion (ppb)

ND = Not Detected

Tr = Trace (less than 2 ppb)

APPENDIX Q
STATISTICAL ANALYSIS OF CONTAMINANT DISTRIBUTION
FOR
SELECTION OF CHEMICALS OF CONCERN

8.87.31
0191.0.0

SITE: CADILLAC AREA GROUNDWATER CONTAMINATION REMEDIAL INVESTIGATION

TASK: SELECTION OF CONTAMINANTS OF CONCERN
STATISTICAL ANALYSIS OF CONTAMINANT DISTRIBUTION

BY: JOHN H. BARKACH
DATE: FEBRUARY 2, 1988

DISK: 210
FILE: DAT_SHAL

PLUME: SHALLOW

1. CHEMICAL: TRICHLOROETHENE

DATA

2	18	3
8300	380	3900
21	4000	260
860	3100	2
3800	4800	44
2100	59	1700
3800	380	17
89	3	1
360	78000	32
12000	240	1300

DISTRIBUTION

INTERVAL	FREQUENCY
1 - 5	5
5 - 10	0
10 - 100	7
100 - 1000	6
1000 - 10000	10
10000 - 100000	2
100000 - 1000000	0

MAXIMUM: 78000 ug/l
ARITHMETIC MEAN: 2879.4 ug/l
NUMBER OF SAMPLES: 45
% OF SAMPLES ABOVE DETECTION: 66.7 %

2. CHEMICAL: 1,1,1-TRICHLOROETHANE

DATA

1900	3600	32000	440	1800
6200	2	76	230	1200
120	1700	150	14000	
20000	28000	1500	300	
4300	5400	54	190	
160	90	180	16000	

DISTRIBUTION

INTERVAL	FREQUENCY
1 - 5	1
5 - 10	0
10 - 100	3
100 - 1000	8
1000 - 10000	9
10000 - 100000	5
100000 - 1000000	0

MAXIMUM: 32000 ug/l
ARITHMETIC MEAN: 3102.0 ug/l
NUMBER OF SAMPLES: 45
% OF SAMPLES ABOVE DETECTION: 57.8 %

3. CHEMICAL: TETRACHLOROETHENE

DISTRIBUTION

			INTERVAL	FREQUENCY
DATA	510	250	1 - 5	2
	3	720	5 - 10	0
	400	430	10 - 100	1
	3	170	100 - 1000	7
	360		1000 - 10000	0
	62		10000 - 100000	0
			100000 - 1000000	0

MAXIMUM: 720 ug/l
 ARITHMETIC MEAN: 64.6 ug/l
 NUMBER OF SAMPLES: 45
 % OF SAMPLES ABOVE DETECTION: 22.2 %

4. CHEMICAL: 1,1-DICHLOROETHENE

DISTRIBUTION

				INTERVAL	FREQUENCY
DATA	280	430	13	1 - 5	1
	710	330	29	5 - 10	2
	280	8	380	10 - 100	5
	4	13	110	100 - 1000	9
	78	6	93	1000 - 10000	0
	1000	460		10000 - 100000	0
				100000 - 1000000	0

MAXIMUM: 1000 ug/l
 ARITHMETIC MEAN: 93.9 ug/l
 NUMBER OF SAMPLES: 45
 % OF SAMPLES ABOVE DETECTION: 37.8 %

5. CHEMICAL: 1,1-DICHLOROETHANE

DISTRIBUTION

			INTERVAL	FREQUENCY
DATA	280		1 - 5	0
	420		5 - 10	1
	6		10 - 100	0
			100 - 1000	2
			1000 - 10000	0
			10000 - 100000	0
			100000 - 1000000	0

MAXIMUM: 420 ug/l
 ARITHMETIC MEAN: 15.7 ug/l
 NUMBER OF SAMPLES: 45
 % OF SAMPLES ABOVE DETECTION: 6.7 %

DISTRIBUTION

6. CHEMICAL: 1,2-DICHLOROETHANE

		INTERVAL	FREQUENCY
DATA	1500	1 - 5	0
	2000	5 - 10	0
	1200	10 - 100	0
		100 - 1000	0
		1000 - 10000	3
		10000 - 100000	0
		100000 - 1000000	0

MAXIMUM: 2000 ug/l
 ARITHMETIC MEAN: 104.4 ug/l
 NUMBER OF SAMPLES: 45
 % OF SAMPLES ABOVE DETECTION: 6.7 %

DISTRIBUTION

7. CHEMICAL: TOLUENE

		INTERVAL	FREQUENCY
DATA	3200 84	1 - 5	1
	2 44	5 - 10	0
	720	10 - 100	3
	35	100 - 1000	2
	140	1000 - 10000	2
	1600	10000 - 100000	0
		100000 - 1000000	0

MAXIMUM: 3200 ug/l
 ARITHMETIC MEAN: 129.4 ug/l
 NUMBER OF SAMPLES: 45
 % OF SAMPLES ABOVE DETECTION: 17.8 %

DISTRIBUTION

8. CHEMICAL: ETHYLBENZENE

		INTERVAL	FREQUENCY
DATA	400	1 - 5	0
	240	5 - 10	0
	670	10 - 100	0
		100 - 1000	3
		1000 - 10000	0
		10000 - 100000	0
		100000 - 1000000	0

MAXIMUM: 670 ug/l
 ARITHMETIC MEAN: 29.1 ug/l
 NUMBER OF SAMPLES: 45
 % OF SAMPLES ABOVE DETECTION: 6.7 %

9. CHEMICAL: XYLENE

DISTRIBUTION

DATA		INTERVAL		FREQUENCY
	1700	1 -	5	0
	1000	5 -	10	0
		10 -	100	0
		100 -	1000	0
		1000 -	10000	2
		10000 -	100000	0
		100000 -	1000000	0

MAXIMUM: 1700 ug/l
 ARITHMETIC MEAN: 60.0 ug/l
 NUMBER OF SAMPLES: 45
 % OF SAMPLES ABOVE DETECTION: 4.4 %

10. CHEMICAL: BENZENE

DISTRIBUTION

DATA		INTERVAL		FREQUENCY
	20	1 -	5	0
		5 -	10	0
		10 -	100	1
		100 -	1000	0
		1000 -	10000	0
		10000 -	100000	0
		100000 -	1000000	0

MAXIMUM: 20 ug/l
 ARITHMETIC MEAN: 0.4 ug/l
 NUMBER OF SAMPLES: 45
 % OF SAMPLES ABOVE DETECTION: 2.2 %

11. CHEMICAL: ACETONE

DISTRIBUTION

DATA		INTERVAL		FREQUENCY
	510	1 -	5	1
	920	5 -	10	1
	500	10 -	100	1
	5	100 -	1000	3
	62	1000 -	10000	0
	7	10000 -	100000	0
		100000 -	1000000	0

MAXIMUM: 920 ug/l
 ARITHMETIC MEAN: 44.5 ug/l
 NUMBER OF SAMPLES: 45
 % OF SAMPLES ABOVE DETECTION: 13.3 %

12. CHEMICAL: METHYLENE CHLORIDE

DISTRIBUTION

		INTERVAL	FREQUENCY
DATA	690	1 - 5	0
	140	5 - 10	0
	410	10 - 100	1
	260	100 - 1000	4
	26	1000 - 10000	0
		10000 - 100000	0
		100000 - 1000000	0

MAXIMUM: 690 ug/l
ARITHMETIC MEAN: 33.9 ug/l
NUMBER OF SAMPLES: 45
% OF SAMPLES ABOVE DETECTION: 11.1 %

13. CHEMICAL: CHLOROFORM

DISTRIBUTION

		INTERVAL	FREQUENCY
DATA	15	1 - 5	1
	4	5 - 10	0
		10 - 100	1
		100 - 1000	0
		1000 - 10000	0
		10000 - 100000	0
		100000 - 1000000	0

MAXIMUM: 15 ug/l
ARITHMETIC MEAN: 0.4 ug/l
NUMBER OF SAMPLES: 45
% OF SAMPLES ABOVE DETECTION: 4.4 %

NOTE: THE DATA COLLECTED ON JUNE AND AUGUST 1987 WAS USED
IN THE STATISTICAL ANALYSIS OF CONTAMINANT
DISTRIBUTION.

SITE: CADILLAC AREA GROUNDWATER CONTAMINATION REMEDIAL INVESTIGATION

TASK: SELECTION OF CONTAMINANTS OF CONCERN
STATISTICAL ANALYSIS OF CONTAMINANT DISTRIBUTION

BY: JOHN H. BARKACH

DATE: FEBRUARY 2, 1988

DISK: 210

FILE: DAT_INTR

PLUME: INTERMEDIATE

1. CHEMICAL: TRICHLOROETHENE

				DISTRIBUTION	
				INTERVAL	FREQUENCY
DATA	10	19	6800	1 - 5	0
	7500	1100	4700	5 - 10	2
	5000	5000	3900	10 - 100	1
	950	1800		100 - 1000	1
	3600	6		1000 - 10000	11
	1600	5300		10000 - 100000	0
				100000 - 1000000	0

MAXIMUM: 7500 ug/l
ARITHMETIC MEAN: 2251.7 ug/l
NUMBER OF SAMPLES: 21
% OF SAMPLES ABOVE DETECTION: 71.4 %

2. CHEMICAL: 1,1,1-TRICHLOROETHANE

				DISTRIBUTION	
				INTERVAL	FREQUENCY
DATA	1200			1 - 5	0
	680			5 - 10	0
	1100			10 - 100	0
	910			100 - 1000	2
				1000 - 10000	2
				10000 - 100000	0
				100000 - 1000000	0

MAXIMUM: 1200 ug/l
ARITHMETIC MEAN: 185.2 ug/l
NUMBER OF SAMPLES: 21
% OF SAMPLES ABOVE DETECTION: 19.0 %

3. CHEMICAL: TETRACHLOROETHENE

DISTRIBUTION

		INTERVAL	FREQUENCY
DATA	17	1 - 5	0
		5 - 10	0
		10 - 100	1
		100 - 1000	0
		1000 - 10000	0
		10000 - 100000	0
		100000 - 1000000	0

MAXIMUM: 17 ug/l
ARITHMETIC MEAN: 0.8 ug/l
NUMBER OF SAMPLES: 21
% OF SAMPLES ABOVE DETECTION: 4.8 %

4. CHEMICAL: 1,1-DICHLOROETHENE

DISTRIBUTION

		INTERVAL	FREQUENCY
DATA		1 - 5	0
	62	5 - 10	0
	99	10 - 100	3
	90	100 - 1000	0
		1000 - 10000	0
		10000 - 100000	0
		100000 - 1000000	0

MAXIMUM: 99 ug/l
ARITHMETIC MEAN: 12.0 ug/l
NUMBER OF SAMPLES: 21
% OF SAMPLES ABOVE DETECTION: 14.3 %

5. CHEMICAL: TRANS-1,2-DICHLOROETHENE

DISTRIBUTION

		INTERVAL	FREQUENCY
DATA	4200 4600	1 - 5	0
		5 - 10	0
		10 - 100	0
		100 - 1000	0
		1000 - 10000	2
		10000 - 100000	0
		100000 - 1000000	0

MAXIMUM: 4600 ug/l
ARITHMETIC MEAN: 419.0 ug/l
NUMBER OF SAMPLES: 21
% OF SAMPLES ABOVE DETECTION: 9.5 %

6. CHEMICAL: METHYLENE CHLORIDE

DISTRIBUTION

		INTERVAL	FREQUENCY
DATA	240	1 - 5	0
	78	5 - 10	0
		10 - 100	1
		100 - 1000	1
		1000 - 10000	0
		10000 - 100000	0
		100000 - 1000000	0

MAXIMUM: 240 ug/l
ARITHMETIC MEAN: 15.1 ug/l
NUMBER OF SAMPLES: 21
% OF SAMPLES ABOVE DETECTION: 9.5 %

7. CHEMICAL: CHLOROFORM

DISTRIBUTION

		INTERVAL	FREQUENCY
DATA	4	1 - 5	1
		5 - 10	0
		10 - 100	0
		100 - 1000	0
		1000 - 10000	0
		10000 - 100000	0
		100000 - 1000000	0

MAXIMUM: 4 ug/l
ARITHMETIC MEAN: 0.2 ug/l
NUMBER OF SAMPLES: 21
% OF SAMPLES ABOVE DETECTION: 4.8 %

8. CHEMICAL: ACETONE

DISTRIBUTION

		INTERVAL	FREQUENCY
DATA	330	1 - 5	1
	420	5 - 10	0
	4	10 - 100	0
		100 - 1000	2
		1000 - 10000	0
		10000 - 100000	0
		100000 - 1000000	0

MAXIMUM: 420 ug/l
ARITHMETIC MEAN: 35.9 ug/l
NUMBER OF SAMPLES: 21
% OF SAMPLES ABOVE DETECTION: 14.3 %

DATA

1

INTERVAL		
1 -	5	1
5 -	10	0
10 -	100	0
100 -	1000	0
1000 -	10000	0
10000 -	100000	0
100000 -	1000000	0

MAXIMUM: 1 ug/l
ARITHMETIC MEAN: 0.0 ug/l
NUMBER OF SAMPLES: 21
% OF SAMPLES ABOVE DETECTION: 4.8 %

NOTE: THE DATA COLLECTED ON JUNE AND AUGUST 1987 WAS USED
IN THE STATISTICAL ANALYSIS OF CONTAMINANT
DISTRIBUTION.

SITE: CADILLAC AREA GROUNDWATER CONTAMINATION REMEDIAL INVESTIGATION

TASK: SELECTION OF CONTAMINANTS OF CONCERN
STATISTICAL ANALYSIS OF CONTAMINANT DISTRIBUTION

BY: JOHN H. BARKACH
DATE: FEBRUARY 2, 1988

DISK: 210
FILE: DAT_NORT

PLUME: NORTHERNAIRE

1. CHEMICAL: TRICHLOROETHENE

				DISTRIBUTION	
				INTERVAL	FREQUENCY
DATA	16	20	460	1 - 5	1
	6	100	58	5 - 10	4
	4	42	7	10 - 100	7
	600	12	310	100 - 1000	5
	61	6	110	1000 - 10000	1
	6	120	2500	10000 - 100000	0
				100000 - 1000000	0

MAXIMUM: 2500 ug/l
ARITHMETIC MEAN: 201.7 ug/l
NUMBER OF SAMPLES: 22
% OF SAMPLES ABOVE DETECTION: 81.8 %

2. CHEMICAL: 1,1,1-TRICHLOROETHANE

				DISTRIBUTION	
				INTERVAL	FREQUENCY
DATA	95			1 - 5	0
	6			5 - 10	1
	440			10 - 100	1
	440			100 - 1000	2
				1000 - 10000	0
				10000 - 100000	0
				100000 - 1000000	0

MAXIMUM: 440 ug/l
ARITHMETIC MEAN: 44.6 ug/l
NUMBER OF SAMPLES: 22
% OF SAMPLES ABOVE DETECTION: 18.2 %

3. CHEMICAL: TRANS-1,2-DICHLOROETHENE

DISTRIBUTION

DATA		INTERVAL		FREQUENCY
	12	1 -	5	0
	40	5 -	10	0
		10 -	100	2
		100 -	1000	0
		1000 -	10000	0
		10000 -	100000	0
		100000 -	1000000	0

MAXIMUM: 40 ug/l
ARITHMETIC MEAN: 2.4 ug/l
NUMBER OF SAMPLES: 22
% OF SAMPLES ABOVE DETECTION: 9.1 %

4. CHEMICAL: 1,1-DICHLOROETHENE

DISTRIBUTION

DATA		INTERVAL		FREQUENCY
	12	1 -	5	0
	26	5 -	10	0
		10 -	100	2
		100 -	1000	0
		1000 -	10000	0
		10000 -	100000	0
		100000 -	1000000	0

MAXIMUM: 26 ug/l
ARITHMETIC MEAN: 1.7 ug/l
NUMBER OF SAMPLES: 22
% OF SAMPLES ABOVE DETECTION: 9.1 %

5. CHEMICAL: TETRACHLOROETHENE

DISTRIBUTION

DATA		INTERVAL		FREQUENCY
	3	1 -	5	1
		5 -	10	0
		10 -	100	0
		100 -	1000	0
		1000 -	10000	0
		10000 -	100000	0
		100000 -	1000000	0

MAXIMUM: 3 ug/l
ARITHMETIC MEAN: 0.1 ug/l
NUMBER OF SAMPLES: 22
% OF SAMPLES ABOVE DETECTION: 4.5 %

6. CHEMICAL: HEXAVALENT CHROMIUM

DISTRIBUTION

DATA			INTERVAL		FREQUENCY
	384	103	1 -	5	0
	76.8	89.6	5 -	10	0
	197	773	10 -	100	5
	75.8	652	100 -	1000	5
	89.2		1000 -	10000	0
	52.6		10000 -	100000	0
			100000 -	1000000	0

MAXIMUM: 773 ug/l
ARITHMETIC MEAN: 103.9 ug/l
NUMBER OF SAMPLES: 24
% OF SAMPLES ABOVE DETECTION: 41.7 %

7. CHEMICAL: CYANIDE

DISTRIBUTION

DATA			INTERVAL		FREQUENCY
			1 -	5	0
			5 -	10	0
			10 -	100	0
			100 -	1000	0
			1000 -	10000	0
			10000 -	100000	0
			100000 -	1000000	0

MAXIMUM: 0.0 ug/l
ARITHMETIC MEAN: 0.0 ug/l
NUMBER OF SAMPLES: 24
% OF SAMPLES ABOVE DETECTION: 0.0 %

NOTE: THE DATA COLLECTED ON JUNE AND AUGUST 1987 WAS USED
IN THE STATISTICAL ANALYSIS OF CONTAMINANT
DISTRIBUTION. TWENTY TWO SAMPLES WERE COLLECTED.

TWO ADDITIONAL SAMPLES WERE INCLUDED IN THE INORGANIC
STATISTICAL ANALYSIS. THESE TWO SAMPLES REPRESENT
DUPLICATE ANALYSES OF MW-9 AND MW-7.

SITE: CADILLAC AREA GROUNDWATER CONTAMINATION REMEDIAL INVESTIGATION

TASK: SELECTION OF CONTAMINANTS OF CONCERN
STATISTICAL ANALYSIS OF CONTAMINANT DISTRIBUTION

BY: JOHN H. BARKACH
DATE: FEBRUARY 1, 1988

DISK: 210
FILE: DAT_REX

PLUME: REXAIR

1. CHEMICAL: TRICHLOROETHENE

							DISTRIBUTION	
							INTERVAL	FREQUENCY
DATA	2	470	1100	3	100000	2	1 - 5	6
	1900	370	1200	3	47000		5 - 10	2
	3600	38	2700	660	130		10 - 100	3
	3	30	1600	510	110		100 - 1000	6
	25000	153000	1100	12	7		1000 - 10000	8
	30000	104000	1600	7	4		10000 - 100000	4
							100000 - 1000000	2

MAXIMUM: 153000 ug/l
ARITHMETIC MEAN: 11073.5 ug/l
NUMBER OF SAMPLES: 43
% OF SAMPLES ABOVE DETECTION: 72.1 %

2. CHEMICAL: 1,1,1-TRICHLOROETHANE

							DISTRIBUTION	
							INTERVAL	FREQUENCY
DATA	17						1 - 5	0
	15						5 - 10	0
							10 - 100	2
							100 - 1000	0
							1000 - 10000	0
							10000 - 100000	0
							100000 - 1000000	0

MAXIMUM: 17 ug/l
ARITHMETIC MEAN: 0.7 ug/l
NUMBER OF SAMPLES: 43
% OF SAMPLES ABOVE DETECTION: 4.7 %

3. CHEMICAL: TOLUENE

DISTRIBUTION

		INTERVAL	FREQUENCY
DATA	4	1 - 5	3
	20	5 - 10	1
	42	10 - 100	2
	5	100 - 1000	0
	7	1000 - 10000	0
	1	10000 - 100000	0
		100000 - 1000000	0

MAXIMUM: 42 ug/l
ARITHMETIC MEAN: 1.8 ug/l
NUMBER OF SAMPLES: 43
% OF SAMPLES ABOVE DETECTION: 14.0 %

4. CHEMICAL: TRANS-1,2-DICHLOROETHENE

DISTRIBUTION

		INTERVAL	FREQUENCY
DATA	38	1 - 5	0
	18	5 - 10	0
	48	10 - 100	3
		100 - 1000	0
		1000 - 10000	0
		10000 - 100000	0
		100000 - 1000000	0

MAXIMUM: 48 ug/l
ARITHMETIC MEAN: 2.4 ug/l
NUMBER OF SAMPLES: 43
% OF SAMPLES ABOVE DETECTION: 7.0 %

5. CHEMICAL: CHLOROFORM

DISTRIBUTION

		INTERVAL	FREQUENCY
DATA	2	1 - 5	2
	3	5 - 10	1
	6	10 - 100	0
		100 - 1000	0
		1000 - 10000	0
		10000 - 100000	0
		100000 - 1000000	0

MAXIMUM: 6 ug/l
ARITHMETIC MEAN: 0.3 ug/l
NUMBER OF SAMPLES: 43
% OF SAMPLES ABOVE DETECTION: 7.0 %

6. CHEMICAL: METHYLENE CHLORIDE

DISTRIBUTION

		INTERVAL	FREQUENCY
DATA	780	1 - 5	2
	2	5 - 10	1
	39	10 - 100	1
	9	100 - 1000	1
	1	1000 - 10000	0
		10000 - 100000	0
		100000 - 1000000	0

MAXIMUM: 780 ug/l
ARITHMETIC MEAN: 19.3 ug/l
NUMBER OF SAMPLES: 43
% OF SAMPLES ABOVE DETECTION: 11.6 %

NOTE: THE DATA COLLECTED ON MARCH 1987 (FOUR SAMPLES) WERE NOT INCLUDED IN THE STATISTICAL ANALYSIS OF CONTAMINANT DISTRIBUTION.

APPENDIX R
FATE AND TRANSPORT CALCULATIONS

8.87.31
0192.0.0

APPENDIX R

FATE AND TRANSPORT CALCULATIONS

This appendix contains an assessment of the fate and transport of contaminants in the various groundwater plumes present in the study area, and an evaluation of the presence of volatile organic compounds remaining in soils at the Kysor facility. The primary focus is on the volatile organic compounds in the groundwater with the addition of the hexavalent chromium plume believed to have originated at Northernair. During the course of earlier investigations, elevated levels of inorganics such as calcium, sodium, potassium, and magnesium were also noted in the shallow and intermediate aquifers. These salts are believed to have been caused by salting of roads for ice and dust control. Although they affect water quality, they are not believed to be associated with any manufacturing facility, and the risk associated with possible consumption of the groundwater is mainly due to the presence of volatile organic compounds (solvents) and hexavalent chromium.

The first section of this appendix presents some general aspects of the contaminant plumes identified at the site and also presents the models that are used in the assessments. The remaining sections consider each identifiable plume and zones of higher concentration within larger plumes (e.g., the chromium plume within a larger and more diffuse volatile organic plume) and estimate concentrations as a function of time at selected receptor locations at the boundary of the industrial park (along the plume pathway) and, in two instances, to the municipal well field.

1.1 GENERAL CONSIDERATIONS AND MODELS

Several major factors control the migration of contaminants in the environment. In this study, the principal focus has been on groundwater, although in one area (at Kysor) some consideration has also been given to contaminants in soils. The major factors controlling migration include:

- o the hydrogeology, including the unsaturated zone,
- o chemical specific factors (e.g., solubility, adsorption potential, volatility, and density),
- o size, mode, date, and frequency of contaminant release, and
- o mobilizing factors (e.g., precipitation, rusting of drums).

The geology and hydrogeology of the site have been described: (1) in the RI report by Jordan, (2) in prior reports by others; (3) the recent pump test performed by Keck for MDNR; and (4) the groundwater flow model prepared for the site by MDNR. Chemical specific factors are available from the literature for the contaminants identified through analysis of groundwater samples. The size, mode, and frequency of contaminant release may be known in some cases, but generally must be inferred from the chemical distributions found in the groundwater and soils. Mobilizing factors such as leaching by precipitation, or repeated spill events, may be apparent, or again may be deduced from the

distribution of the contaminants in the groundwater, e.g., a continuous plume or a series of pulses.

Contaminants in the groundwater in the Cadillac Industrial Park appear to be present as the result of single or multiple slug source inputs, such as at Northernnaire and Kysor, respectively, or as an areal source, such as from the shallow to the intermediate aquifers in an area near Northernnaire. The equation used to model the instantaneous slug source is:

$$C = \frac{M_o}{8 (\pi t)^{3/2} \sqrt{D_x D_y D_z}} \exp - \frac{X^2}{2D_x t} - \frac{Y^2}{2D_y t} - \frac{Z^2}{2D_z t} \quad (\text{Eq. - 1})$$

where:

C = concentration at a point in time and space [M/L³]

M_o = mass in the input slug [M]

t_o = time since the input [T]

X, Y, Z = coordinated directions of an advected cartesian coordinate system (the origin moves at the average groundwater velocity along a stream line) [L]

D_x, D_y, D_z = dispersion coefficients which are a measure of the three dimensional gaussian (normal) spread of the solute as the plume moves downgradient [L²/T]

Uniform one-dimensional flow in a homogeneous aquifer is assumed, and the dispersion coefficients are assumed constant. The dispersion coefficient is the product of the average groundwater velocity and the dispersivity, which is a measure of the capacity of the soil matrix to cause spreading of the plume. The solute is assumed to be conservative (does not react or degrade) in the aquifer and to move with the groundwater (does not sorb). The growth of the plume as it migrates from the point source can be used to estimate the dispersivities or diffusion coefficients if the size of the plume and the date of the spill are known or can be estimated. The concentration distribution of the contaminant can be used to estimate the mass of contaminant in the spill. With these parameters known, the concentrations as a function of location and time can be calculated. Usually, the maximum concentration within the plume is a concern. Since this occurs at the centroid (X=Y=Z=0), equation can be simplified to:

$$C_{\max} = \frac{M_o}{8 (\pi t)^{3/2} \sqrt{D_x D_y D_z}} \quad (\text{Eq. - 2})$$

If C_{max} is known at one point in time (t = t₁), then it can be determined at any other (t = t₂) by the relationship:

$$C_2 = C_1 \frac{t_1^{3/2}}{t_2^{3/2}} \quad (\text{Eq. - 3})$$

Equation 1 describes the concentration within the ellipsoid (football shaped) plume as a function of time. Along any major axis of the plume, the concentration distribution is gaussian or normal. Hence a mean and standard deviation can be calculated for the distribution. The dispersion coefficients are related to the standard deviation ($\sigma = \sqrt{2DT}$), and the mass of the slug within the plume at a distance from the centroid can be readily calculated. At three standard deviations from the centroid, the ellipsoid will contain 99.7% of the total original mass of the slug. The surface of this ellipsoid with major axes of three standard deviations is called the 3-sigma envelope. When times for arrival and final effect are referenced in the subsequent analyses, they will refer to the leading and trailing edge of this envelope. This "initial" and "final" impact may or may not be measurable depending on the initial maximum concentration within the plume and how far it has spread. Likewise, one can reference the 1-sigma and 2-sigma envelopes. The 1-sigma envelope contains 68.3% of the original mass in about 3.7% of the total (3 sigma envelope) volume. The 2-sigma envelope contains 95.4% of the original mass in 29.6% of the total volume. Hence relative average concentrations of the core of the plume to that in each successive annulus are about 300 to 17 to 1.

For a source which is introduced over an area and over a relatively long time, the slug point source equation is inappropriate. For this situation, the USEPA vertical and horizontal spread (VHS) model is more applicable. The equation (reference: 50 FR 48897) is:

$$C_y = C_o \operatorname{erf} Z / (2(\alpha_z Y)^{0.5}) \operatorname{erf} X / (4(\alpha_t Y)^{0.5}) \quad (\text{Eq. - 4})$$

where:

C_y = concentration at the selected receptor $[M/L^3]$
 C_o = initial or leachate concentration $[M/L^3]$
 erf = error function
 Z = penetration depth of leachate into the aquifer $= (\alpha_z Y)^{0.5} [L]$
 where Y = length of the disposal area in the direction of flow
 α_t, α_z = lateral and vertical dispersivities $[L]$
 Y^t = distance to the receptor $[L]$
 X = width of the disposal area (perpendicular to flow) $[L]$

The parameter Z is also called the mixing zone, and is the depth to which the leachate has migrated into the aquifer at the downgradient edge of the disposal area.

The VHS model allows for dilution through dispersion laterally and vertically in the aquifer. The model is steady state and predicts the maximum concentration on the center line of the plume as a function of distance only. Times for arrival must be calculated separately from the average groundwater velocity and the longitudinal dispersion coefficient. The equation assumes a constant source; however, some accommodation for a decreasing source can be applied to be more realistic over time provided the kinetics of the decreasing source term are known. Most of the other assumptions for the slug source analysis apply to the VHS model as well.

Both models tend to be very conservative in that they do not directly account for degradation (chemical or biochemical) of the contaminant or retardation, nor do they provide for dilution through the addition of infiltrating precipitation to the aquifer. The deeper the plume in the aquifer, the less appreciable this latter factor would be. Although the models are conservative, the conservative approach is probably appropriate where assessments of public health are being made.

Evaluation of the potential for impact of residual volatile organic contaminants in soil was done using EPA's organic leaching model (OLM) for impacts on groundwater while estimating losses of volatiles to the atmosphere by a model described in the EPA guidance "Superfund Exposure Assessment Manual" (OSWER directive 92855-1, Sept. 22, 1987). The OLM (51 FR 41088), an empirical expression, is:

$$C_l = 0.00221 C_w^{0.678} S^{0.373} \quad (\text{Eq. - 5})$$

where:

C_l = predicted concentration in leachate, mg/l
 C_w = concentration of contaminant in soil, ppm
 S = contaminant solubility in water, mg/l

The volatilization model is:

$$E_i = DC_S A / \sqrt{d^2 + 2Dt} \quad (\text{Eq. - 6})$$

where:

E_i = average emission rate of component i over time t (g/sec)
 D = phase transfer coefficient (cm²/sec)
 C_S = the liquid-phase concentration of component i in the soil, (g/cm³)
 A = contaminated surface area (cm²)
 d = depth of dry zone at sampling time (cm)
 t = time measured from sampling time (seconds)

To be conservative from the standpoint of impact on groundwater, the sampling time is taken as the time of the spill.

1.2 PLUME ANALYSES

In the following sections, the general approach is to estimate, where possible, the mass of contaminants in a plume, select an appropriate transport model, estimate groundwater velocity and plume dispersion coefficients, estimate a time of spill, identify an appropriate receptor location, and calculate time and concentrations of the plume as it passes by the receptor location. In two instances, for the Northernair hexavalent chromium plume and the Rexair volatile organic plume, estimates of the potential for impact on the municipal water supply are also made. The following major plumes are addressed: (1) a

VOA plume in the shallow aquifer; (2) the Rexair plume in the intermediate aquifer; and (3) the general VOA plume in the intermediate aquifer. The lesser plumes associated with Four Star and Northernair are considered as part of the intermediate aquifer plume. The Four Winns plume merges with the shallow Kysor plume and also possibly with some small contribution from the Ingraham property. While some groundwater data are available downgradient of Ingraham, and some remediation of soils on the Ingraham property is reported, a complete evaluation of the Ingraham property is not possible due to lack of soils data. This report recommends soils testing at the Ingraham property, but will not presently consider the Ingraham plume further since it appears that concentrations in groundwater due to the Ingraham plume are less in that vicinity than in the intermingling Kysor and Four Winns plumes.

1.2.1 The Shallow Aquifer Plume

In this section, the residual contaminants in soils at Kysor are considered first, then the general shallow Kysor plume, and then the Four Winns plume.

1.2.1.1 Evaluation of Soils at Kysor

In the years prior to about 1980, Kysor disposed of some waste solvents in a pit area at the rear of the plant. This resulted in migration of solvents through the unsaturated soils to the groundwater. It is believed that some of the denser-than-water solvents may have passed into the groundwater table as stringers of free product descending through the aquifer. The potential exists for these solvent stringers to have also seeped through the upper aquitard and into the intermediate aquifer. At this point it is uncertain whether free product would still have persisted, or if the subsequent contaminant migration would have occurred strictly as solubilized components in a groundwater plume. This scenario of intermittent disposal and/or solubilization by precipitation events helps to explain the spotty nature and relatively deep vertical spread of the Kysor plumes in the shallow and intermediate aquifers, but makes their analysis by transport modeling more difficult to perform. Partial remediation of the soils has reduced the potential for contamination of groundwater, but the impact is still significant.

Based on MDNR and the Cadillac RI soil and groundwater data for this assessment the following observations have been made:

1. Two areas of contaminated soils remain, one near boring KB-3 and KB-7, and the other near KB-2.
2. The first area is estimated to contain total volatiles at a concentration of about 400 mg/kg, yielding a solvent mass in soils of about 900# or 410 kg.
3. The second area contains an estimated 3 pounds of total solvents.
4. The distribution of contaminants is about 67% xylenes, 10% each of toluene, TCE and ethylbenzene, and about 2 to 3% of 1,1,1-TCA.

A model was run to estimate what the rate of loss of the solvent would be to both leaching and volatilization. The EPA organic leaching model (OLM) and a volatilization model from EPA's site assessment guidance manual were applied for both TCE and xylene to assess the respective potentials and as a check with observed concentrations in groundwater.

The modeling suggested that, due to the depth of the contamination, the loss rate of solvents would be only about 2% per year, still mostly by volatilization (more than 99% of the loss). The OLM predicted xylene concentrations of about 600 to 700 ppb in the leachate, while in SLW-25 the xylene concentration was about 1,000 ppb. While some dilution through dispersion would occur between the source and SLW-25, the numbers are reasonably close. The source term may conceivably be more than the originally estimated 900 pounds, or there may be releases of more concentrated contaminants to groundwater from the unsaturated matrix due to unusually strong precipitation events or the unstable nature of the flow in the unsaturated zone. This latter factor may also account for the very high TCE observed in one sampling event in SLW-25. The model was run for 5 years and indicated only slight reductions in the concentration and a total loss from the soil of about 10% of the originally estimated mass. This has implications for a potential long-term impact on groundwater due to the presence of the remaining solvents in soils.

1.2.1.2 Shallow Plume (Originating in the Vicinity of Kysor)

Several factors make analysis of the shallow plume difficult:

- 1) The nature of one major source, the contaminated soils at Kysor, has been radically altered with partial remediation of the source.
- 2) The influence of the probable presence (at least in the past) of free product migrating downward through the upper aquifer and possibly through the upper aquitard.
- 3) The probable presence of more pervious zones through the upper aquitard (e.g., near SLW-49 and K-6).
- 4) The presence of other sources with similar contaminants that comingle with the Kysor plume.
- 5) The dynamics of the source at Kysor which probably creates pulses of contaminants mobilized into the upper aquifer.

In this section, attention is focused on the transport in groundwater of solvents released from the soils at Kysor as they migrate through the trailer park, and through and beyond the industrial park. The Four Winns plume comingles with the shallow aquifer plume. This area of higher concentration will be addressed in Section 1.2.1.3.

As a basis for calculations, an instantaneous release of VOA from the soils behind Kysor is taken as equivalent to a gallon of mixed solvents. The slug release equation is used with the calculated dispersion coefficients of $D_x = 4,590 \text{ ft}^2/\text{yr}$; $D_y = 570 \text{ ft}^2/\text{yr}$; and $D_z = .66 \text{ ft}^2/\text{yr}$. The dispersivities were

assumed the same as those derived by analysis of the Northernnaire plume. The velocity in the shallow aquifer is taken as 300 ft/yr, based on hydraulic conductivities and gradients calculated from field data.

The maximum concentrations through the aquifer with distance are calculated from Equation 1 as:

Table 1.1
Estimated Concentrations with
Distance from Kysor

<u>Distance</u>	<u>Concentration</u>
300 ft.	10.7 mg/l
600	3.8
900	2.05
1200	1.34
1500	0.95
1800	0.72
3000	0.34

At a distance of 1500 to 1800 feet, the model predicts almost an order of magnitude lower concentrations than observed, suggesting some other mechanism may also be or have been active to release larger amounts of contaminant to the groundwater. Another possibility is that the source area needs to be considered as larger in applying the equation. Taking the distance to the edge of the aquitard (3000 ft) and using the dispersion coefficients as derived for the shallow aquifer, one may calculate the passage of the slug as:

Time to arrival to edge of aquitard = 7.39 yrs.
Time for centroid = 10
Time to last effect = 13.52

The smearing of the slug is apparent and the concentration at the centroid would be about 0.34 mg/l. The plume still needs to travel about another 1000 feet to get to 13th St. The times involved are governed by migration in the intermediate aquifer (i.e., velocity and dispersion become about 1/3 of what they were before):

Time to arrival = 5.95 years
Time to centroid = 10 years
Time to last effect = 16.8 years

The total travel then is: (from 1985 (date of release))
Time to arrival = 13.34 years
Time for centroid = 20 years
Time to last effect = 30.32 years

The final concentration of the centroid at 13th St. would then be about the same as if the plume had been controlled by the greater velocity, but only

travelled 1/3 the time in the intermediate aquifer. Continuing the analysis with the assumed source as a basis for calculation, the final concentration would be approximately (using distances in place of times in Equation 3):

$$C = 10.7 \frac{300}{3333}^{3/2} = .29 \text{ mg/l}$$

In summary, it appears that simple solubilization and mobilization of contaminants in soils by leaching of infiltration appears to be insufficient to explain some of the higher values of concentration observed in the shallow aquifer. However, there exist mechanisms whereby small sized releases with extremely high concentrations may occur, as was observed in SLW-25. Alternately, there may be a larger source than present data have defined. In either case, the release is subject to dilutions of about 30 from SLW-25 to 13th St. The resultant concentrations in the aquifer would be about 10 times as shown on Table 1.1, with a maximum of about 3.4 mg/l for an observed concentration of about 115 mg/l in SLW-25. The distribution across the aquifer and time of exposure at 13th St. are shown, respectively, on Figures R-1 and R-2 for the 115 mg/l observed in SLW-25. Maximum values predicted for this situation match reasonably well with some of the higher values in the aquifer as one moves north-northeast along the plume migration pathway.

Note that the distribution of contaminants in groundwater in SLW-25 (> 95% TCE) is not compatible with the distribution in the soil (about 67% xylene) and that the OLM predicted a concentration of xylene nearly twice that of TCE. This supports the presumption of some other mechanisms of release of pockets of contaminants within the soil matrix (primarily in the past).

For the contaminant distribution observed at SLW-25 the resultant concentrations for the individual VOA arriving at 13th St. would be:

1,1-DCE	11 ppb
1,1,1-TCA	1066 ppb
TCE	2600 ppb
PCE	12 ppb
Toluene	53 ppb
Ethylbenzene	22 ppb
Xylenes	33 ppb

1.2.1.3 Four Winns

Plume Characterization

The Four Winns plume has the characteristic shape of a slug input. Analysis under this assumption yields a slug source size of about 30 kg for acetone and of about 50 kg for chlorinated solvents (including degradation products). The dispersivities calculated for this plume (based on the present distribution, assumed source location, and calculated groundwater velocity) were: $\alpha_x = 24$ feet; $\alpha_y = 3.6$ feet; and $\alpha_z = 0.009$ feet. These values are higher than for the Northernair and Rexair plumes, probably due to the presence of interbedded silts and clays which produce a greater heterogeneity in the soil matrix in this area. The analysis for this plume has been done on the assumption of a

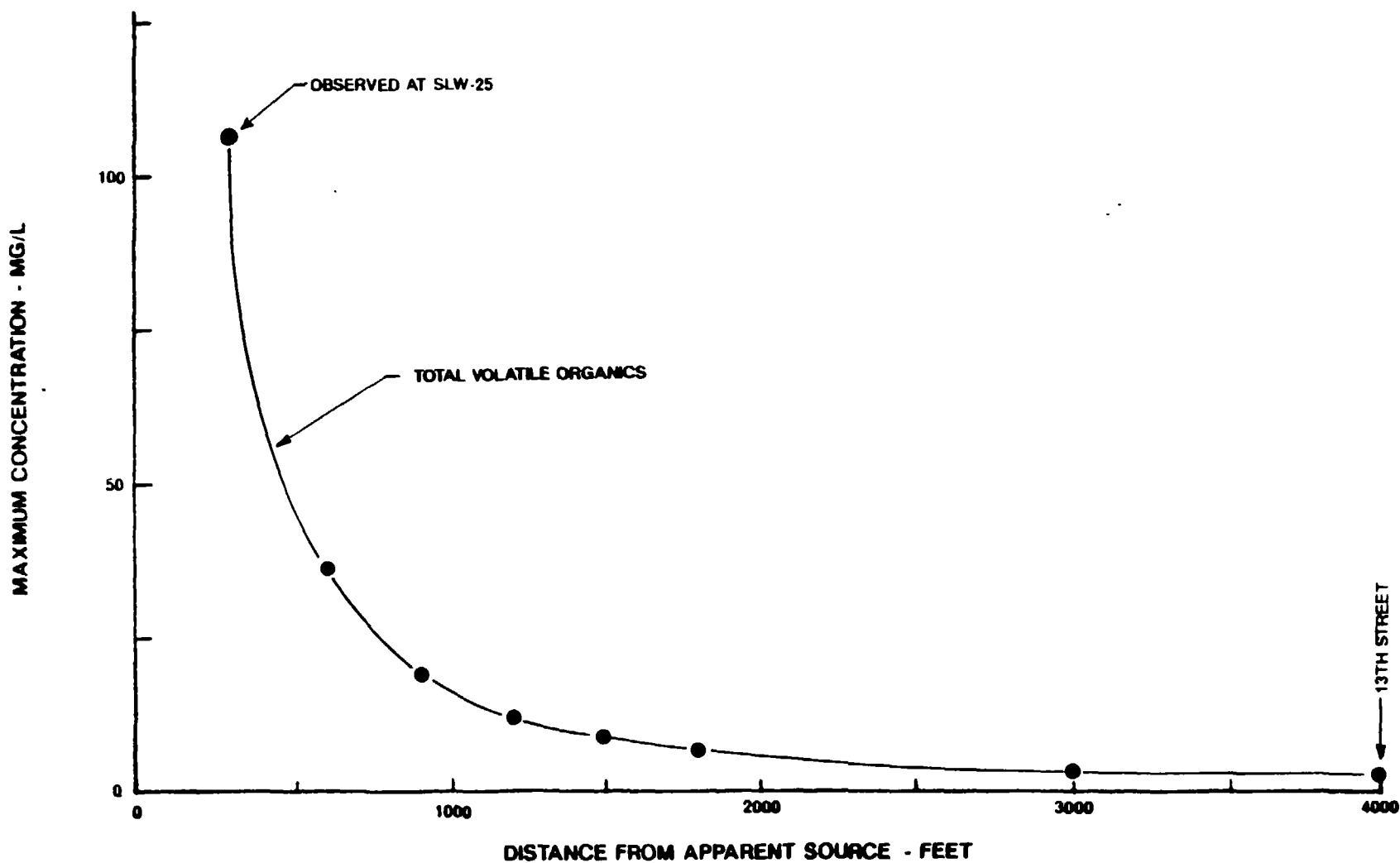


FIGURE R-1
CONCENTRATION/DISTANCE
KYSOR SHALLOW PLUME
CADILLAC AREA #1
CADILLAC, MICHIGAN

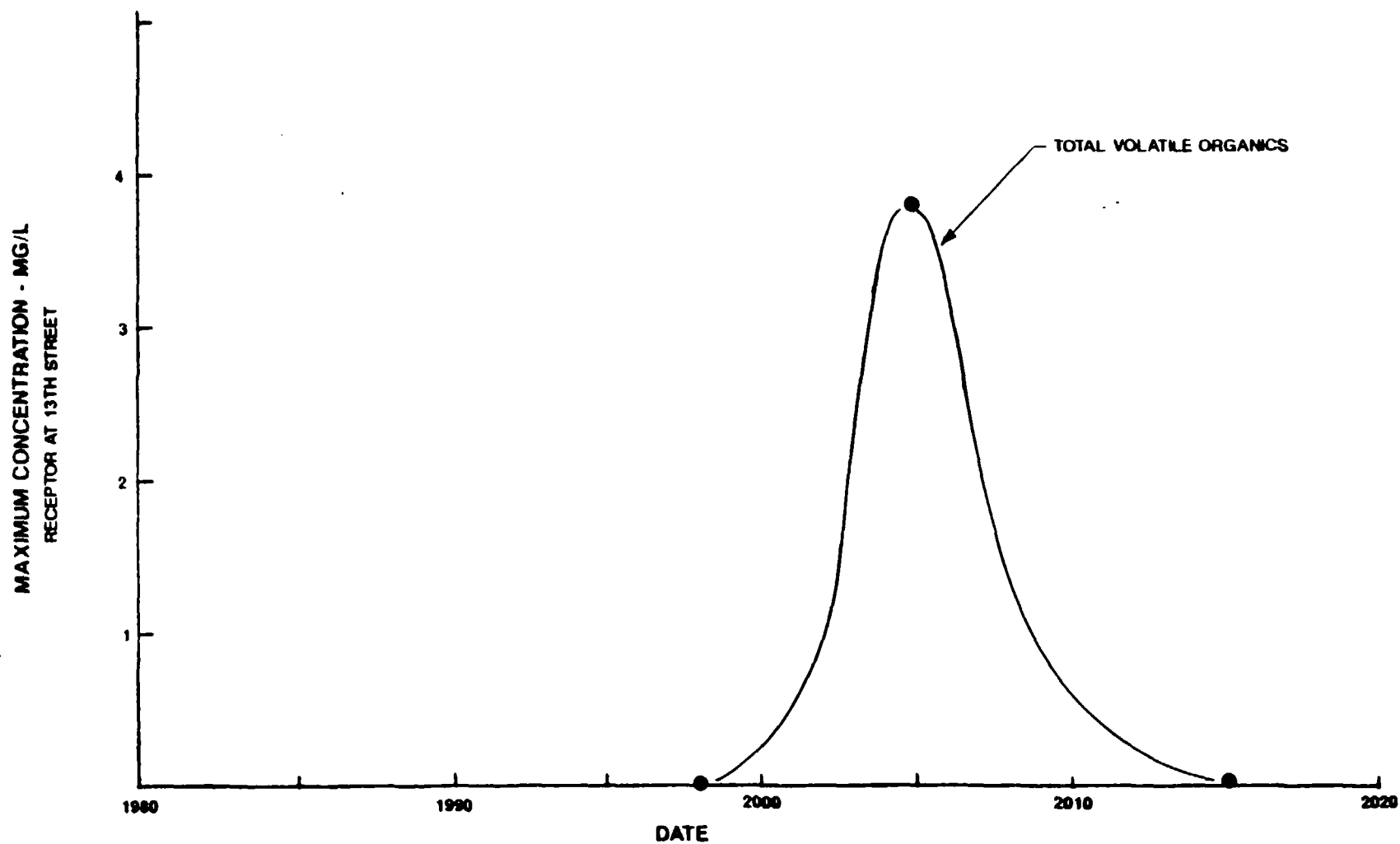


FIGURE R-2
CONCENTRATION/TIME
KYSOR SHALLOW PLUME
CADILLAC AREA RI
CADILLAC, MICHIGAN

Plume Fate

The plume, particularly the acetone constituent, is expected to move at approximately the same rate as the groundwater, calculated as 232 feet/year. When the plume reaches 13th St. and the intermediate aquifer, it will tend to move more slowly and more to the northwest. It may then descend through the intermediate aquifer and gradually seep through the lower aquitard, reaching the deep municipal well aquifer. The MDNR modeling effort suggests that this seepage will occur beyond the range of influence of the municipal well field and not be drawn to the wells. The plume would continue to migrate north-northwest, becoming more dilute, primarily through mechanical dispersion.

A hypothetical receptor well at 13th Street (worst case location) would begin to intercept the Four Winns plume in about 3.5 years, with the centroid arriving in 6.0 years, and the last vestige (the trailing 3 sigma envelope) of the plume passing in 10.4 years from early 1985, the estimated time of the spill (based on the estimated groundwater flow velocity and the presumption of a single short term spill). The concentration of 1,1,1-TCA seen by the receptor is plotted as Figure R-3. The maximum concentration of 1,1,1-TCA with distance from Four Winns is shown on Figure R-4. The concentrations of other contaminants within the plume would be directly proportional, and the maximum concentrations expected at the receptor location are shown in Table 1.2. These values represent point values. A real well would draw from an interval and thus the resultant concentration would likely be less than the maximum value.

Table 1.2

Projected Maximum Four Winns Plume Concentrations at 13th Street

<u>Compound</u>	<u>Maximum Concentration ppb</u>
Acetone	1792
1,1,1-TCA	2139
PCE	89
TCE	248
1,1-DCE	61
1,2-DCE	5
1,1-DCA	210
1,2-DCA	260
1,1,2-TCA	5

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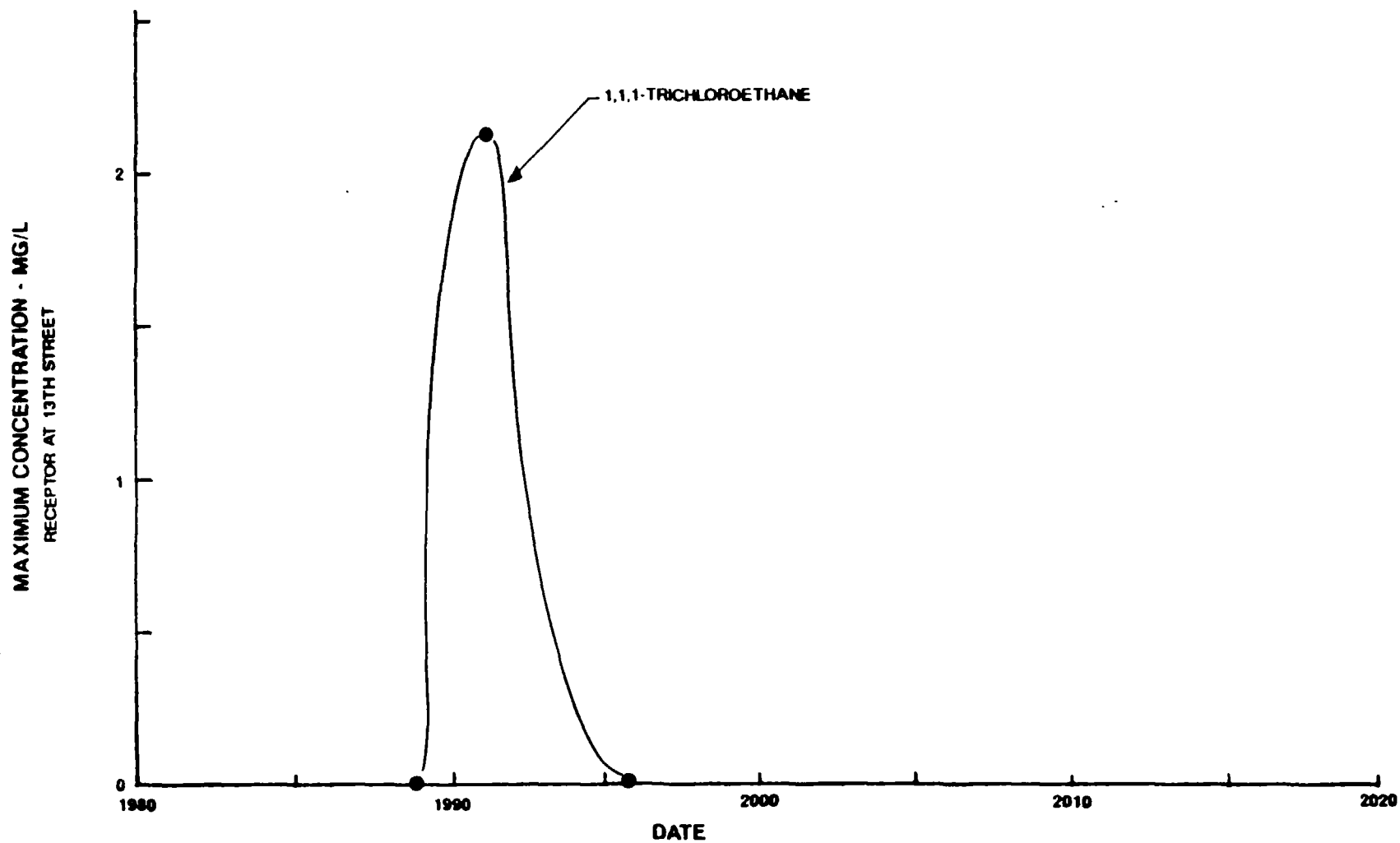


FIGURE R-3
CONCENTRATION/TIME
FOUR WINNS PLUME
CADILLAC AREA RI
CADILLAC, MICHIGAN

EC JORDANCO

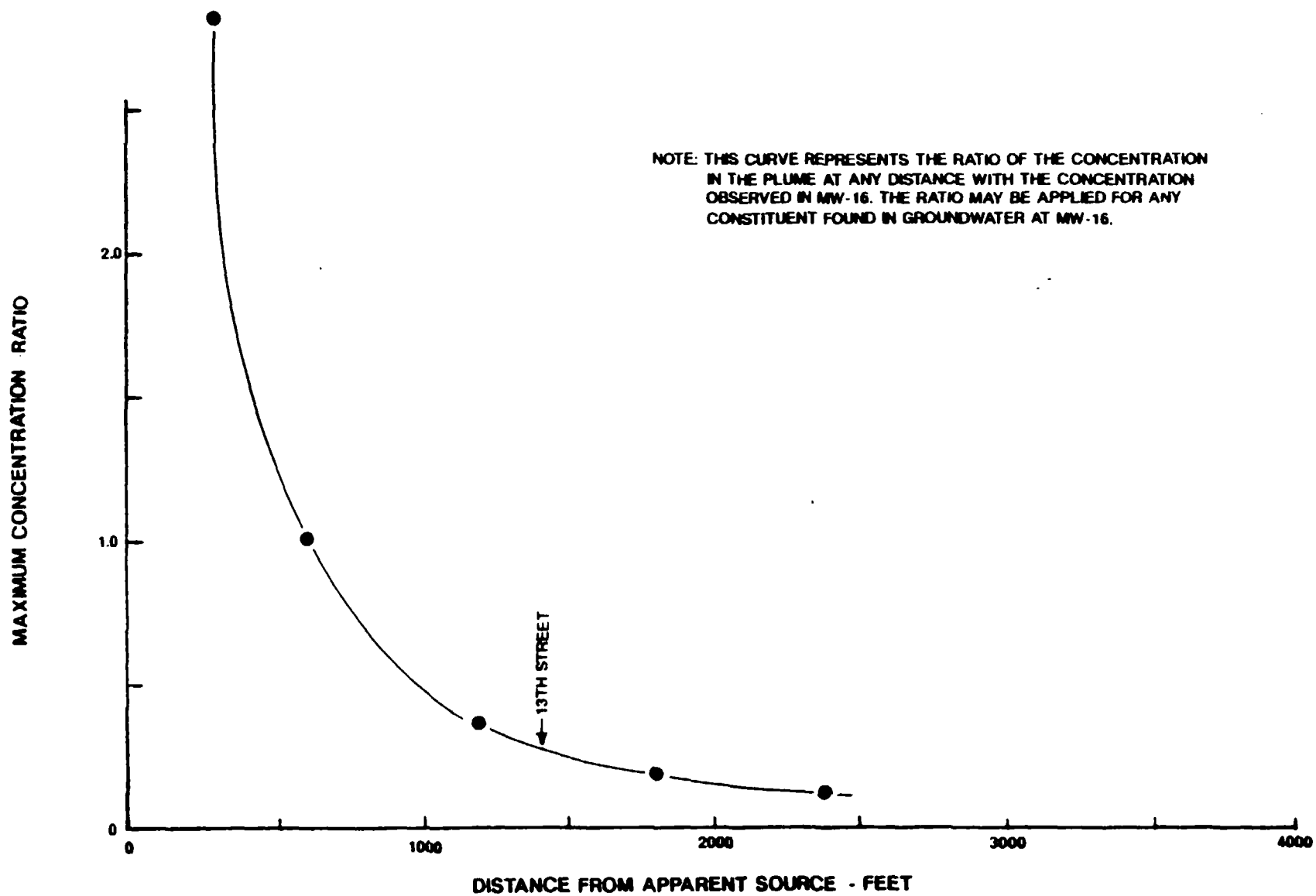


FIGURE R-4
CONCENTRATION/DISTANCE
FOUR WINNS PLUME
CADILLAC AREA RI
CADILLAC, MICHIGAN

Example Calculations Contaminants in MW-16

	<u>Most Recent Analysis (7/9/87)</u>	<u>Other Maximum Concentration</u>
1,1,1-TCA	4.15 mg/l	
PCE	0.317	0.331 mg/l
TCE	0.887	
1,1-DCE	0.218	
1,2-DCE	0.019	
1,1-DCA	0.750	
1,2-DCA	0.930	
1,1,2-TCA	0.016	0.021
ACETONE	6.4	

$$V = \frac{(34000 \text{ ft/yr}) (.00225)}{0.3 (1.1)} = 232 \text{ ft/yr}$$

$$\text{approximate time of spill} \sim \frac{600}{232} = 2.6 \text{ yrs}$$

or $\sim 2/85$

$$3\sigma_y = 200' = 3 \sqrt{2D_y t}$$

$$67' = \sqrt{(2)(2.6)D_y}$$

$$D_y = 863 \text{ ft}^2/\text{yr}$$

$$\alpha_y = 3.6 \text{ ft}$$

$$3\sigma_x = 570' = 3 \sqrt{2D_x t}$$

$$D_x = 5558 \text{ ft}^2/\text{yr}$$

$$\alpha_x = 24 \text{ ft}$$

$$3\sigma_z = 10' = 3 \sqrt{2D_z t}$$

$$D_z = 2.09 \text{ ft}^2/\text{yr}$$

$$\alpha_z = 0.009 \text{ ft}$$

$$C = C_{\max} \frac{600}{\text{dist}}^{3/2}$$

1.2.2 Rexair Plume

The Rexair plume is migrating northwesterly in the intermediate aquifer. The leading edge has nearly reached the location of the more easterly municipal wells. Vertical definition of the plume is not complete, but it appears that effects of pumping have drawn part of the plume downward in the aquifer. It is unknown if the plume has contacted or penetrated the lower aquitard, but the potential for it to do so exists.

2.88.143
0010.0.0

The mass of contaminants present in the Rexair plume is first estimated based on the contaminants distribution of TCE and an assumption of an ellipsoidal shape of the plume (equivalent to a slug release in a homogeneous aquifer). The masses of other contaminants in the plume are estimated as proportional to the TCE concentration (153 µg/l) as determined in SLW-50. Next, the groundwater velocity and dispersivities are estimated, followed by considerations of potential impacts to receptors at 13th St., or to the municipal well water supply.

VOLUME

$$\begin{aligned}
 @ 125 \text{ ppm} \quad V_1 &= (4\pi/3) (25)(5)(100) = 52,360 \text{ ft}^3 \\
 @ 33 \text{ ppm} \quad V_2 &= (4\pi/3) (350)(75)(40) - V_1 = 4,345,880 \text{ ft}^3 \\
 @ 3.3 \text{ ppm} \quad V_3 &= (4\pi/3) (450)(175)(60) - V_1 - V_2 = 54,978,000 \text{ ft}^3 \\
 @ .33 \text{ ppm} \quad V_4 &= (4\pi/3) (650)(250)(80) - V_3 - V_2 - V_1 = 103,986,944
 \end{aligned}$$

MASS

$$\begin{aligned}
 V_{1m} &= (52360)(.3)(62.4)(125)/10^6 = 122 \text{ \# [Volume x porosity x density x conc]} \\
 V_{2m} &= (4345880)(.3)(6.24)(33)/10^6 = 2678 \text{ \# [Concentrations are taken as the geometric mean of consecutive contours]} \\
 V_{3m} &= (54978000)(.3)(62.4)(3.3)/10^6 = 3390 \text{ \#} \\
 V_{4m} &= (103,986,944)(.3)(62.4)(.33)(10^6) = 638 \text{ \#} \\
 &6825 \text{ \# TCE}
 \end{aligned}$$

The average concentration of TCE in the plume is about 2.2 mg/l based on this estimate. Assuming simultaneous migration or uniform degradation, then the plume would contain approximately the following amounts of other identified chemicals (based on a ratio of maximum detected concentrations relative to TCE)

$$\begin{aligned}
 1,1,1\text{-TCA} &= 6825 (17/153000) = 0.8 \text{ \#} \\
 \text{Toluene} &= 6825 (42/153000) = 1.88 \\
 \text{Trans-1,2-DCE} &= 6825 (48/153000) = 2.1 \\
 \text{Chloroform} &= 6825 (6/153000) = 0.3 \\
 \text{Methylene Chloride} &= 6825 (780/153000) = 35
 \end{aligned}$$

The groundwater velocity near Rexair is calculated as:

$$v = Ki/n = \frac{(40,000 \text{ ft/yr})(1/1100)}{.3} \cong 120 \text{ ft/yr}$$

The centroid of the plume is about 1300 feet from the old portion of the Rexair facility. This represents a travel time of about 10 to 12 years assuming little or no retardation of the solute.

Estimated dispersivity values are (using $t = 11$ years):

$$36_x = 600' = 3 \sqrt{2 D_x t} = 3 \sqrt{(2)(D_x)(11)}$$

$$D_x = 1818 \text{ ft}^2/\text{yr} \quad \alpha_x = 15.2 \text{ ft}$$

$$36_y = 250' = 3 \sqrt{2 D_y t} = 3 \sqrt{(2)(D_y)(11)}$$

$$D_y = 313 \text{ ft}^2/\text{yr} \quad \alpha_y = 2.6 \text{ ft}$$

$$36_z = 65' = 3 \sqrt{2 D_z t} = 3 \sqrt{(2)(D_z)(11)}$$

$$D_z = 22 \text{ ft}^2/\text{yr} \quad \alpha_z = 0.18 \text{ ft}$$

While α_x and α_y are close to those values calculated for the Northernnaire plume, α_z is much larger, probably indicating the effects of the pumping of the municipal wells in creating downward gradients and also of the presence of gravels in this area. However, to consider maximum downgradient concentrations, the smaller value of α_z (0.0022 feet) will be used.

Two effects are considered: (1) Continued migration to 13th Street; and (2) seepage into the municipal well aquifer.

- (1) Continued migration - part of the plume could possibly escape being drawn down through the aquitard (which gets thicker directly over the well field) and continues on towards 13th St.

A slug source of about 6825 # (3102 kg) is assumed (suggested by plume shape), and the three dimensional slug source equation is applied to estimate transport. Note that source concentrations and volume (about 550 gallons of TCE) would indicate the likelihood for free product and/or a longer period of time for source introduction. Nevertheless, the present shape suggests a slug (or past short-term release), and future transport can be adequately modeled with this equation. Some slight degradation (suggested by the presence of trans-1,2-DCE) is indicated, but further degradation within the aquifer is not considered significant, and further degradation will not be considered in estimates of concentrations at receptors.

The release of 6825 # of contaminant as a slug, results in a calculated point maximum concentration at the present (1987) centroid location of about 375 mg/l. Given the overall reasonable match with the observed plume and the uncertainty of the mode of initial release, the equation seems appropriate to use for estimation of dilution factors in the aquifer as the plume moves downgradient, i.e., the dispersivities appear consistent with those determined for other plumes. Hence, dilution factors will be generated using the slug input equation and the dispersivities $\alpha_x = 15.3 \text{ ft}$, $\alpha_y = 1.9 \text{ ft}$ and $\alpha_z = 0.0022 \text{ ft}$, the latter value being conservative as the pumping seems to have a spreading effect in the vertical direction.

Then for a receptor at 13th Street (distance = 2700 feet from Rexair), one calculates a time to arrival (of the 3 sigma envelope), time to centroid, and time to last effect (trailing edge of the 3 sigma envelope) of about:

$$\begin{aligned} t(\text{arr}) &= 16.27 \text{ yrs} \\ t(\text{cent}) &= 22.3 \quad (\text{from 1976}) \\ t(\text{final}) &= 30.56 \end{aligned}$$

Maximum concentrations are calculated based on the slug equation, which reduces, for a reference point of 153 mg/l, to

$$C_{\text{max}} = 153 \frac{11 \text{ years}}{t_2}^{3/2} \text{ mg/l}$$

Therefore at 13th Street, the maximum expected concentration occurs when the centroid passes in 1998 and is 53 mg/l. Again, this represents a point maximum concentration. A well actually penetrating the plume would be subject to lower concentrations due to withdrawal of water from less contaminated zones. Figures R-5 and R-6 show the maximum concentrations as a function of distance downgradient of Rexair, and the maximum concentrations a receptor well at 13th Street might intercept as a function of time, respectively.

Maximum contaminant levels at 13th Street of other plume constituents are estimated as being subject to the same dilution factors as the TCE. Hence the maximums at 13th Street become:

1,1,1-TCA	17(.346) =	6 ppb
Toluene	41(.346) =	15
Trans-1,2-DCE	48(.346) =	17
Chloroform	6(.346) =	2
Methylene Chloride	780(.346) =	270

- (2) Keck, in their pump test evaluation, estimated flow from the plume area through the aquitard to be about 100 gpm (Keck, 1987). Assuming that the plume moves vertically through the aquitard at this rate and that the contaminant distribution is uniform in the vertical direction, then the resultant concentration of TCE in the municipal water supply (2200 gpm) would eventually be about

$$c = \frac{(100 \text{ gpm})(2.2 \text{ mg/l})}{2200 \text{ gpm}} = 0.100 \text{ mg/l} = 100 \text{ ppb}$$

This analysis assumes (conservatively) no retardation or further dilution of the plume as it completes its descent through the intermediate aquifer, passes through the aquitard, and migrates to the municipal supply well(s). However, variations in the TCE concentration across the plume, and the pumping sequences might cause concentrations to be higher at times.

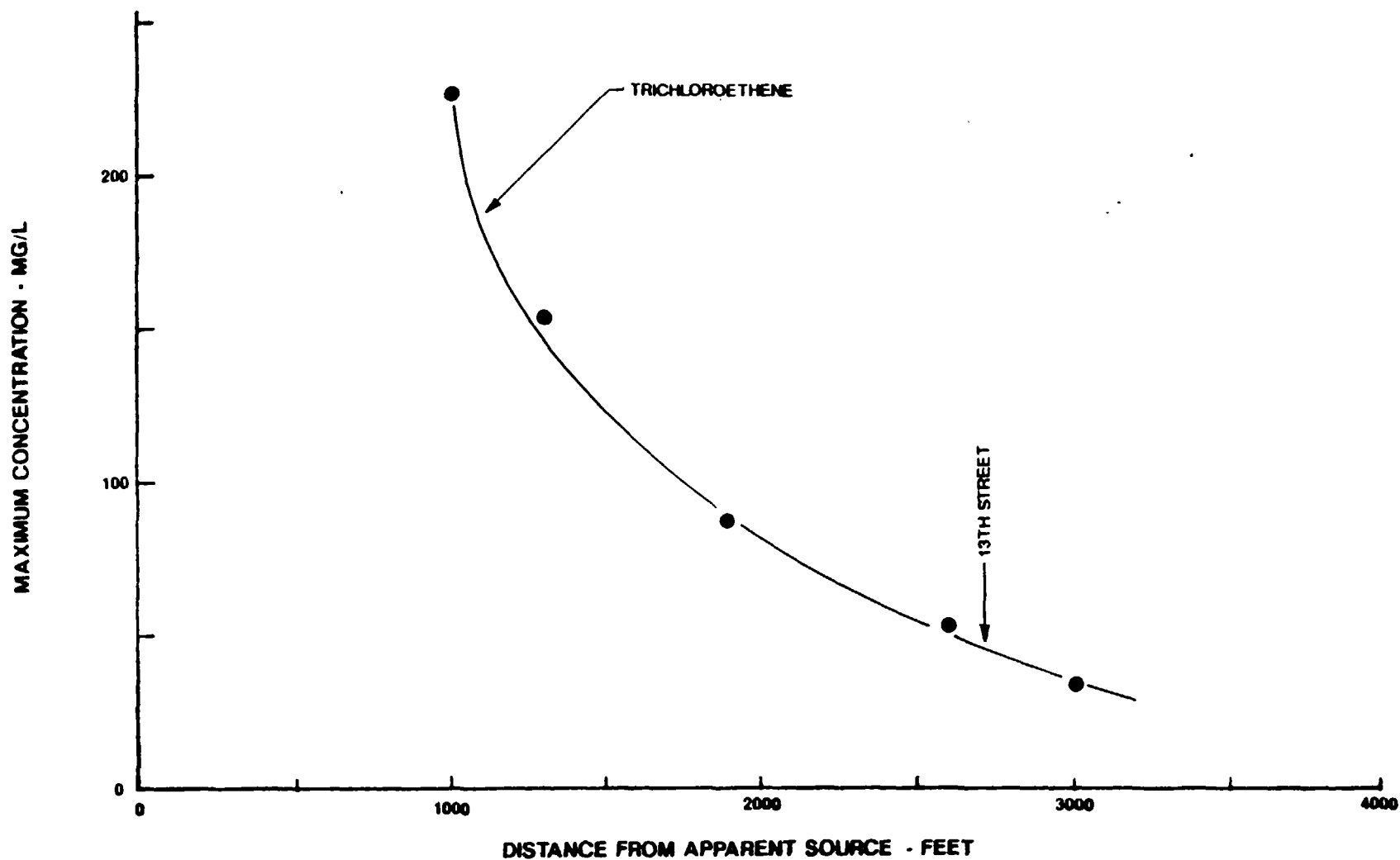


FIGURE R-5
CONCENTRATION/DISTANCE
REXAIR PLUME
CADILLAC AREA RI
CADILLAC, MICHIGAN

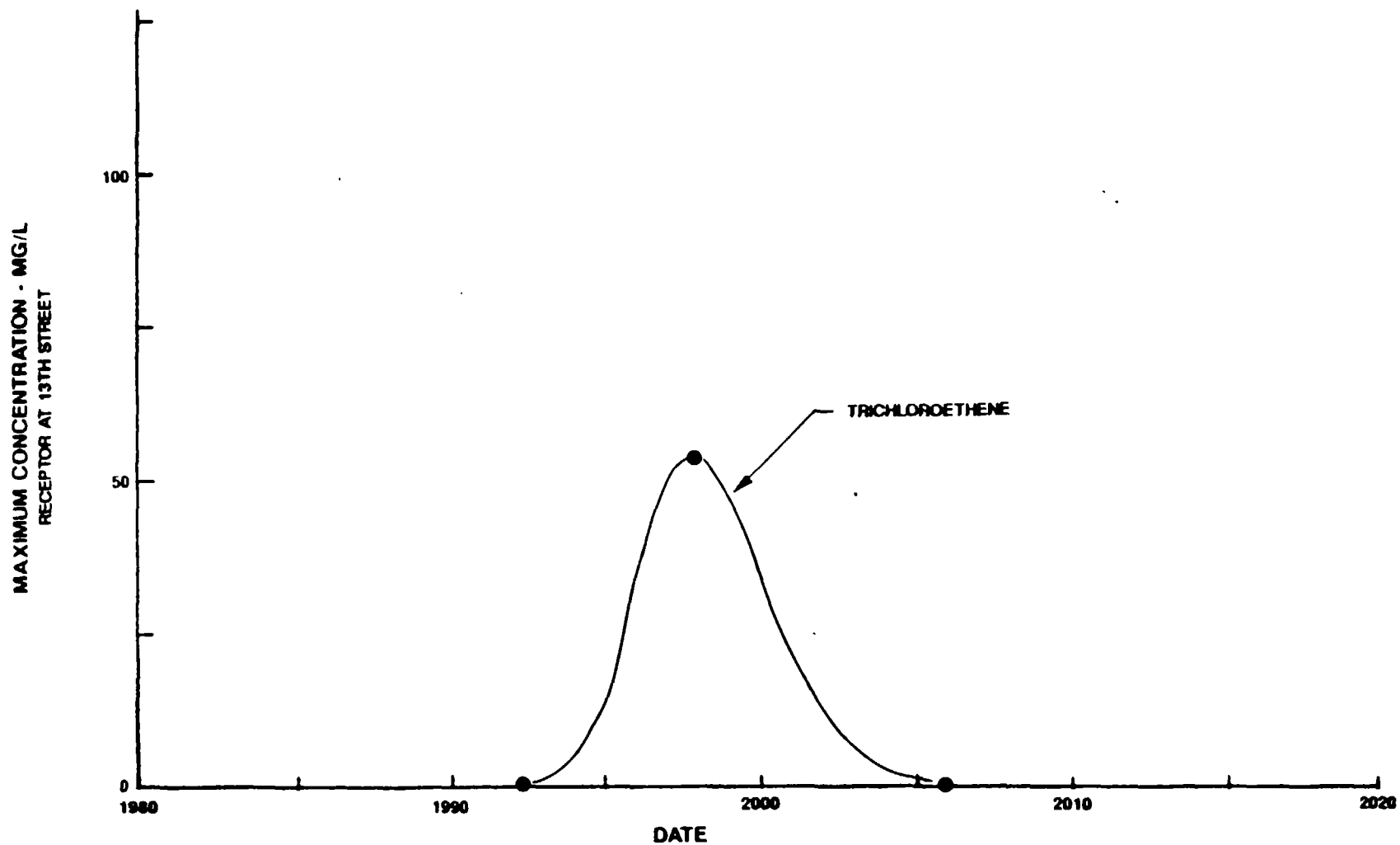


FIGURE R-6
CONCENTRATION/TIME
REXAIRE PLUME
CADILLAC AREA RI
CADILLAC, MICHIGAN

1.2.3 Intermediate Aquifer Plumes

In this section, the contaminants present in the intermediate aquifer west of 6th Ave. are considered. First, the generalized plume, due largely to releases from Kysor, is considered. Then concentrated areas represented by Four Star and Northernair releases are also presented.

1.2.3.1 Intermediate Aquifer Plume (Kysor and Others)

The interpretation of the widespread VOA plume in the intermediate aquifer west of Sixth Avenue is subject to some of the same difficulties listed for interpretation of the shallow plume in this area. The contaminant distribution in the intermediate aquifer arises predominantly from seepage through the upper aquitard along an area reaching from Kysor to the former Four Star facility. The contribution of the Four Star facility, while it cannot be completely separated, is focused upon more closely in Section 1.2.3.2. The seeping VOA contaminants emerge into the intermediate aquifer and move (as dissolved constituents) generally north-westerly at about 100 feet/year. It is difficult to estimate the dispersion, for the source concentrations and date of introduction are not precisely known. However, in this analysis, we will try to back-calculate a probable date of introduction into the intermediate aquifer. This date would have varied along the migration pathway in the shallow aquifer due to the rate of movement and variability in the effectiveness of the upper aquitard.

Instead of a receptor at 13th St., one will be placed along Leeson Avenue near the intersection of Frisbee Street as this appears to be the point where effects may be noted earliest. Also, since the zone of introduction is no longer considered a point slug source injection, it will be more appropriate to calculate dilutions using the VHS model which employs an areal zone of introduction, but which is a steady state model (constant source) predicting maximum concentrations at a distance. For the analysis, the initial concentration used will be 12.0 mg/l, the highest value observed in the intermediate aquifer in this area (in SLW-10D). The time to reach Leeson Avenue is calculated based on the fastest velocity on the 3-sigma envelope or leading edge. No time to final effect is predicted since the model presumes a constant source. However, as the source degrades at Kysor, this should eventually result in declining concentrations at the hypothetical receptor location.

Some seepage of the VOA plume through the lower aquitard into the municipal water aquifer is possible. Part of this VOA plume comingles with the chromium plume (Section -2.3.3 discusses this area in greater detail). Resultant concentrations of VOA in the municipal well might be expected to be in the same range as that calculated for the chromium (1 ppb). Due to the uncertainty of actual occurrence, and the probable low resultant concentrations, the potential for impact on the water supply is not assessed further here.

The VHS model predicts a maximum concentration of 3.4 mg/l at Leeson Ave. The vertical thickness of the plume is expected to be relatively thin (< 10 feet) so that this represents a limited risk to supply wells (which may be above the plume). If a well were to intersect the plume, however, chances are that the resultant contaminant concentration would be significant. The dilution factor

to this point is only about 3.5. This can be applied to the distribution of contaminants in SLW-10D (a total of 12 mg/l VOA) assuming a source area of 2500' by 400':

	<u>Max Observed</u>	<u>Estimated At Receptor</u>
TCE	7.5 mg/l	2.14 mg/l
Trans-1,2-DCE	4.2	1.20
Methylene Chloride	.24	0.07

Figure R-7 shows the profile of total VOA arriving at the hypothetical receptor. The time to first effect is estimated to be about 6 years from the present (1986) position of the plume.

Approximate time of introduction to time of arrival at Leeson.

The distance between the 1.0 mg/l contour of the plume and Leeson St. in the area of concern is about 1,200 feet. The average flow rate is about 100 feet per year and the longitudinal dispersivity is about 15 feet. Therefore time to first arrival is calculated as 7.5 years and a date for emergence into the intermediate aquifer is estimated as 1978 about to 1979 in this location. The distance to Leeson (from the downgradient edge of the major seepage area) is about 1,400 feet. The time to first arrival and time to maximum concentration are 9.05 and 14 years respectively.

1.2.3.2 FOUR STAR PLUME

Plume characterization/transport and fate

Four Star employed vapor degreasers in its operations in 1978 and 1979, at which time the facility closed (Gosling - Czubak, 1983). Inspections by the MDNR in 1983 indicated loss of solvents, primarily TCE, had occurred to the ground, and MDNR requested a soils and hydrogeologic investigation be conducted. No significant concentrations of solvents, nor of metals, were detected in soils, but water quality sampling conducted by Gosling-Czubak for Four Star, and later in a more extensive study by MDNR, indicated the presence of a plume of solvents in the groundwater traveling almost due north at the facility. The plume subsequently turned toward the northwest as the flow in the intermediate aquifer became dominant (the upper aquitard terminates at the facility). The average groundwater flow rate is estimated to be about 100 ft/yr and relatively constant from the source along the plume pathway. The plume appears to be descending in the aquifer and is comingling with apparent sources from Kysor. The Four Star plume may reach and pass through the lower aquitard, but it is not clear if the plume would emerge in the municipal supply aquifer within the radius of influence of the municipal wells.

Analysis

The shape of the plume suggests the applicability of the slug source equation to analyze and predict contaminant transport. Based on a probable date for the release of the solvents (early 1979), the contamination distribution in 1983,

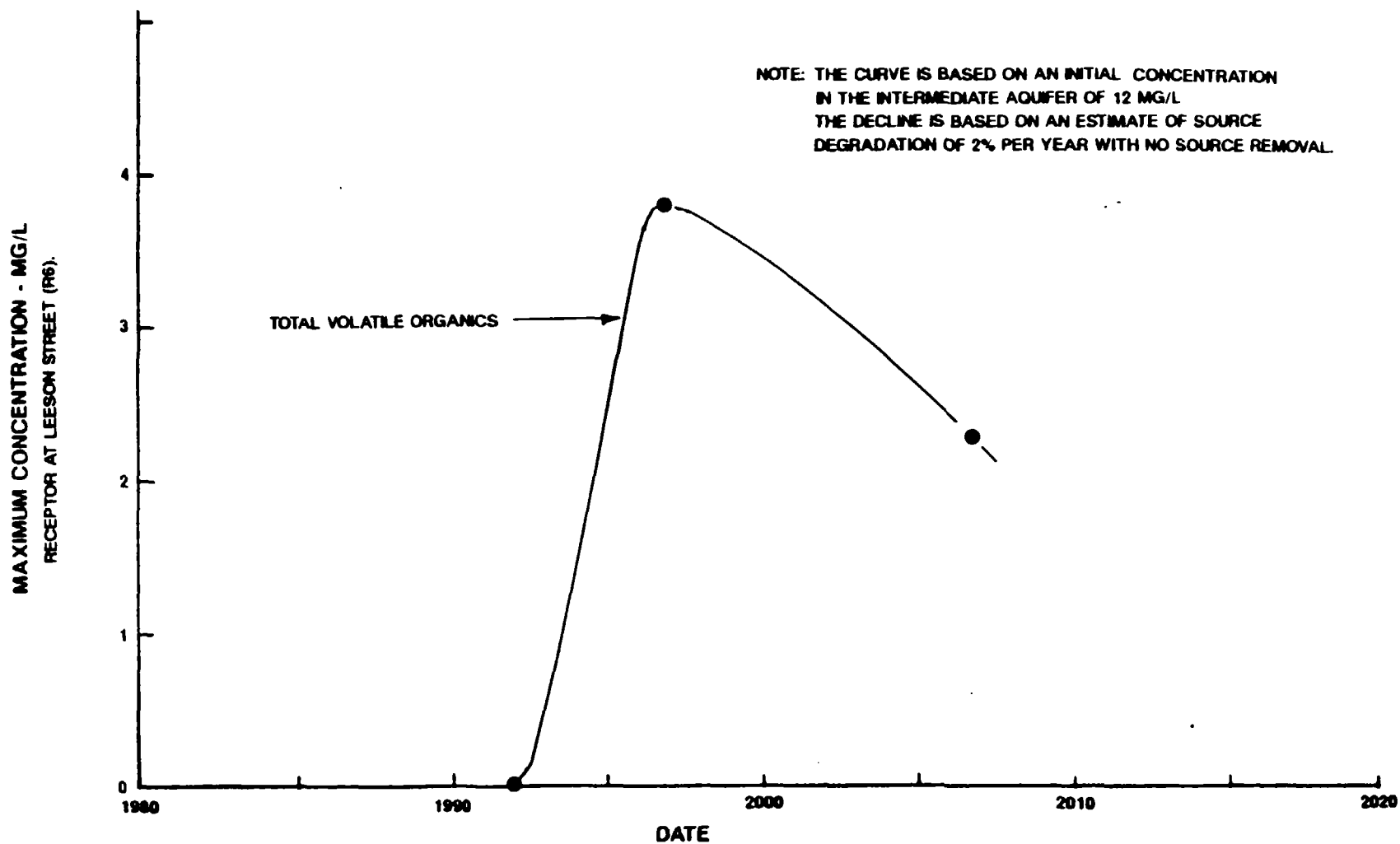


FIGURE R-7
CONCENTRATION/TIME
INTERMEDIATE AQUIFER PLUME
CADILLAC AREA R6
CADILLAC, MICHIGAN

and a reasonable estimate of 100 ft/yr for the groundwater flow, dispersion coefficients and dispersivities could be calculated. These values were: $\alpha_x = 12.5$ ft; $\alpha_y = 1.36$ ft; and $\alpha_z = 0.0035$ ft. These dispersivities are reasonable and in close agreement with those calculated for other plumes in the aquifers.

From the groundwater flow rate, dispersivities and times, dilution factors in the aquifer due to dispersion alone could be calculated. Maximum concentrations at a hypothetical receptor well at 13th St. were calculated as 6.73 mg/l TCE; 1.74 mg/l PCE; 0.132 mg/l 1,1,1-TCA; and 0.056 mg/l 1,1-DCA. These are point values, and actual concentrations intercepted by a water supply well would be lower since the well screen would present a wide interval for receiving lesser concentrations, especially if the well location were above the plume. The deepest water supply wells along 13th St. go 90 to 100 feet deep, with an average closer to 50 to 60 feet. Concentrations in the plume as a function of distance are shown on Figure R-8, and concentrations through time for the hypothetical receptor at 13th St. are shown on Figure R-9.

Data and Calculations

Spill about 1979

First plume determination by Gosling-Czubak 12/83

MDNR determination @ 5/19/86

Assume groundwater velocity of 100 ft/yr as for other intermediate aquifer locations

<u>at t = 4 yrs</u>	<u>Dispersion Coeff.</u>	<u>Dispersivities</u>
$300' = 3 \sqrt{2 Dxt}$	1250	12.5 ft.
$100' = 3 \sqrt{2 Dyt}$	136.12 ft ² /yr	1.36
$5' = 3 \sqrt{2 Dzt}$	0.35	0.0035

Estimate TCE in Plume (assume thickness of 5 feet)

$$V_1 = \pi a_1 b_1 h = \pi(60)(15)(5) = 14,137 \text{ ft}^3$$

$$V_2 = \pi a_2 b_2 h - V_1 = \pi(165)(50)(5) - V_1 = 12,9591 - 14,137 = 115,454$$

$$V_3 = \pi a_3 b_3 h - V_2 = \pi(220)(70)(5) - V_2 - V_1 = 241,903 - 129,591 = 505,769$$

$$V_4 = \pi a_4 b_4 h - V_3 = \pi(320)(100)(5) - V_3 - V_2 - V_1 = 502,656 - 241,903 = 1,034,011$$

$$M_1 = (14,137)(.3)(62.4)(42)/10^6 = 11.1 \# \quad [\text{Volume} \times \text{porosity} \times \text{density} \times \text{concentration}]$$

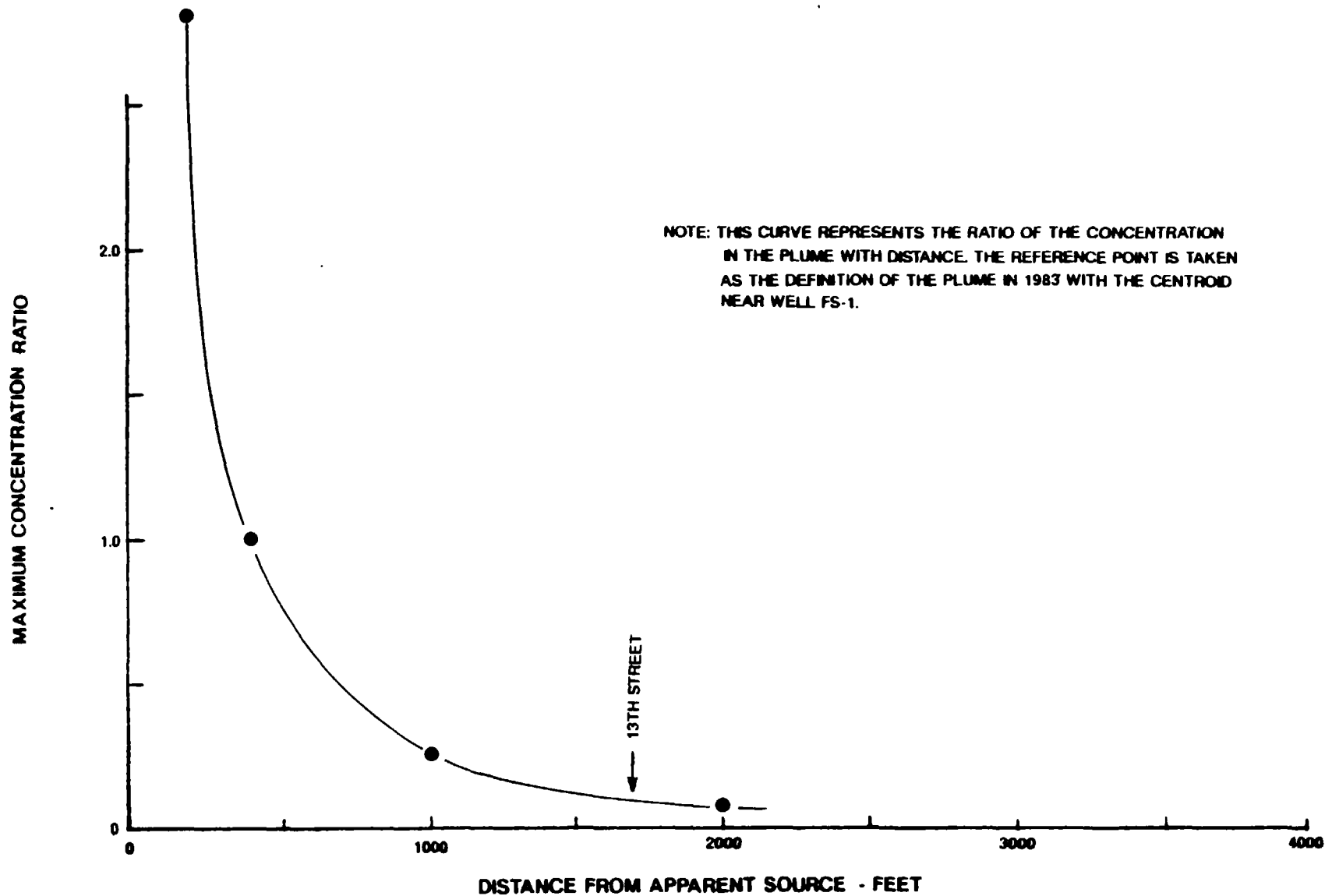


FIGURE R-8
CONCENTRATION/DISTANCE
FOUR STAR PLUME
CADILLAC AREA PI
CADILLAC, MICHIGAN

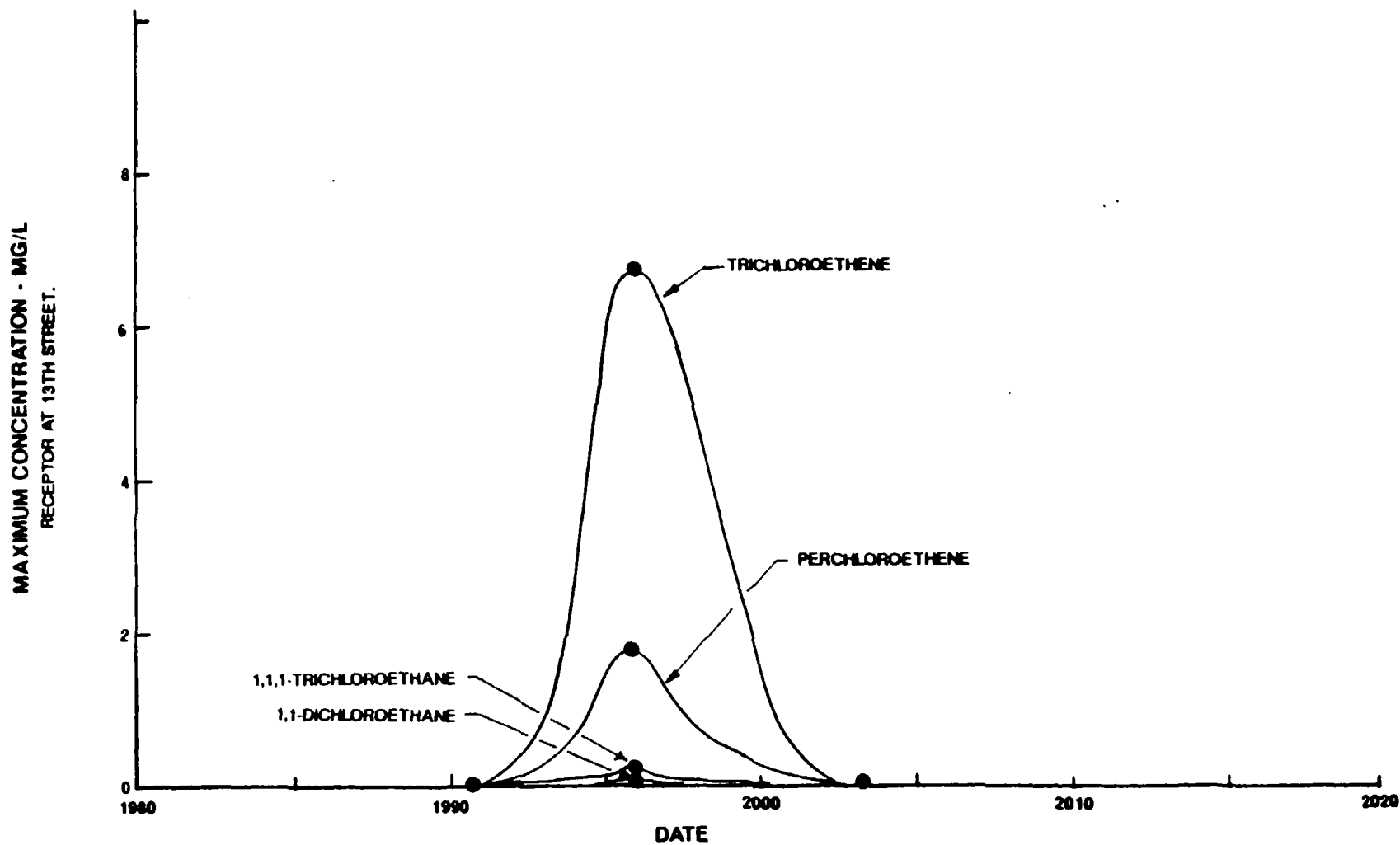


FIGURE R-9
CONCENTRATION/TIME
FOUR STAR PLUME
CADILLAC AREA RI
CADILLAC, MICHIGAN

$$\begin{aligned}
 M_2 &= (115,454)(.3)(62.4)(24.5)/10^6 = 52.9 \# \\
 M_3 &= (112,312)(.3)(62.4)(14.1)/10^6 = 29.6 \# \\
 M_4 &= (260,753)(.3)(62.4)(3.2)/10^6 = \frac{15.6 \#}{109.2 \#} \\
 &\quad \text{TCE} \\
 \text{Approx mass of PCE}(\frac{1}{4}\text{TCE}) & 27.3 \# \\
 \text{Approx mass of 1,1,1-TCA}(1/10\text{PCE}) & 2.7 \# \\
 \text{Approx mass of 1,1-DCA}(1/3\text{TCA}) & 0.9 \# \\
 \text{Total VOA} & \underline{140.1 \#}
 \end{aligned}$$

[Concentrations were taken as geometric means between each successive concentration contour from a plot of concentration distribution by Gosling - Czubac.]

Distance (from approximate point of spill) to 13th St = 1700 ft

$$\begin{aligned}
 \text{Time to arrival} &= 11.8 \text{ years (dispersive effects)} \\
 \text{Time to centroid} &= 17 \\
 \text{Time to final effect} &= 24.4
 \end{aligned}$$

$$C_{\max} = C_4 \frac{4}{17}^{3/2} = 0.114 C_4$$

<u>Comp.</u>	<u>C_{max}</u> (t=4)	<u>C_{max}</u> (t=17)
TCE	59 mg/l	6.73 mg/l
PCE	15.3	1.74
1,1,1-TCA	1.16	0.132
1,1-DCA	0.49	0.056

These, again, are point concentrations. The concentration in a supply well would likely be lower depending on the position of the well screen relative to the plume and the screen length.

1.2.3.3 Northernnaire Chromium Plume

Plume Characterization

The hexavalent chromium plume analyzed here was defined within the intermediate aquifer in the central portion of the industrial park during the supplemental remedial investigation conducted for the Northernnaire CERCLA site. While a shallow plume to the northeast of the facility was also detected, it is believed that this detection, in one well, is a minor plume of low concentration which is being quickly removed by groundwater flow from the shallow aquifer. The main plume, believed to have originated from Northernnaire about 1977, proceeded northeasterly with the shallow groundwater flow in that vicinity, but made its way into the intermediate aquifer by passing over the edge of the upper aquitard and/or seeping through the aquitard which becomes progressively thinner toward the northeast. The groundwater flow velocity is estimated as about 300 feet per year in the shallow aquifer and about 100 feet per year in the intermediate aquifer, which flows toward the northwest in the vicinity of the plume.

Based on the interpretation of the distribution of the hexavalent chromium in the plume from the supplemental RI data, the plume contains about 30 pounds or 14 kilograms of hexavalent chromium (as chromium). Almost all of the total chromium present in the plume is in the hexavalent state indicating little reduction of the chromium within the plume. Given that the aquifer is mostly sands and gravels, little reduction would be expected. The present (1986 data) maximum concentration within the plume was between 1 and 2 parts per million. Since the plume is very narrow vertically, the screen length of 5 feet actually intercepts a wide portion of the plume and thus the sample at the well represents an average portion of the plume and not a true maximum. The centroid of the plume has been interpreted as being slightly to the north-northwest of well K-15.

Fate and Transport

Little reduction (to trivalent chromium) and loss of the hexavalent chromium is anticipated within the sandy aquifer as it continues to migrate to the north-west and beyond the industrial park. However, it is possible that the plume may continue to descend in the aquifer and eventually come in contact with the silt/clay aquitard between the intermediate and municipal water supply aquifers. The aquitard appears, based on pump test data at the wellfield, to be permeable enough, and a downward vertical gradient exists, to permit seepage of the plume to the lower aquifer. Modeling done by MDNR of the aquifer systems indicates that in its present position, the plume, if it were to seep into the lower aquifer, could be drawn back to the wellfield. An analysis was done to examine the concentration that might be expected as the plume migrates along its present path, and also the potential impact on the municipal wellfield should it seep through the aquitard within the capture zone of the wellfield.

The analysis was performed primarily using the slug source equation. By assuming a slug input source at Northernnaire in 1977 (residential wells some 600 feet away were noted as being contaminated in 1979 and the groundwater velocity is about 300 feet per year), a reasonable fit to the interpreted plume was obtained when accounting for the change in velocity when the plume arrives at the intermediate aquifer. The slug input was assumed to be 14 kilograms and a maximum of 2 mg/l at the present location was calculated from the equation. The dispersivities were calculated as 15.3 feet in the longitudinal direction, 1.9 feet in the lateral horizontal direction, and 0.0022 feet in the vertical direction. Note that the dispersivities are quite small everywhere due to the relative homogeneity of the aquifer, and that the spread in the vertical direction is particularly small.

Based on the velocity distributions and the calculated maximum centroid concentrations through time (or distance), a time of arrival, maximum concentration and time to final impact of the plume at a receptor at 13th Street could be estimated. The maximum concentration in the plume at this location was estimated as about 0.60 mg/l. However, the concentration that a well might yield at this point along the center line of the plume might be quite different. The plume is narrow and deep. If a well were to intercept the plume at all, it is probable that water would be drawn from only part of the plume, yielding a much lower concentration in the well. The expected maximum concentration within the

plume at a 13th Street receptor directly on the center line of the plume is shown in Figure R-10.

Next, the potential impact of the chromium plume on the municipal water supply was estimated. A simple mass balance approach was used with the rate of chromium potentially received by a well calculated by assuming a uniform seepage of the plume through the lower aquitard. The aquitard thickness was assumed to be about 10 feet based on information obtained from mud rotary borings through the lower aquitard at various points, particularly MB-2. The hydraulic conductivity was estimated from the data generated through the pump test and the thickness assumption. The analysis suggested that the plume could pass through the aquifer in about three years assuming no retardation or degradation of the hexavalent chromium. It is recognized that some further reduction could take place due to the organic content of the aquitard, but a worst case scenario is being described here; hence no such transformation is assumed. A pumping rate at the wellfield of 2200 gpm was assumed as well as a uniform delivery to the pumping well. No consideration of the effects of alternating the pumps was taken. The analysis indicated a concentration of about 1 microgram per liter (1 ppb) of hexavalent chromium in the water supply well as a result. This is far below the 50 microgram per liter drinking water standard for chromium.

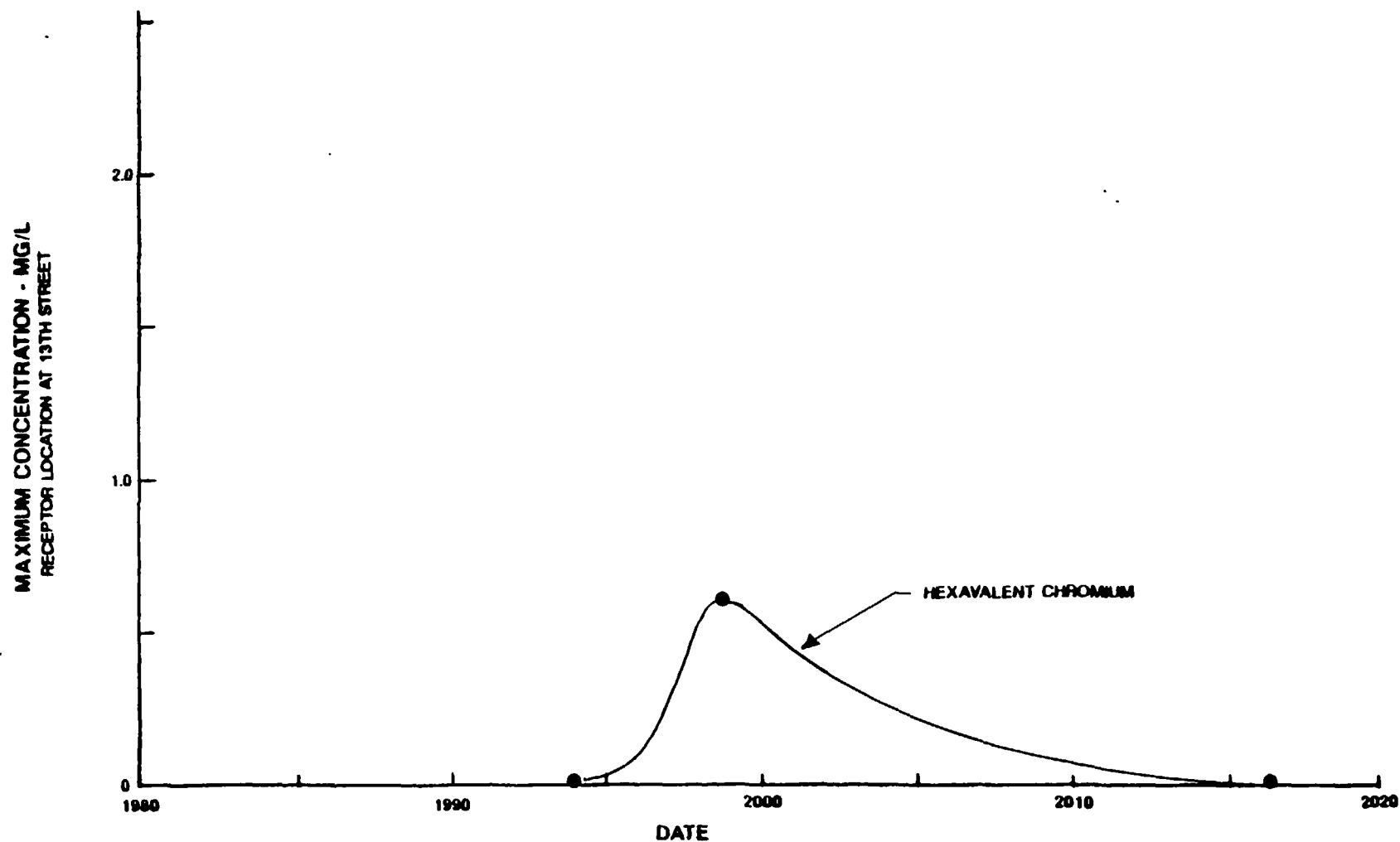


FIGURE R-10
CONCENTRATION/TIME
NORTHERNAIRE PLUME
CADILLAC AREA RI
CADILLAC, MICHIGAN

APPENDIX S
EXPOSURE AND RISK CALCULATIONS
FOR THE
BASELINE RISK ASSESSMENT

8.87.31
0193.0.0

DISK: 211
FILE: CAD_SHAL
DATE: 16-Jun-88
SITE: Cadillac
Shallow Plume (Present)
SUMMARY TABLE: TABLE B-9

Exposure Variables	Most Probable	Realistic Worst	Most Probable	Realistic Worst
	Child	Child	Adult	Adult
Amount Ingested (l/day)	1.0	1.0	2.0	2.0
Number of Exposures (per year)	365.0	365.0	365.0	365.0
Number of Years of Exposure	10.0	10.0	70.0	70.0

Body Weight	Kilograms
Child	35
Adult	70

[illegible]

Cadillac Shallow Plume (Present)
Groundwater Ingestion Most Probable Case
Noncarcinogenic Effects Child

Compound	Concentration (ug/l)	Amount Ingested (l/day)	Toxicokinetic Factor (unitless)	Body Weight (kg)	Number of Exposures (per year)	Body Dose (ug/kg/day)	Standard/ Guideline (ug/kg/day)	Hazard Ratio
1,1,1-TRICHLOROETHANE	3102.0	1.0	1.0	35.0	365.0	8.06E-02	8.60E-02	1.03E+00
TOLUENE	129.4	1.0	1.0	35.0	365.0	3.70E-03	3.00E-01	1.23E-02
ACETONE	44.5	1.0	1.0	35.0	365.0	1.27E-03	1.00E-01	1.27E-02

SUMMARY HAZARD
INDEX 1.06E+00

Cadillac Shallow Plume (Present)
Groundwater Ingestion Realistic Worst Case
Noncarcinogenic Effects Child

Compound	Concentration (ug/l)	Amount Ingested (l/day)	Toxicokinetic Factor (unitless)	Body Weight (kg)	Number of Exposures (per year)	Body Dose (ug/kg/day)	Standard/ Guideline (ug/kg/day)	Hazard Ratio
1,1,1-TRICHLOROETHANE	3200.0	1.0	1.0	35.0	365.0	9.14E-01	8.60E-02	1.06E+01
TOLUENE	3290.0	1.0	1.0	35.0	365.0	9.14E-02	3.00E-01	3.05E-01
ACETONE	920.0	1.0	1.0	35.0	365.0	2.63E-02	1.00E-01	2.63E-01

SUMMARY HAZARD
INDEX 1.12E+01

Cadillac Shallow Plume (Present)
Groundwater Ingestion Most Probable Case
Noncarcinogenic Effects Adult

Compound	Concentration (ug/l)	Amount Ingested (l/day)	Toxicokinetic Factor (unitless)	Body Weight (kg)	Number of Exposures (per year)	Body Dose (ng/kg/day)	Standard/ Guideline (ng/kg/day)	Hazard Ratio
1,1,1-TRICHLOROETHANE	3102.0	2.0	1.0	70.0	365.0	8.86E-02	8.60E-02	1.03E+00
TOLUENE	129.4	2.0	1.0	70.0	365.0	3.70E-03	3.00E-01	1.23E-02
ACETONE	44.5	2.0	1.0	70.0	365.0	1.27E-03	1.00E-01	1.27E-02

SUMMARY HAZARD
INDEX 1.06E+00

Cadillac Shallow Plume (Present)
Groundwater Ingestion Realistic Worst Case
Noncarcinogenic Effects Adult

Compound	Concentration (ug/l)	Amount Ingested (l/day)	Toxicokinetic Factor (unitless)	Body Weight (kg)	Number of Exposures (per year)	Body Dose (ng/kg/day)	Standard/ Guideline (ng/kg/day)	Hazard Ratio
1,1,1-TRICHLOROETHANE	32665.0	2.0	1.0	70.0	365.0	9.14E-01	8.60E-02	1.06E+01
TOLUENE	3200.0	2.0	1.0	70.0	365.0	9.14E-02	3.00E-01	3.05E-01
ACETONE	920.0	2.0	1.0	70.0	365.0	2.63E-02	1.00E-01	2.63E-01

SUMMARY HAZARD
INDEX 1.12E+01

Cadillac
Groundwater Ingestion
Carcinogenic Effects

Shallow Plume (Present)
Most Probable Case
Child

Compound	Concentration (ug/l)	Amount Ingested (l/day)	Toxicokinetic Factor (unitless)	Body Weight (kg)	Number of Exposures (per year)	Number of Years of Exposure 70 yr lifetime	Body Dose (ug/kg/day)	CAS Potency Value (ug/kg/day) ⁻¹	Incremental Cancer Risk
1,1-DICHLOROETHENE	93.9	1.0	1.0	35.0	365.0	10.0	3.83E-04	5.80E-01	2.22E-04
TETRACHLOROETHENE	64.6	1.0	1.0	35.0	365.0	10.0	2.64E-04	5.10E-02	1.34E-05
TRICHLOROETHENE	2879.4	1.0	1.0	35.0	365.0	10.0	1.18E-02	1.10E-02	1.29E-04
CHLOROFORM	0.4	1.0	1.0	35.0	365.0	10.0	1.63E-06	8.10E-02	1.32E-07
1,2-DICHLOROBENZENE	104.4	1.0	1.0	35.0	365.0	10.0	4.26E-04	9.10E-02	3.88E-05
METHYLENE CHLORIDE	33.9	1.0	1.0	35.0	365.0	10.0	1.38E-04	7.50E-03	1.04E-06

Summary Incremental
Cancer Risk 4.05E-04

Cadillac
Groundwater Ingestion
Carcinogenic Effects

Shallow Plume (Present)
Realistic worst Case
Child

Compound	Concentration (ug/l)	Amount Ingested (l/day)	Toxicokinetic Factor (unitless)	Body Weight (kg)	Number of Exposures (per year)	Number of Years of Exposure 70 yr lifetime	Body Dose (ug/kg/day)	CAS Potency Value (ug/kg/day) ⁻¹	Incremental Cancer Risk
1,1-DICHLOROETHENE	1000.0	1.0	1.0	35.0	365.0	10.0	4.08E-03	5.80E-01	2.37E-03
TETRACHLOROETHENE	720.0	1.0	1.0	35.0	365.0	10.0	2.94E-03	5.10E-02	1.50E-04
TRICHLOROETHENE	78060.0	1.0	1.0	35.0	365.0	10.0	3.18E-01	1.10E-02	3.50E-03
CHLOROFORM	15.0	1.0	1.0	35.0	365.0	10.0	6.12E-05	8.10E-02	4.96E-06
1,2-DICHLOROBENZENE	2050.0	1.0	1.0	35.0	365.0	10.0	8.16E-03	9.10E-02	7.43E-04
METHYLENE CHLORIDE	590.0	1.0	1.0	35.0	365.0	10.0	2.82E-03	7.50E-03	2.11E-05

Summary Incremental
Cancer Risk 6.79E-03

Cadillac
Groundwater Ingestion
Carcinogenic Effects

Shallow Plume (Present)
Most Probable Case
Adult

Compound	Concentration (ug/l)	Amount Ingested (l/day)	Toxicokinetic Factor (unitless)	Body Weight (kg)	Number of Exposures (per year)	Number of Years of Exposure 70 yr lifetime	Body Dose (ug/kg/day)	CAS Potency Value (ug/kg/day) ⁻¹	Incremental Cancer Risk
1,1-DICHLOROETHENE	93.9	2.0	1.0	70.0	365.0	70.0	2.68E-03	5.80E-01	1.56E-03
TETRACHLOROETHENE	64.6	2.0	1.0	70.0	365.0	70.0	1.83E-03	5.10E-02	9.41E-05
TRICHLOROETHENE	2879.4	2.0	1.0	70.0	365.0	70.0	8.23E-02	1.10E-02	9.05E-04
CHLOROFORM	0.4	2.0	1.0	70.0	365.0	70.0	1.14E-05	8.10E-02	9.26E-07
1,2-DICHLOROETHANE	104.4	2.0	1.0	70.0	365.0	70.0	2.98E-03	9.10E-02	2.71E-04
METHYLENE CHLORIDE	33.9	2.0	1.0	70.0	365.0	70.0	9.69E-04	7.90E-03	7.26E-06

Summary Incremental
Cancer Risk 2.83E-03

Cadillac
Groundwater Ingestion
Carcinogenic Effects

Shallow Plume (Present)
Realistic Worst Case
Adult

Compound	Concentration (ug/l)	Amount Ingested (l/day)	Toxicokinetic Factor (unitless)	Body Weight (kg)	Number of Exposures (per year)	Number of Years of Exposure 70 yr lifetime	Body Dose (ug/kg/day)	CAS Potency Value (ug/kg/day) ⁻¹	Incremental Cancer Risk
1,1-DICHLOROETHENE	1000.0	2.0	1.0	70.0	365.0	70.0	2.84E-02	5.80E-01	1.66E-02
TETRACHLOROETHENE	720.0	2.0	1.0	70.0	365.0	70.0	2.04E-02	5.10E-02	1.05E-03
TRICHLOROETHENE	78000.0	2.0	1.0	70.0	365.0	70.0	2.23E+00	1.10E-02	2.45E-02
CHLOROFORM	15.0	2.0	1.0	70.0	365.0	70.0	4.29E-04	8.10E-02	3.47E-05
1,2-DICHLOROETHANE	2000.0	2.0	1.0	70.0	365.0	70.0	5.71E-02	9.10E-02	5.20E-03
METHYLENE CHLORIDE	690.0	2.0	1.0	70.0	365.0	70.0	1.97E-02	7.90E-03	1.48E-04

Summary Incremental
Cancer Risk 4.75E-02

TABLE B-9

RISK CALCULATION SUMMARY
 Cadillac
 Shallow Plume (Present)

Groundwater Ingestion

Noncarcinogenic Effects

Compounds	Most Probable Case		Realistic Worst Case		Most Probable Case		Realistic Worst Case	
	Child	Percent Total Risk	Child	Percent Total Risk	Adult	Percent Total Risk	Adult	Percent Total Risk
1,1,1-TRICHLOROETHANE	1.03	97.6%	10.63	94.9%	1.03	97.6%	10.63	94.9%
TOLUENE	0.01	1.2%	0.30	2.7%	0.01	1.2%	0.30	2.7%
ACETONE	0.01	1.2%	0.26	2.3%	0.01	1.2%	0.26	2.3%
SUMMARY HAZARD INDEX		1.06		11.20		1.06		11.20

Groundwater Ingestion

Carcinogenic Effects

Compounds	Most Probable Case		Realistic Worst Case		Most Probable Case		Realistic Worst Case	
	Child	Percent Total Risk	Child	Percent Total Risk	Adult	Percent Total Risk	Adult	Percent Total Risk
1,1-DICHLOROETHENE	2.22E-04	34.9%	2.37E-03	34.9%	1.56E-03	34.9%	1.66E-02	34.9%
TETRACHLOROETHENE	1.34E-05	3.3%	1.90E-04	2.2%	9.41E-05	3.3%	1.05E-03	2.2%
TRICHLOROETHENE	1.29E-04	31.9%	3.90E-03	51.6%	9.09E-04	31.9%	2.45E-02	51.6%
CHLOROFORM	1.32E-07	0.0%	4.96E-06	0.1%	9.26E-07	0.0%	3.47E-05	0.1%
1,2-DICHLOROBENZENE	3.88E-05	9.6%	7.43E-04	10.9%	2.71E-04	9.6%	9.20E-03	10.9%
METHYLENE CHLORIDE	1.04E-06	0.3%	2.11E-05	0.3%	7.26E-06	0.3%	1.40E-04	0.3%
SUMMARY INCREMENTAL CANCER RISK		4.09E-04		6.79E-03		2.83E-03		4.75E-02

DISK: 212
FILE: SHAL CFU

DISK: 212

FILE: SHAL CFU

DATE: 15-Jun-88

SITE: Cadillac: Shallow Kysor Plume
Receptor: 13th Street (Ch:10)

SUMMARY TABLE: TABLE 8-10

Exposure Variables	Most Probable	Realistic Worst	Most Probable	Realistic Worst
	Child	Child	Adult	Adult
Amount Ingested (l/day)	1.0	1.0	NA	NA
Number of Exposures (per year)	365.0	365.0	NA	NA
Number of Years of Exposure	10.0	10.0	NA	NA

Body Weight	Kilograms
Child	35
Adult	70

[illegible]

Noncarcinogenic Effects	Child
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Noncarcinogenic Effects Child

[illegible]

Facilities: Shallow lysor PIReceptor: 13th Street (Child)
Groundwater Ingestion Most Probable Case
Carcinogenic Effects Child

[illegible]

Cadillac: Shallow Kysor FIReceptor: 13th Street (Child,
Groundwater Ingestion Realistic worst Case
Carcinogenic Effects Child

[illegible]

RISK CALCULATION SUMMARY
Cadillac: Shallow Kysor Plume
Receptor: 13th Street (Child)

Noncarcinogenic Effects

Groundwater Ingestion

Carcinogenic Effects

Compounds	Most Probable Case		Realistic Worst Case		Most Probable Case		Realistic Worst Case	
	Child	Percent Total Risk	Child	Percent Total Risk	Adult	Percent Total Risk	Adult	Percent Total Risk
1,1-DICHLORoETHENE	1.18E-05	17.8%	2.60E-05	17.9%				
TRICHLORoETHENE	5.34E-05	80.5%	1.17E-04	80.4%				
TETRACHLORoETHENE	1.14E-06	1.7%	2.50E-06	1.7%				
SUMMARY INCREMENTAL CANCER RISK:		6.64E-05		1.45E-04				

DISK: 212
FILE: SHAL FUA

DISC: 213

FILE: SHAL FUA

DATE: 15-Jun-88

SITE: Cadillac: Shallow Kysor Plume
Receptor: 13th Street (Adults)

SUMMARY TABLE: TABLE 8-10

Exposure Variables	Most Probable	Realistic Worst	Most Probable	Realistic Worst
	Child	Child	Adult	Adult
Amount Ingested (l/day)	NA	NA	2.0	2.0
Number of Exposures (per year)	NA	NA	365.0	365.0
Number of Years of Exposure	NA	NA	15.0	70.0

Body Weight	Kilograms
Child	35
Adult	70

[illegible]

Cadillac: Shallow Kysor PI Receptor: 13th Street (adults)
Groundwater Ingestion Most Probable Case
Noncarcinogenic Effects Adult

[illegible]

Cadillac: Shallow lysor Pikeceptor: 13th Street :Adults
Groundwater Ingestion Realistic: Worst Case
Noncarcinogenic Effects Adult

[illegible]

Carcinogenic Effects	Adult
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Summary Incremental      :
Cancer Risk              6.83E-05 :

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Carcinogenic Effects	Adult
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:Summary Incremental      :
:Cancer Risk              1.02E-03 :

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TABLE B-10

Groundwater Ingestion
Noncarcinogenic Effects

RISK CALCULATION SUMMARY
Cadillac: Shallow Kysor Plume
Receptor: 13th Street (Adults)

Compounds	Most Probable Case		Realistic Worst Case		Most Probable Case		Realistic Worst Case	
	Child	Percent Total Risk	Child	Percent Total Risk	Adult	Percent Total Risk	Adult	Percent Total Risk
1,1,1-TRICHLOROETHANE					0.11	98.6%	0.33	98.6%
TOLUENE					0.00	1.4%	0.01	1.4%
SUMMARY HAZARD INDEX					0.11		0.36	

Groundwater Ingestion
Carcinogenic Effects

Compounds	Most Probable Case		Realistic Worst Case		Most Probable Case		Realistic Worst Case	
	Child	Percent Total Risk	Child	Percent Total Risk	Adult	Percent Total Risk	Adult	Percent Total Risk
1,1-DICHLOROETHENE					1.21E-05	17.7%	1.82E-04	17.9%
TRICHLOROETHENE					3.91E-05	80.6%	8.17E-04	80.4%
TETRACHLOROETHENE					1.19E-06	1.7%	1.79E-05	1.7%
SUMMARY INCREMENTAL CANCER RISK					6.83E-05		1.02E-03	

DISK: 212
FILE: 4WIN_FUT

DATE: 15-Jun-88

SITE: Cadillac Four Minns Plume
Receptor: 13th Street (Future)

SUMMARY TABLE: TABLE B-11

Exposure Variables	Most Probable	Realistic Worst	Most Probable	Realistic Worst
	Child	Child	Adult	Adult
Amount Ingested (l/day)	1.0	1.0	2.0	2.0
Number of Exposures (per year)	365.0	365.0	365.0	365.0
Number of Years of Exposure	8.0	10.0	8.0	70.0

Body Weight	Kilograms
Child	35
Adult	70

[illegible]

Noncarcinogenic Effects	Child
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Noncarcinogenic Effects Child

[illegible]

Groundwater Ingestion	Most Probable Case
Noncarcinogenic Effects	Adult

Cadillac: Four Minns Plume Receptor: 13th Street (Future)
Groundwater Ingestion Realistic Worst Case
Sarcinogenic Effects Adult

[illegible]

Cadillac Four Winns Plume Receptor: 13th Street (Future)

Groundwater Ingestion Most Probable Case
Carcinogenic Effects Child

Compound	Concentration (ug/l)	Amount Ingested (l/day)	Toxicokinetic Factor (unitless)	Body Weight (kg)	Number of Exposures (per year)	Number of Years of Exposure (70 yr lifetime)	Body Dose (ug/kg/day)	CAS Potency Value (ug/kg/day) ⁻¹	Incremental Cancer Risk
1,1-DICHLOROETHENE	14.0	1.0	1.0	35.0	365.0	5.0	4.57E-05	5.80E-01	2.65E-05
TRICHLOROETHENE	57.0	1.0	1.0	35.0	365.0	8.0	1.92E-04	1.10E-02	2.12E-05
TETRACHLOROETHENE	20.0	1.0	1.0	35.0	365.0	8.0	1.00E-04	5.10E-02	5.10E-05
1,1-DICHLORODIFLUOROMETHANE	35.5	1.0	1.0	35.0	365.0	8.0	1.00E-04	5.10E-02	5.10E-05

Carcinogenic Effects	Adult

Carcinogenic Effects	Adult

[illegible]

TABLE B-11

Groundwater Ingestion

Noncarcinogenic Effects

RISK CALCULATION SUMMARY

Cadillac: Four Minns Plume

Receptor: 13th Street (Future)

Compounds	Most Probable Case Child	Percent Total Risk	Realistic Worst Case Child	Percent Total Risk	Most Probable Case Adult	Percent Total Risk	Realistic Worst Case Adult	Percent Total Risk
1,1,1-TRICHLOROETHANE	0.16	53.6%	0.71	54.9%	0.16	53.6%	0.71	54.9%
ACETONE	0.12	39.0%	0.51	39.6%	0.12	39.0%	0.51	39.6%
TRANS-1,2-DICHLOROETHENE	0.02	7.5%	0.07	5.5%	0.02	7.5%	0.07	5.5%
SUMMARY HAZARD INDEX		0.31		1.29		0.31		1.29

Groundwater Ingestion

Carcinogenic Effects

Compounds	Most Probable Case Child	Percent Total Risk	Realistic Worst Case Child	Percent Total Risk	Most Probable Case Adult	Percent Total Risk	Realistic Worst Case Adult	Percent Total Risk
1,1-DICHLOROETHENE	2.65E-05	53.5%	1.44E-04	53.4%	2.65E-05	53.5%	1.01E-03	53.4%
TRICHLOROETHENE	2.05E-06	4.1%	1.11E-05	4.1%	2.05E-06	4.1%	7.79E-05	4.1%
TETRACHLOROETHENE	3.33E-06	6.7%	1.89E-05	6.8%	3.33E-06	6.7%	1.30E-04	6.8%
1,2-DICHLOROETHANE	1.77E-05	35.7%	9.66E-05	35.7%	1.77E-05	35.7%	6.76E-04	35.7%
SUMMARY INCREMENTAL CANCER RISK		4.96E-05		2.71E-04		4.96E-05		1.09E-03

DISK: 211
FILE: CAD REX

SITE: Cadillac
Repair Plume (Present)

Exposure Variables	Most Probable	Realistic Worst	Most Probable	Realistic Worst
	Child	Child	Adult	Adult
Amount Ingested (l/day)	1.0	1.0	2.0	2.0
Number of Exposures (per year)	365.0	365.0	365.0	365.0
Number of Years of Exposure	10.0	10.0	70.0	70.0

Body Weight	Kilograms
Child	35
Adult	70

[illegible]

Cadillac	Repair Plus (Present)
Groundwater Ingestion	Most Probable Case
Noncarcinogenic Effects	Child

[illegible]

Facillac	Repair Phase (Present)
Groundwater Ingestion	Realistic Worst Case
Noncarcinogenic Effects	Child

Compound	Concentration (ug/l)	Amount Ingested (l/day)	Toxicokinetic Factor (unitless)	Body Weight (kg)	Number of Exposures (per year)	Body Dose (mg/kg/day)	Standard/ Guideline (mg/kg/day)	Hazard Ratio
1,1,1-TRICHLOROETHANE	17.0	1.0	1.0	35.0	365.0	4.86E-04	8.60E-02	5.65E-03
TOLUENE	42.0	1.0	1.0	35.0	365.0	1.20E-03	3.00E-01	4.00E-03
TRANS-1,2-DICHLOROETHENE	48.0	1.0	1.0	35.0	365.0	1.37E-03	2.00E-03	6.86E-01
:	:	:	:	:	:	:	:	:
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:	:	:	:	:	:	:	:	:
							SUMMARY HAZARD INDEX	6.95E-01

DISK: 214
FILE:MORT_FUT

TABLE 8-19
RISK CALCULATION SUMMARY

**Cadillac: Northernair Chronium Plume
Receptor: 13th Street (Future)**

This table calculates estimated body doses and risk ratios.

The equations to calculate body dose level and risk ratio are:

$$\text{Body Dose} = \frac{\text{Groundwater Concentration} \times \text{Amount Ingested} \times \text{Fraction Absorbed}}{\text{Body Weight}}$$

Risk Ratio = Body Dose

Standard or Guideline

Compound	Concentration (ug/l)	Amount Ingested (l/day)	Fraction Absorbed	Body Weight (kg)	Number of Exposures (Per Year)	Body Dose (ug/kg/day)	Standard/ Guideline (ug/kg/day)	Risk Ratio
CHROMIUM VI								
Most Probable								
Child	295	1	1	35	365	8.43	5	1.69
Adult	122	2	1	70	365	3.49	5	0.70
Realistic Worst								
Child	600	1	1	35	365	17.14	5	3.43
Adult	600	2	1	70	365	17.14	5	3.43

Four Star Plume Receptor: 13th Street (Adult)

ter Ingestion Most Probable Case

rogenic Effects Adult

Compound	Concentration (ug/l)	Amount Ingested (l/day)	Toxicokinetic Factor (unitless)	Body Weight (kg)	Number of Exposures (per year)	Body Dose (ug/kg/day)	Standard/ Guideline (ug/kg/day)	Hazard Ratio
CHLOROETHANE	33.7	2.0	1.0	70.0	365.0	9.63E-04	8.60E-02	1.12E-02

SUMMARY HAZARD

INDEX

1.12E-02

Four Star Plume Receptor: 13th Street (Adult)

ter Ingestion Realistic Worst Case

rogenic Effects Adult

Compound	Concentration (ug/l)	Amount Ingested (l/day)	Toxicokinetic Factor (unitless)	Body Weight (kg)	Number of Exposures (per year)	Body Dose (ug/kg/day)	Standard/ Guideline (ug/kg/day)	Hazard Ratio
CHLOROETHANE	132.0	2.0	1.0	70.0	365.0	3.77E-03	8.60E-02	4.39E-02

SUMMARY HAZARD

INDEX

4.39E-02

Carcinogenic Effects	Adult

Carcinogenic Effects	Adult
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Compound	Concentration (ug/l)	Avg amt Ingested (l/day)	Toxicokinetic Factor (unitless)	Body Weight (kg)	Number of Exposures (per year)	Number of Years of Exposure 70 yr lifetime	Body Dose (mg/kg/day)	CAG Potency Value (mg/kg/day) ⁻¹	Incremental Cancer Risk
TETRACHLOROETHENE	1740.0	2.0	1.0	70.0	365.0	70.0	4.97E-02	5.10E-02	2.54E-03
TRICHLOROETHENE	6730.0	2.0	1.0	70.0	365.0	70.0	1.92E-01	1.10E-02	2.12E-03
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								:Summary Incremental :Cancer Risk	4.65E-03

RISK CALCULATION SUMMARY
Cadillac: Four Star Plus
Receptor: 13th Street (Adult)

Noncarcinogenic Effects

Compounds	Most Probable Case		Realistic Worst Case		Most Probable Case		Realistic Worst Case	
	Child	Percent Total Risk	Child	Percent Total Risk	Adult	Percent Total Risk	Adult	Percent Total Risk
1,1,1-TRICHLOROETHANE					0.01	100.0%	0.04	100.0%
SUMMARY HAZARD INDEX						0.01		0.04

Carcinogenic Effects

Compounds	Most Probable Case		Realistic Worst Case		Most Probable Case		Realistic Worst Case	
	Child	Percent Total Risk	Child	Percent Total Risk	Adult	Percent Total Risk	Adult	Percent Total Risk
TETRACHLOROETHENE					1.36E-04	55.1%	2.94E-03	94.3%
TRICHLOROETHENE					1.11E-04	44.9%	2.12E-03	45.3%
SUMMARY INCREMENTAL CANCER RISK:						2.48E-04		4.65E-03

DISK: 214
FILE: 4STAR.CF

FILE: 4STAR_CF

DATE: 15-Jun-88

SITE: Cadillac: Four Star Plus
Receptor: 13th Street (Child)

SUMMARY TABLE: TABLE B-1B

Exposure Variables	Most Probable Child	Realistic Worst Child	Most Probable Adult	Realistic Worst Adult
Amount Ingested (l/day)	1.0	1.0	NA	NA
Number of Exposures (per year)	365.0	365.0	NA	NA
Number of Years of Exposure	10.0	10.0	NA	NA

Body Weight	Kilograms
Child	35
Adult	70

Noncarcinogenic Compound(s)	Most Probable Concentration (ug/l)	Realistic Worst Case Concentration (ug/l)	Standard or Guideline (mg/kg/day)	Toxicokinetic Factor (unitless)
1,1,1-TRICHLOROETHANE	49.0	132.0	8.60E-02	1.0
TETRACHLOROETHENE	585.0	1740.0	5.10E-02	1.0
TRICHLOROETHENE	2450.0	6730.0	1.10E-02	1.0

Cadillac: Four Star Plume Receptor: 13th Street (Child)

Groundwater Ingestion Most Probable Case

Noncarcinogenic Effects Child

Compound	Concentration (ug/l)	Amount Ingested (l/day)	Toxicokinetic Factor (unitless)	Body Weight (kg)	Number of Exposures (per year)	Body Dose (ng/kg/day)	Standard/ Guideline (ng/kg/day)	Hazard Ratio
1,1,1-TRICHLOROETHANE	49.0	1.0	1.0	35.0	365.0	1.40E-03	8.60E-02	1.63E-02

SUMMARY HAZARD

INDEX 1.63E-02

Cadillac: Four Star Plume Receptor: 13th Street (Child)

Groundwater Ingestion Realistic Worst Case

Noncarcinogenic Effects Child

Compound	Concentration (ug/l)	Amount Ingested (l/day)	Toxicokinetic Factor (unitless)	Body Weight (kg)	Number of Exposures (per year)	Body Dose (ng/kg/day)	Standard/ Guideline (ng/kg/day)	Hazard Ratio
1,1,1-TRICHLOROETHANE	132.0	1.0	1.0	35.0	365.0	3.77E-03	8.60E-02	4.39E-02

SUMMARY HAZARD

INDEX 4.39E-02

DISK: 214
FILE: 4STAR_AF

DATE: 15-Jun-88

SITE: Cadillac: Four Star Plume
Receptor: 13th Street (Adult)

Exposure Variables	Most Probable	Realistic Worst	Most Probable	Realistic Worst
	Child	Child	Adult	Adult
Amount Ingested (l/day)	NA	NA	2.0	2.0
Number of Exposures (per year)	NA	NA	365.0	365.0
Number of Years of Exposure	NA	NA	15.0	70.0

Body Weight	Kilograms
Child	35
Adult	70

[illegible]

Cadillac
Groundwater Ingestion
Noncarcinogenic Effects

Repair Plume (Present)
Most Probable Case
Adult

Compound	Concentration (ug/l)	Amount Ingested (l/day)	Toxicokinetic Factor (unitless)	Body Weight (kg)	Number of Exposures (per year)	Body Dose (mg/kg/day)	Standard/ Guideline (mg/kg/day)	Hazard Ratio
1,1,1-TRICHLOROETHANE	0.7	2.0	1.0	70.0	365.0	2.00E-05	8.60E-02	2.33E-04
TOLUENE	1.8	2.0	1.0	70.0	365.0	5.14E-05	3.00E-01	1.71E-04
TRANS-1,2-DICHLORoETHENE	2.4	2.0	1.0	70.0	365.0	6.86E-05	2.00E-03	3.43E-02

SUMMARY HAZARD
INDEX 3.47E-02

Cadillac
Groundwater Ingestion
Noncarcinogenic Effects

Repair Plume (Present)
Realistic Worst Case
Adult

Compound	Concentration (ug/l)	Amount Ingested (l/day)	Toxicokinetic Factor (unitless)	Body Weight (kg)	Number of Exposures (per year)	Body Dose (mg/kg/day)	Standard/ Guideline (mg/kg/day)	Hazard Ratio
1,1,1-TRICHLOROETHANE	17.0	2.0	1.0	70.0	365.0	4.86E-04	8.60E-02	5.65E-03
TOLUENE	42.0	2.0	1.0	70.0	365.0	1.20E-03	3.00E-01	4.00E-03
TRANS-1,2-DICHLORoETHENE	48.0	2.0	1.0	70.0	365.0	1.37E-03	2.00E-03	6.86E-01

SUMMARY HAZARD
INDEX 6.95E-01

Cadillac	Repair Plume (Present)
Groundwater Ingestion	Most Probable Case
Carcinogenic Effects	Child

[illegible]

Cadillac	Federal Floor (Present)
Groundwater Ingestion	Realistic worst Case
Carcinogenic Effects	Child

[illegible]

111ac
 groundwater ingestion
 carcinogenic Effects

Repair Plume (Present)
 Most Probable Case
 Adult

Compound	Concentration (ug/l)	Amount Ingested (l/day)	Toxicokinetic Factor (unitless)	Body Weight (kg)	Number of Exposures (per year)	Number of years of Exposure 70 yr lifetime	Body Dose (ug/kg/day)	CAS Potency Value (ug/kg/day) ⁻¹	Incremental Cancer Risk
DICHLOROMETHANE	0.3	2.0	1.0	70.0	365.0	70.0	8.57E-06	8.10E-02	6.94E-07
ETHYLENE CHLORIDE	19.3	2.0	1.0	70.0	365.0	70.0	5.51E-04	7.50E-03	4.14E-06
CHLOROETHENE	11073.5	2.0	1.0	70.0	365.0	70.0	3.16E-01	1.10E-02	3.48E-03

Summary Incremental
 Cancer Risk 3.49E-03

1111ac
 groundwater ingestion
 carcinogenic Effects

Repair Plume (Present)
 Realistic Worst Case
 Adult

Compound	Concentration (ug/l)	Amount Ingested (l/day)	Toxicokinetic Factor (unitless)	Body Weight (kg)	Number of Exposures (per year)	Number of years of Exposure 70 yr lifetime	Body Dose (ug/kg/day)	CAS Potency Value (ug/kg/day) ⁻¹	Incremental Cancer Risk
DICHLOROMETHANE	6.0	2.0	1.0	70.0	365.0	70.0	1.71E-04	8.10E-02	1.39E-05
ETHYLENE CHLORIDE	780.0	2.0	1.0	70.0	365.0	70.0	2.23E-02	7.50E-03	1.67E-04
CHLOROETHENE	153000.0	2.0	1.0	70.0	365.0	70.0	4.37E+00	1.10E-02	4.81E-02

Summary Incremental
 Cancer Risk 4.83E-02

TABLE B-12

Groundwater Ingestion
Noncarcinogenic Effects

RISK CALCULATION SUMMARY
Cadillac
Repair Plume (Present)

Compounds	Most Probable Case Child	Percent Total Risk	Realistic Worst Case Child	Percent Total Risk	Most Probable Case Adult	Percent Total Risk	Realistic Worst Case Adult	Percent Total Risk
1,1,1-TRICHLOROETHANE	0.00	0.7%	0.01	0.8%	0.00	0.7%	0.01	0.8%
TOLUENE	0.00	0.5%	0.00	0.6%	0.00	0.5%	0.00	0.6%
TRANS-1,2-DICHLOROETHENE	0.03	98.8%	0.69	98.6%	0.03	98.8%	0.69	98.6%
SUMMARY HAZARD INDEX		0.03		0.70		0.03		0.70

Groundwater Ingestion
Carcinogenic Effects

Compounds	Most Probable Case Child	Percent Total Risk	Realistic Worst Case Child	Percent Total Risk	Most Probable Case Adult	Percent Total Risk	Realistic Worst Case Adult	Percent Total Risk
CHLOROFORM	9.92E-08	0.0%	1.98E-06	0.0%	6.94E-07	0.0%	1.39E-05	0.0%
METHYLENE CHLORIDE	5.91E-07	0.1%	2.39E-05	0.3%	4.14E-06	0.1%	1.67E-04	0.3%
TRICHLOROETHENE	4.97E-04	99.9%	6.87E-03	99.6%	3.40E-03	99.9%	4.81E-02	99.6%
SUMMARY INCREMENTAL CANCER RISK:		4.98E-04		6.90E-03		3.49E-03		4.83E-02

DISK: 213
FILE: RE1 CFU

FILE: REX CFU

DATE: 17-Jun-88

SITE: Cadillac: Rexair Plume
Receptor: 13th Street Child

SUMMARY TABLE: TABLE 2-13

Exposure Variables	Most Probable	Realistic Worst	Most Probable	Realistic Worst
	Child	Child	Adult	Adult
Amount Ingested (l/day)	1.0	1.0	NA	NA
Number of Exposures (per year)	365.0	365.0	NA	NA
Number of years of Exposure	10.0	10.0	NA	NA

Body Weight	Kilograms
Child	35
Adult	70

[illegible]

air Plume	Receptor: 13th Street (Child)
Ingestion	Most Probable Case
Acute Effects	Child

[illegible]

Repair Plume	Receptor: 13th Street (Child)
r Ingestion	Realistic Worst Case
sonic Effects	Child

[illegible]

Cadillac: Repair Plume	Receptor: 13th Street (Child)
Groundwater Ingestion	Most Probable Case
Carcinogenic Effects	Child

Compound	Concentration (ug/l)	Amount Ingested (l/day)	Toxicokinetic Factor (unitless)	Body Weight (kg)	Number of Exposures (per year)	Number of Years: of Exposure (70 yr lifetime)	Body Dose (mg/kg/day)	CAS Potency Value (mg/kg/day) ⁻¹	Incremental Cancer Risk
CHLOROFORM	0.9	1.0	1.0	35.0	365.0	10.0	3.67E-06	8.10E-02	2.98E-07
TRICHLOROETHENE	22000.0	1.0	1.0	35.0	365.0	10.0	8.98E-02	1.10E-02	9.88E-04
METHYLENE CHLORIDE	112.5	1.0	1.0	35.0	365.0	10.0	4.59E-04	7.50E-03	3.44E-06
								Summary Incremental Cancer Risk	9.91E-04

Labellans: Hexair Plume	Receptor: 13th Street (Child)
Brownwater Ingestion	Realistic Worst Case
Carcinogenic Effects	Child

[illegible]

RISK CALCULATION SUMMARY
Cadillac: Repair Plume
Receptor: 13th Street (Child)

Noncarcinogenic Effects

Compounds	Most Probable Case Child	Percent Total Risk	Realistic Worst Case Child	Percent Total Risk	Most Probable Case Adult	Percent Total Risk	Realistic Worst Case Adult	Percent Total Risk
1,1,1-TRICHLOROETHENE	0.00	0.9%	0.00	0.8%				
TOLUENE	0.00	0.6%	0.00	0.6%				
TRANS-1,2-DICHLOROETHENE	0.11	98.5%	0.24	98.6%				
SUMMARY HAZARD INDEX		0.11		0.25				

Groundwater Ingestion

Carcinogenic Effects

Compounds	Most Probable Case		Realistic Worst Case		Most Probable Case		Realistic Worst Case	
	Child	Percent Total Risk	Child	Percent Total Risk	Adult	Percent Total Risk	Adult	Percent Total Risk
CHLOROFORM	2.98E-07	0.02:	6.61E-07	0.02:	:	:	:	:
TRICHLOROETHENE	9.88E-04	99.62:	2.38E-03	99.62:	:	:	:	:
METHYLENE CHLORIDE	3.44E-06	0.32:	8.27E-06	0.32:	:	:	:	:
	:	:	:	:	:	:	:	:
	:	:	:	:	:	:	:	:
	:	:	:	:	:	:	:	:
	:	:	:	:	:	:	:	:
	:	:	:	:	:	:	:	:
SUMMARY INCREMENTAL CANCER RISK:	9.91E-04	:	2.39E-03	:	:	:	:	:

GROUNDWATER INGESTION RISK SCENARIOS

DISK: 213
FILE: REI_AFU

DATE: 15-Jun-88

SITE: Cadillac: Repair Plume
Receptor: 13th Street (Adult)

SUMMARY TABLE: TABLE B-13

Exposure Variables	Most Probable	Realistic Worst	Most Probable	Realistic Worst
	Child	Child	Adult	Adult
Amount Ingested (l/day)	NA	NA	2.0	2.0
Number of Exposures (per year)	NA	NA	365.0	365.0
Number of Years of Exposure	NA	NA	15.0	70.0

Body Weight	kilograms
Child	35
Adult	70

SUMMARY TABLE: TABLE B-13

[illegible]

Cadillac: Repair Pileup Receptor: 13th Street (Adult)
 Groundwater Ingestion Most Probable Case
 Noncarcinogenic Effects Adult

Compound	Concentration (ug/l)	Amount Ingested (l/day)	Toxicokinetic Factor (unitless)	Body Weight (kg)	Number of Exposures (per year)	Body Dose (ug/kg/day)	Standard/ Guideline (ug/kg/day)	Hazard Ratio
1,1,1-TRICHLOROETHENE	2.0	2.0	1.0	70.0	365.0	5.71E-05	8.60E-02	6.64E-04
TOLUENE	4.5	2.0	1.0	70.0	365.0	1.29E-04	3.00E-01	4.29E-04
TRANS-1,2-DICHLOROETHENE	9.2	2.0	1.0	70.0	365.0	1.49E-04	2.00E-03	7.43E-02

SUMMARY HAZARD
 INDEX 7.54E-02

Cadillac: Repair Pileup Receptor: 13th Street (Adult)
 Groundwater Ingestion Realistic Worst Case
 Noncarcinogenic Effects Adult

Compound	Concentration (ug/l)	Amount Ingested (l/day)	Toxicokinetic Factor (unitless)	Body Weight (kg)	Number of Exposures (per year)	Body Dose (ug/kg/day)	Standard/ Guideline (ug/kg/day)	Hazard Ratio
1,1,1-TRICHLOROETHENE	6.0	2.0	1.0	70.0	365.0	1.71E-04	8.60E-02	1.99E-03
TOLUENE	15.0	2.0	1.0	70.0	365.0	4.29E-04	3.00E-01	1.43E-03
TRANS-1,2-DICHLOROETHENE	17.0	2.0	1.0	70.0	365.0	4.66E-04	2.00E-03	2.43E-01

SUMMARY HAZARD
 INDEX 2.46E-01

Receptor: 13th Street (Adult)
 Most Probable Case
 Adult

Concentration (ug/l)	Amount Ingested (l/day)	Toxicokinetic Factor (unitless)	Body Weight (kg)	Number of Exposures (per year)	Number of Years of Exposure 70 yr lifetime	Body Dose (ng/kg/day)	CAS Potency Value (ng/kg/day) ⁻¹	Incremental Cancer Risk
0.6	2.0	1.0	70.0	365.0	15.0	3.67E-06	8.10E-02	2.98E-07
15130.0	2.0	1.0	70.0	365.0	15.0	9.26E-02	1.10E-02	1.02E-03
77.0	2.0	1.0	70.0	365.0	15.0	4.71E-04	7.50E-03	3.54E-06

Summary Incremental
 Cancer Risk 1.02E-03

Receptor: 13th Street (Adult)
 Realistic Worst Case
 Adult

Concentration (ug/l)	Amount Ingested (l/day)	Toxicokinetic Factor (unitless)	Body Weight (kg)	Number of Exposures (per year)	Number of Years of Exposure 70 yr lifetime	Body Dose (ng/kg/day)	CAS Potency Value (ng/kg/day) ⁻¹	Incremental Cancer Risk
2.0	2.0	1.0	70.0	365.0	70.0	5.71E-05	8.10E-02	4.63E-06
53000.0	2.0	1.0	70.0	365.0	70.0	1.51E+00	1.10E-02	1.67E-02
270.0	2.0	1.0	70.0	365.0	70.0	7.71E-03	7.50E-03	5.79E-03

Summary Incremental
 Cancer Risk 1.67E-02

RISK CALCULATION SUMMARY
Cadillac: Renair Place
Receptor: 13th Street (Adult)

[illegible]

RISK: 213
FILE: RXX_000

TABLE 8-14

RISK CALCULATION SUMMARY
Cadillac: Hexair Plume
Receptor: Municipal Well (Future)

This table calculates estimated body doses and incremental cancer risks.

The equations to calculate body dose level and incremental cancer risks are:

$$\text{Body Dose} = \text{Concentration} \times \text{Amount Ingested} \times \text{Fraction Absorbed} \times \frac{1}{\text{Body Weight (kg)}} \times \frac{\text{Frequency of Exposure}}{365 \text{ days}} \times \frac{\text{No. of Years of Exposure}}{70 \text{ yr. Lifetime}}$$

(ug/kg/day)

$$\text{Incremental Cancer Risk} = \text{Body Dose} \times \text{Carcinogenic Potency Factor}$$

Compound	Concentration (ug/l)	Amount Ingested (l/day)	Fraction Absorbed	Body Weight (kg)	Frequency of Ingestion (days)	Fraction of Year Exposed	Years of Exposure (years)	Fraction of Lifetime Exposed	Body Dose (ug/kg/day)	Micrograms per milligram	CAC Potency Estimate (ug/kg/day)-1	Risk Estimate
TCX	100	2	1	70	365	1.00	70	1.00	2.857143	1000	0.011	3.14E-05

GROUNDWATER INGESTION RISK SCENARIOS

DISK: 210
FILE: CAD INTR

DATE: 15-Jun-68

SITE: Cadillac
Intermediate Plus (Present)

SUMMARY TABLE: TABLE 8-15

Exposure Variables	Most Probable	Realistic Worst	Most Probable	Realistic Worst
	Child	Child	Adult	Adult
Amount Ingested (l/day)	1.0	1.0	2.0	2.0
Number of Exposures (per year)	365.0	365.0	365.0	365.0
Number of Years of Exposure	10.0	10.0	70.0	70.0

Body Weight	Kilograms
Child	35
Adult	70

[illegible]

Cadillac Intermediate Plume (Present)
Groundwater Ingestion Most Probable Case
Noncarcinogenic Effects Adult

Compound	Concentration (ug/l)	Amount Ingested (l/day)	Toxicokinetic Factor (unitless)	Body Weight (kg)	Number of Exposures (per year)	Body Dose (ug/kg/day)	Standard/ Guideline (ug/kg/day)	Hazard Ratio
1,1,1-TRICHLOROETHANE	185.2	2.0	1.0	70.0	365.0	5.29E-03	8.60E-02	6.15E-02
TOLUENE	0.0	2.0	1.0	70.0	365.0	0.00E+00	3.00E-01	0.00E+00
ACETONE	35.9	2.0	1.0	70.0	365.0	1.03E-03	1.00E-01	1.03E-02
TRANS-1,2-DICHLOROETHENE	419.0	2.0	1.0	70.0	365.0	1.20E-02	2.00E-03	5.99E+00

SUMMARY HAZARD
INDEX 6.06E+00

Cadillac Intermediate Plume (Present)
Groundwater Ingestion Realistic Worst Case
Noncarcinogenic Effects Adult

Compound	Concentration (ug/l)	Amount Ingested (l/day)	Toxicokinetic Factor (unitless)	Body Weight (kg)	Number of Exposures (per year)	Body Dose (ug/kg/day)	Standard/ Guideline (ug/kg/day)	Hazard Ratio
1,1,1-TRICHLOROETHANE	1200.0	2.0	1.0	70.0	365.0	3.43E-02	8.60E-02	3.99E-01
TOLUENE	420.0	2.0	1.0	70.0	365.0	1.20E-02	3.00E-01	4.00E-02
ACETONE	1.0	2.0	1.0	70.0	365.0	2.88E-05	1.00E-01	2.88E-04
TRANS-1,2-DICHLOROETHENE	4600.0	2.0	1.0	70.0	365.0	1.31E-01	2.00E-03	6.57E+01

SUMMARY HAZARD
INDEX 6.62E+01

Saltillo
Groundwater Ingestion
Carcinogenic Effects

Intermediate Plume (Present)
Most Probable Case
Child

Compound	Concentration (ug/l)	Amount Ingested (l/day)	Toxicokinetic Factor (unitless)	Body Weight (kg)	Number of Exposures (per year)	Number of Years of Exposure (70 yr lifetime)	Body Dose (ug/kg/day)	CAS Potency Value (ug/kg/day) ⁻¹	Incremental Cancer Risk
1,1-DICHLOROETHENE	12.0	1.0	1.0	35.0	365.0	10.0	4.90E-05	5.80E-01	2.84E-05
TETRACHLOROETHENE	0.8	1.0	1.0	35.0	365.0	10.0	3.27E-06	5.10E-02	1.67E-07
TRICHLOROETHENE	2251.7	1.0	1.0	35.0	365.0	10.0	9.19E-03	1.10E-02	1.01E-04
CHLOROFORM	0.2	1.0	1.0	35.0	365.0	10.0	8.16E-07	8.10E-02	6.61E-08
METHYLENE CHLORIDE	15.1	1.0	1.0	35.0	365.0	10.0	6.16E-05	7.50E-03	4.62E-07

Summary Incremental
Cancer Risk 1.36E-04

Saltillo
Groundwater Ingestion
Carcinogenic Effects

Intermediate Plume (Present)
Realistic worst Case
Child

Compound	Concentration (ug/l)	Amount Ingested (l/day)	Toxicokinetic Factor (unitless)	Body Weight (kg)	Number of Exposures (per year)	Number of Years of Exposure (70 yr lifetime)	Body Dose (ug/kg/day)	CAS Potency Value (ug/kg/day) ⁻¹	Incremental Cancer Risk
1,1-DICHLOROETHENE	99.0	1.0	1.0	35.0	365.0	10.0	4.04E-04	5.80E-01	2.34E-04
TETRACHLOROETHENE	17.0	1.0	1.0	35.0	365.0	10.0	6.94E-05	5.10E-02	3.54E-06
TRICHLOROETHENE	7500.0	1.0	1.0	35.0	365.0	10.0	3.06E-02	1.10E-02	3.37E-04
CHLOROFORM	4.0	1.0	1.0	35.0	365.0	10.0	1.63E-05	8.10E-02	1.32E-06
METHYLENE CHLORIDE	240.0	1.0	1.0	35.0	365.0	10.0	9.80E-04	7.50E-03	7.35E-06

Summary Incremental
Cancer Risk 5.83E-04

iliac
 ndwater Ingestion
 inogenic Effects

Intermediate Plume (Present)
 Most Probable Case
 Adult

Compound	Concentration (ug/l)	Amount Ingested (l/day)	Toxicokinetic Factor (unitless)	Body Weight (kg)	Number of Exposures (per year)	Number of Years of Exposure (70 yr lifetime)	Body Dose (ug/kg/day)	CAS Potency Value (ug/kg/day) ⁻¹	Incremental Cancer Risk
DICHLOROETHENE	12.0	2.0	1.0	70.0	365.0	70.0	3.43E-04	5.80E-01	1.99E-04
ACHLOROETHENE	0.8	2.0	1.0	70.0	365.0	70.0	2.29E-05	5.10E-02	1.17E-06
HLOROETHENE	2251.7	2.0	1.0	70.0	365.0	70.0	6.43E-02	1.10E-02	7.08E-04
ROFORN	0.2	2.0	1.0	70.0	365.0	70.0	5.71E-06	8.10E-02	4.63E-07
YLENE CHLORIDE	15.1	2.0	1.0	70.0	365.0	70.0	4.31E-04	7.50E-03	3.24E-04

Summary Incremental
 Cancer Risk 9.11E-04

iliac
 ndwater Ingestion
 inogenic Effects

Intermediate Plume (Present)
 Realistic Worst Case
 Adult

Compound	Concentration (ug/l)	Amount Ingested (l/day)	Toxicokinetic Factor (unitless)	Body Weight (kg)	Number of Exposures (per year)	Number of Years of Exposure (70 yr lifetime)	Body Dose (ug/kg/day)	CAS Potency Value (ug/kg/day) ⁻¹	Incremental Cancer Risk
-DICHLOROETHENE	99.0	2.0	1.0	70.0	365.0	70.0	2.83E-03	5.80E-01	1.64E-03
PACHLOROETHENE	17.0	2.0	1.0	70.0	365.0	70.0	4.86E-04	5.10E-02	2.48E-05
CHLOROETHENE	7500.0	2.0	1.0	70.0	365.0	70.0	2.14E-01	1.10E-02	2.36E-03
ROFORN	4.0	2.0	1.0	70.0	365.0	70.0	1.14E-04	8.10E-02	9.26E-06
YLENE CHLORIDE	240.0	2.0	1.0	70.0	365.0	70.0	6.86E-03	7.50E-03	5.14E-05

Summary Incremental
 Cancer Risk 4.08E-03

TABLE B-13

RISK CALCULATION SUMMARY
Cadillac
Intermediate Plume (Present)

Underwater Ingestion
Carcinogenic Effects

Compounds	Most Probable Case		Realistic Worst Case		Most Probable Case		Realistic Worst Case	
	Child	Percent Total Risk	Child	Percent Total Risk	Adult	Percent Total Risk	Adult	Percent Total Risk
1,1-TRICHLOROETHANE	0.06	1.0%	0.40	0.6%	0.06	1.0%	0.40	0.6%
BENZENE	0.00	0.0%	0.04	0.1%	0.00	0.0%	0.04	0.1%
1,1-DICHLOROETHANE	0.01	0.2%	0.00	0.0%	0.01	0.2%	0.00	0.0%
1,1,2,2-TETRACHLOROETHANE	3.99	98.8%	45.71	99.3%	3.99	98.8%	45.71	99.3%
SUMMARY HAZARD INDEX	6.06		66.15		6.06		66.15	

Underwater Ingestion
Carcinogenic Effects

Compounds	Most Probable Case		Realistic Worst Case		Most Probable Case		Realistic Worst Case	
	Child	Percent Total Risk	Child	Percent Total Risk	Adult	Percent Total Risk	Adult	Percent Total Risk
1,1-DICHLOROETHANE	2.84E-05	21.8%	2.34E-04	40.2%	1.99E-04	21.8%	1.64E-03	40.2%
1,1,2-TRICHLOROETHANE	1.67E-07	0.1%	3.54E-06	0.6%	1.17E-06	0.1%	2.48E-05	0.6%
1,1,1-TRICHLOROETHANE	1.01E-04	77.6%	3.27E-04	57.7%	7.08E-04	77.6%	2.36E-03	57.7%
1,1,1,2-TETRACHLOROETHANE	6.61E-08	0.1%	1.32E-06	0.2%	4.63E-07	0.1%	9.26E-06	0.1%
ETHYLENE CHLORIDE	4.52E-07	0.4%	7.35E-06	1.3%	3.24E-06	0.4%	5.14E-05	1.3%
SUMMARY INCREMENTAL CANCER RISK:	1.30E-04		5.83E-04		9.11E-04		4.00E-03	

Cadillac	Intermediate Plume (Present)
Groundwater Ingestion	Most Probable Case
Noncarcinogenic Effects	Child

[illegible]

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SUMMARY HAZARD
INDEX          6.06E+00

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Cadillac	Intermediate Plume (Present)
Groundwater Ingestion	Realistic Worst Case
Noncarcinogenic Effects	Child

[illegible]

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SUMMARY HAZARD
INDEX          4.42E+01

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DISK: 211
FILE: CAD_MORT

FILE: CAD MORT

SITE: Cadillac
Northern Pine (Present)

Exposure Variables	Most Probable Child	Realistic Worst Child	Most Probable Adult	Realistic Worst Adult
Amount Ingested (l/day)	1.0	1.0	2.0	2.0
Number of Exposures (per year)	365.0	365.0	365.0	365.0
Number of Years of Exposure	10.0	10.0	70.0	70.0

Body Weight	Kilograms
Child	35
Adult	70

[illegible]

Well: Northern Plume (Present)
 Groundwater Ingestion: Most Probable Case
 Carcinogenic Effects: Child

Compound	Concentration (ug/l)	Amount Ingested (l/day)	Toxicokinetic Factor (unitless)	Body Weight (kg)	Number of Exposures (per year)	Body Dose (ug/kg/day)	Standard/ Guideline (ug/kg/day)	Hazard Ratio
RONILP	103.9	1.0	1.0	35.0	365.0	2.97E-03	5.00E-03	5.94E-01
ANIDE	0.0	1.0	1.0	35.0	365.0	0.00E+00	2.90E-02	0.00E+00
1,1-TRICHLOROETHANE	44.6	1.0	1.0	35.0	365.0	1.27E-03	8.60E-02	1.48E-02
ANS-1,2-DICHLOROETHENE	2.4	1.0	1.0	35.0	365.0	6.86E-05	2.00E-03	3.43E-02

SUMMARY HAZARD
 INDEX 6.43E-01

Well: Northern Plume (Present)
 Groundwater Ingestion: Realistic Worst Case
 Carcinogenic Effects: Child

Compound	Concentration (ug/l)	Amount Ingested (l/day)	Toxicokinetic Factor (unitless)	Body Weight (kg)	Number of Exposures (per year)	Body Dose (ug/kg/day)	Standard/ Guideline (ug/kg/day)	Hazard Ratio
RONILP	773.0	1.0	1.0	35.0	365.0	2.21E-02	5.00E-03	4.42E+00
ANIDE	22.0	1.0	1.0	35.0	365.0	6.29E-04	2.90E-02	2.17E-02
1,1-TRICHLOROETHANE	440.0	1.0	1.0	35.0	365.0	1.26E-02	8.60E-02	1.46E-01
ANS-1,2-DICHLOROETHENE	40.0	1.0	1.0	35.0	365.0	1.14E-03	2.00E-03	5.71E-01

SUMMARY HAZARD
 INDEX 3.16E+00

Cadillac
Groundwater Ingestion
Noncarcinogenic Effects

Northernaire Plume (Present)
Most Probable Case
Adult

Compound	Concentration (ug/l)	Amount Ingested (l/day)	Toxicokinetic Factor (unitless)	Body Weight (kg)	Number of Exposures (per year)	Body Dose (mg/kg/day)	Standard/ Guideline (mg/kg/day)	Hazard Ratio
CHROMIUM	103.9	2.0	1.0	70.0	365.0	2.97E-03	5.00E-03	5.94E-01
CYANIDE	0.0	2.0	1.0	70.0	365.0	0.00E+00	2.90E-02	0.00E+00
1,1,1-TRICHLOROETHANE	44.6	2.0	1.0	70.0	365.0	1.27E-03	8.60E-02	1.40E-02
TRANS-1,2-DICHLOROETHENE	2.4	2.0	1.0	70.0	365.0	6.86E-05	2.00E-03	3.43E-02

SUMMARY HAZARD
INDEX 6.43E-01

Cadillac
Groundwater Ingestion
Noncarcinogenic Effects

Northernaire Plume (Present)
Realistic Worst Case
Adult

Compound	Concentration (ug/l)	Amount Ingested (l/day)	Toxicokinetic Factor (unitless)	Body Weight (kg)	Number of Exposures (per year)	Body Dose (mg/kg/day)	Standard/ Guideline (mg/kg/day)	Hazard Ratio
CHROMIUM	773.0	2.0	1.0	70.0	365.0	2.21E-02	5.00E-03	4.42E+00
CYANIDE	22.0	2.0	1.0	70.0	365.0	6.29E-04	2.90E-02	2.17E-02
1,1,1-TRICHLOROETHANE	440.0	2.0	1.0	70.0	365.0	1.26E-02	8.60E-02	1.46E-01
TRANS-1,2-DICHLOROETHENE	40.0	2.0	1.0	70.0	365.0	1.14E-03	2.00E-03	5.71E-01

SUMMARY HAZARD
INDEX 5.16E+00

Compound	Concentration (ug/l)	Amount Ingested (l/day)	Toxicokinetic Factor (unitless)	Body Weight (kg)	Number of Exposures (per year)	Number of Years of Exposure 70 yr lifetime	Body Dose (mg/kg/day)	CAS Potency Value (mg/kg/day) ⁻¹	Incremental Cancer Risk
1,1-DICHLOROETHENE	1.7	1.0	1.0	35.0	365.0	10.0	6.94E-06	5.80E-01	4.02E-06
TETRACHLOROETHENE	0.1	1.0	1.0	35.0	365.0	10.0	4.08E-07	5.10E-02	2.36E-08
TRICHLOROETHENE	261.7	1.0	1.0	35.0	365.0	10.0	8.23E-04	1.10E-02	9.06E-06
:	:	:	:	:	:	:	:	:	:
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								:Summary Incremental	:
								:Cancer Risk	1.31E-05 :

Compound	Concentration (ug/l)	Amount Ingested (l/day)	Toxicokinetic Factor (unitless)	Body Weight (kg)	Number of Exposures (per year)	Number of Years of Exposure (70 yr lifetime)	Body Dose (mg/kg/day)	CAS Potency Value (mg/kg/day) ⁻¹	Incremental Cancer Risk
1,1-DICHLOROETHENE	26.0	1.0	1.0	35.0	365.0	10.0	1.08E-04	5.80E-01	6.16E-05
TETRACHLOROETHENE	3.0	1.0	1.0	35.0	365.0	10.0	1.22E-05	9.10E-02	6.24E-07
TRICHLOROETHENE	2500.0	1.0	1.0	25.0	365.0	10.0	1.02E-02	1.10E-02	1.12E-04
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								:Summary Incremental :Cancer Risk	1.74E-04

Cadillac	Northern Pine (Present)
Groundwater Ingestion	Most Probable Case
Carcinogenic Effects	Adult

[illegible]

Faciliac	Northernaire Pluce (Present)
Groundwater Ingestion	Realistic Worst Case
Carcinogenic Effects	Adult

[illegible]

TABLE 8-16

Groundwater Ingestion
Noncarcinogenic Effects

RISK CALCULATION SUMMARY
Cadillac
Northern Plume (Present)

Compounds	Most Probable Case Child	Percent Total Risk	Realistic Worst Case Child	Percent Total Risk	Most Probable Case Adult	Percent Total Risk	Realistic Worst Case Adult	Percent Total Risk
CHROMIUM	0.59	92.4%	4.42	85.7%	0.59	92.4%	4.42	85.7%
CYANIDE	0.00	0.0%	0.02	0.4%	0.00	0.0%	0.02	0.4%
1,1,1-TRICHLOROETHANE	0.01	2.3%	0.15	2.8%	0.01	2.3%	0.15	2.8%
TRANS-1,2-DICHLOROETHENE	0.03	5.3%	0.57	11.1%	0.03	5.3%	0.57	11.1%
SUMMARY HAZARDOUS INGESTION		0.64		5.16		0.64		5.16

Groundwater Ingestion

Carcinogenic Effects

Compounds	Most Probable Case Child	Percent Total Risk	Realistic Worst Case Child	Percent Total Risk	Most Probable Case Adult	Percent Total Risk	Realistic Worst Case Adult	Percent Total Risk
1,1-DICHLOROETHENE	4.02E-06	30.7%	6.16E-05	35.3%	2.82E-05	30.7%	4.31E-04	35.3%
TETRACHLOROETHENE	2.08E-08	0.2%	6.24E-07	0.4%	1.46E-07	0.2%	4.37E-06	0.4%
TRICHLOROETHENE	9.06E-06	69.1%	1.12E-04	64.4%	6.34E-05	69.1%	7.86E-04	64.4%
SUMMARY INCREMENTAL CANCER RISK:		1.31E-05		1.74E-04		9.17E-05		1.22E-03

DISK: 213
FILE: INT_FUT

DISK: 213
FILE: INT_FUT

SITE: Cadillac: Inter. Kysor Plume
Receptor: Leeson Ave. (Future)

Exposure Variables	Most Probable	Realistic Worst	Most Probable	Realistic Worst
	Child	Child	Adult	Adult
Amount Ingested (l/day)	1.0	1.0	2.0	2.0
Number of Exposures (per year)	365.0	365.0	365.0	365.0
Number of Years of Exposure	10.0	10.0	70.0	70.0

Body Weight	Kilograms
Child	35
Adult	70

[illegible]

Noncarcinogenic Effects Child

Noncarcinogenic Effects Child

[illegible]

Cadillac: Inter. Kysor PluReceptor: Leeson Ave. (Future);
Groundwater Ingestion Most Probable Case
Noncarcinogenic Effects Adult

[illegible]

Cadillac Inter. Lysr Plu Receptor: Leeson Ave. (Future)
 Groundwater Ingestion Realistic Worst Case
 Noncarcinogenic Effects Adult

[illegible]

Carcinogenic Effects	Child
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Carcinogenic Effects:	Child
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[illegible]

Groundwater Ingestion	Most Probable Case
Carcinogenic Effects	Adult

Cadillac: Inter. Xysor Pluceptor: Leeson Ave. (Future)
 Groundwater Ingestion Realistic worst Case
 Carcinogenic Effects Adult

[illegible]

RISK CALCULATION SUMMARY
Cadillac: Inter. Kysor Plume
Receptor: Leeson Ave. (Future)

RISK CALCULATION SUMMARY
Cadillac: Inter. Kysor Plume
Receptor: Leeson Ave. (Future)

RISK CALCULATION SUMMARY
Cadillac: Inter. Kysor Plume
Receptor: Leeson Ave. (Future)

RISK CALCULATION SUMMARY
Cadillac: Inter. Kysor Plume
Receptor: Leeson Ave. (Future)

RISK CALCULATION SUMMARY
Cadillac: Inter. Kysor Plume
Receptor: Leeson Ave. (Future)

Cadillac: Four Star Plume Receptor: 13th Street (Child)

Groundwater Ingestion Most Probable Case

Carcinogenic Effects Child

Compound	Concentration (ug/l)	Amount Ingested (l/day)	Toxicokinetic Factor (unitless)	Body Weight (kg)	Number of Exposures (per year)	Number of Years of Exposure (70 yr lifetime)	Body Dose (mg/kg/day)	CAS Potency Value (mg/kg/day) ⁻¹	Incremental Cancer Risk
TETRACHLORDETHENE	585.0	1.0	1.0	35.0	365.0	10.0	2.39E-03	5.10E-02	1.22E-04
TRICHLORDETHENE	2450.0	1.0	1.0	35.0	365.0	10.0	1.00E-02	1.10E-02	1.10E-04

Summary Incremental

Cancer Risk 2.32E-04

Cadillac: Four Star Plume Receptor: 13th Street (Child)

Groundwater Ingestion Realistic Worst Case

Carcinogenic Effects Child

Compound	Concentration (ug/l)	Amount Ingested (l/day)	Toxicokinetic Factor (unitless)	Body Weight (kg)	Number of Exposures (per year)	Number of Years of Exposure (70 yr lifetime)	Body Dose (mg/kg/day)	CAS Potency Value (mg/kg/day) ⁻¹	Incremental Cancer Risk
TETRACHLORDETHENE	1740.0	1.0	1.0	35.0	365.0	10.0	7.10E-03	5.10E-02	3.62E-04
TRICHLORDETHENE	6730.0	1.0	1.0	35.0	365.0	10.0	2.75E-02	1.10E-02	3.02E-04

Summary Incremental

Cancer Risk 6.64E-04

RISK CALCULATION SUMMARY
Cadillac: Four Star Plus
Receptor: 13th Street (Child)

Noncarcinogenic Effects

Compounds	Most Probable Case		Realistic Worst Case		Most Probable Case		Realistic Worst Case	
	Child	Percent Total Risk	Child	Percent Total Risk	Adult	Percent Total Risk	Adult	Percent Total Risk
1,1,1-TRICHLOROETHANE	0.02	100.0%	0.04	100.0%				
SUMMARY HAZARD INDEX		0.02		0.04				

Groundwater Ingestion

Carcinogenic Effects

Compounds	Most Probable Case		Realistic Worst Case		Most Probable Case		Realistic Worst Case	
	Child	Percent Total Risk	Child	Percent Total Risk	Adult	Percent Total Risk	Adult	Percent Total Risk
TETRACHLOROETHENE	1.22E-04	52.5%	3.62E-04	54.5%				
TRICHLOROETHENE	1.10E-04	47.5%	3.02E-04	45.5%				
SUMMARY INCREMENTAL CANCER RISK:	2.32E-04		6.64E-04					